

# Prombot\_st125923\_A2

September 13, 2025

## 1 Task 1 - Implementation

Some notes about this assignment.

- I declare some of the type hints that basically tells what the data type should be. (Python does not enforce it even if I put it in)
- Some of the code in this jupyter notebook is copied into `utils` folder which I make it as a module so it makes some part of the experiments compatible with the module naming and type hinting dependency in pickle file when saving the model. For instance, saving the model with `mlflow` which its class initialized in jupyter notebook and loading it later with `mlflow` will cause the model class to error because the model class will be `__main__.LinearRegression` which the `__main__` is a namespace from Jupyter Notebook itself, when `mlflow` tries to use the pickled model, it will try to find module named `__main__` to import the class `LinearRegression` from which is impossible to do. So I have to copy the classes from this jupyter notebook to be in another module folder `utils` that can be imported inside a function to train and save the model. After doing this, the model module path will be `utils.model.LinearRegression`. Also I have to replicate the module folder again in backend server as well

### 1.1 Import dependencies and declaring some type hints

```
[47]: import itertools
      from abc import ABC, abstractmethod
      from dataclasses import dataclass
      from typing import Literal

      import mlflow
      import numpy as np
      import pandas as pd
      from numpy.typing import NDArray
      from sklearn.base import BaseEstimator, RegressorMixin
      from sklearn.compose import ColumnTransformer
      from sklearn.impute import SimpleImputer
      from sklearn.model_selection import KFold, train_test_split
      from sklearn.pipeline import Pipeline
      from sklearn.preprocessing import OneHotEncoder, OrdinalEncoder, \
      ↪PolynomialFeatures, StandardScaler

      # from utils import data_cleaning, get_preprocessor
```

```
[48]: # np.number is a base class for all integers, floating numbers in numpy
# In Python 3.12, type keyword was introduced to replace typing.TypeAlias
# read more about this https://docs.astral.sh/ruff/rules/non-pep695-type-alias/
type Vector = NDArray[np.integer | np.floating]
type Matrix = NDArray[np.integer | np.floating]
type Scalar = int | float | np.integer | np.floating

# These are old alternatives
# Vector: TypeAlias = NDArray[np.integer | np.floating]
# Matric: TypeAlias = NDArray[np.integer | np.floating]
# Scalar: TypeAlias = int | float | np.integer | np.floating

type TrainMethod = Literal["batch", "stochastic", "mini-batch"]
type WeightInitMethod = Literal["zero", "xavier"]

@dataclass
class KFoldScore:
    train_loss: float
    val_loss: float
    val_r2: float
```

## 1.2 Connecting to MLFlow

```
[3]: mlflow.set_tracking_uri("http://localhost:5000")
mlflow.set_experiment("Car Price Prediction")
```

```
[3]: <Experiment: artifact_location='mlflow-artifacts:/769694705593400528',
creation_time=1757233222067, experiment_id='769694705593400528',
last_update_time=1757233222067, lifecycle_stage='active', name='Car Price
Prediction', tags={}>
```

## 1.3 Declare Penalty Classes

```
[ ]: class BasePenalty(ABC):
    @abstractmethod
    def __call__(self, theta: Vector) -> Scalar:
        pass

    @abstractmethod
    def derivation(self, theta: Vector) -> Vector:
        pass

class LassoPenalty(BasePenalty):
    def __init__(self, l: Scalar) -> None:
```

```

        self.l = 1 # lambda value

# __call__ allows call class as method
def __call__(self, theta: Vector) -> Scalar:
    return self.l * np.sum(np.abs(theta[1:])) # do not regularize bias term

def derivation(self, theta: Vector) -> Vector:
    g = np.zeros_like(theta)
    g[1:] = self.l * np.sign(theta[1:]) # do not regularize bias term
    return g

class RidgePenalty(BasePenalty):
    def __init__(self, l: Scalar) -> None:
        self.l = l

    def __call__(self, theta: Vector) -> Scalar:
        return self.l * np.sum(np.square(theta[1:])) # do not regularize bias
↳ term

    def derivation(self, theta: Vector) -> Vector:
        g = np.zeros_like(theta)
        g[1:] = self.l * 2 * (theta[1:]) # do not regularize bias term
        return g

```

```

[ ]: # This is not used in the assignment but I implemented it anyway

# class ElasticPenalty(BasePenalty):
#     def __init__(self, l: Scalar = 0.1, l_ratio: Scalar = 0.5) -> None:
#         self.l = l
#         self.l_ratio = l_ratio

#     def __call__(self, theta: Vector) -> Scalar:
#         # The coefficient 1 is to avoid double counting of lambda
#         # Also, do not regularize the bias term but this is handled in Lasso
↳ and Ridge
#         # So just pass the full theta
#         l1_contribution = self.l_ratio * LassoPenalty(1)(theta)
#         l2_contribution = (1 - self.l_ratio) * RidgePenalty(1)(theta)
#         return self.l * (l1_contribution + l2_contribution)

#     def derivation(self, theta: Vector) -> Vector:
#         # The coefficient 1 and passing full theta is here as well
#         l1_derivation = self.l_ratio * LassoPenalty(1).derivation(theta)
#         l2_derivation = (1 - self.l_ratio) * RidgePenalty(1).derivation(theta)
#         return self.l * (l1_derivation + l2_derivation)

```

## 1.4 Declare the Model Class and its variants

```
[ ]: class LinearRegression(BaseEstimator, RegressorMixin):
    cv = KFold(n_splits=3, shuffle=True, random_state=42)

    def __init__(
        self,
        regularization: BasePenalty | None,
        momentum: float,
        train_method: TrainMethod,
        weight_init_method: WeightInitMethod,
        lr: Scalar,
        num_epochs: int,
        batch_size: int | None,
        params_for_logging: dict,
    ) -> None:
        if (train_method in {"batch", "stochastic"}) and batch_size is not None:
            print("Error: Batch size should be None for batch or stochastic_
↪gradient descent.")
            return
        if train_method == "mini-batch" and (batch_size is None or batch_size_
↪<= 0):
            print("Error: Please provide a valid batch size for mini-batch_
↪gradient descent.")
            return
        if regularization is not None and not issubclass(type(regularization),_
↪BasePenalty):
            print("Error: regularization should be a subclass of BasePenalty or_
↪None.")
            return

        self.regularization = regularization
        self.momentum = momentum
        self.train_method = train_method
        self.weight_init_method = weight_init_method
        self.lr = lr
        self.num_epochs = num_epochs
        self.batch_size = batch_size
        self.params_for_logging = params_for_logging

    def mse(self, ytrue: Vector, ypred: Vector) -> Scalar:
        return ((ypred - ytrue) ** 2).sum() / ytrue.shape[0]

    def r_squared(self, ytrue: Vector, ypred: Vector) -> Scalar:
        # Formula is 1 - SSres / SStot
        # SSres = Sum of squared residual
        # SStot = Sum of squared totsl (compared to the mean or ybar)
```

```

        return 1 - np.sum(np.square(ytrue - ypred)) / np.sum(np.square(ytrue -
↪ytrue.mean()))

    def xavier_weight_init(self, m: int, n: int) -> Vector:
        # Since this method use number of samples to calculate a new uniform
↪range,
        # it requires both m and n to initialize theta (n, )
        upper = 1 / np.sqrt(m)
        lower = -(upper)
        numbers = np.random.rand(n)
        # The scaled output is calculated by
        # 1. streching U(0, 1) into U(0, upper - lower)
        #    with upper - lower = length between lower and upper
        # 2. Add lower to shift all the numbers back to U(lower, upper)
        return lower + numbers * (upper - lower)

    def zero_weight_init(self, n: int) -> Vector:
        return np.zeros(n)

    def weight_init(self, X: Matrix) -> Vector:
        if self.weight_init_method == "zero":
            self.theta = np.zeros(X.shape[1])
        elif self.weight_init_method == "xavier":
            self.theta = self.xavier_weight_init(X.shape[0], X.shape[1])

    def add_intercept(self, X: Matrix) -> Matrix:
        return np.hstack([np.ones((X.shape[0], 1)), X])

    def fit(self, X_train: Matrix, y_train: Vector) -> None:
        # create a list of kfold scores
        self.kfold_scores: list[KFoldScore] = []

        # kfold.split in the sklearn.....
        for fold, (train_idx, val_idx) in enumerate(self.cv.split(X_train)):
            # reset val loss at the start of each fold, not outside the fold
↪loop
            self.val_loss_old = np.inf
            self.val_r2_old = -np.inf

            X_cross_train = X_train[train_idx]
            y_cross_train = y_train[train_idx]
            X_cross_val = X_train[val_idx]
            y_cross_val = y_train[val_idx]

            # add intercept column to training set as well as validation set
            # so that theta[0] is the bias / intercept
            # X_cross_train has size (m, n+1)

```

```

# X_cross_val has size (m, n+1)
X_cross_train = self.add_intercept(X_cross_train)
X_cross_val = self.add_intercept(X_cross_val)

# define X_cross_train as only a subset of the data
# how big is this subset? => mini-batch size ==> 50

# initialize theta for each fold
self.weight_init(X_cross_train)
# This should be vector of same shape as theta
# but initialized to zero as it will be broadcasted anyway
self.prev_step = 0 # for momentum

# one epoch will exhaust the WHOLE training set
with mlflow.start_run(run_name=f"Fold-{fold}", nested=True):
    mlflow.log_params(self.params_for_logging)

    for epoch in range(self.num_epochs):
        # with replacement or no replacement
        # with replacement means just randomize
        # with no replacement means 0:50, 51:100, 101:150, .....

        # shuffle your index
        # https://docs.astral.sh/ruff/rules/numpy-legacy-random/
        perm = np.random.permutation(X_cross_train.shape[0])

        X_cross_train = X_cross_train[perm]
        y_cross_train = y_cross_train[perm]

        train_loss = self.start_training(X_cross_train,
        ↪ y_cross_train)

        yhat_val = self.predict(X_cross_val, add_intercept=False)
        val_loss_new = self.mse(y_cross_val, yhat_val)
        val_r2_new = self.r_squared(y_cross_val, yhat_val)
        mlflow.log_metrics(
            {
                "train_loss": train_loss,
                "val_loss": val_loss_new,
                "val_r2": val_r2_new,
            },
            step=epoch,
        )

        # early stopping
        if np.allclose(val_loss_new, self.val_loss_old):
            break

```

```

        self.val_loss_old = val_loss_new
        self.val_r2_old = val_r2_new

        # Log all the metrics for all folds
        self.kfold_scores.append(
            KFoldScore(
                train_loss=train_loss,
                val_loss=val_loss_new,
                val_r2=val_r2_new,
            ),
        )
        print(f"Fold {fold} Train Loss: {train_loss}")
        print(f"Fold {fold} Val Loss: {val_loss_new}")
        print(f"Fold {fold} Val R2: {val_r2_new}")

    def start_training(self, X_train: Matrix, y_train: Vector) -> float:
        if self.train_method == "stochastic":
            for batch_idx in range(X_train.shape[0]):
                # (11,) ==> (1, 11) ==> (m, n)
                X_method_train = X_train[batch_idx].reshape(1, -1)
                y_method_train = y_train[batch_idx].reshape(1) # So this has
↪shape (1,) -> not scalar
                train_loss = self._train(X_method_train, y_method_train)
            elif self.train_method == "mini-batch":
                for batch_idx in range(0, X_train.shape[0], self.batch_size):
                    # batch_idx = 0, 50, 100, 150
                    X_method_train = X_train[batch_idx : batch_idx + self.
↪batch_size, :]
                    y_method_train = y_train[batch_idx : batch_idx + self.
↪batch_size]
                    train_loss = self._train(X_method_train, y_method_train)
            elif self.train_method == "batch":
                X_method_train = X_train
                y_method_train = y_train
                train_loss = self._train(X_method_train, y_method_train)
            else:
                msg = "Error: Please select correct training method."
                raise ValueError(msg)
        return train_loss

    def _train(self, X: Matrix, y: Vector) -> Scalar:
        yhat = self.predict(X, add_intercept=False)
        m = X.shape[0]
        grad_loss = (2 / m) * X.T @ (yhat - y)
        if self.regularization is not None: # noqa: SIM108
            grad_penalty = self.regularization.derivation(self.theta) / m
        else:

```

```

        grad_penalty = 0
        grad = grad_loss + grad_penalty
        step = self.lr * grad
        # If momentum is 0, second term is 0, so no momentum
        self.theta = self.theta - step + self.momentum * self.prev_step
        self.prev_step = step

    return self.mse(y, yhat)

    # Added automatic adding intercept so that when calling the model have
    ↪ nicer interface to work with
    def predict(self, X: Matrix, *, add_intercept: bool = True) -> Vector:
        if add_intercept:
            X = self.add_intercept(X)
        return X @ self.theta # ==>(m, n) @ (n, )

    def _coef(self) -> Vector:
        return self.theta[1:] # remind that theta is (w0, w1, w2, w3, w4.....
    ↪ wn)
        # w0 is the bias or the intercept
        # w1....wn are the weights / coefficients / theta

    def _bias(self) -> Scalar:
        return self.theta[0]

    def get_average_kfold_scores(self) -> KFoldScore:
        # return the average of the kfold scores
        return KFoldScore(**pd.DataFrame(self.kfold_scores).mean().to_dict())

    def plot_feature_importance(self, feature_names: list[str]) -> None:
        import matplotlib.pyplot as plt

        coef = self._coef()
        if len(coef) != len(feature_names):
            print("Error: Length of feature names must match number of features.
    ↪ ")
            return

        feature_importance = pd.Series(coef, index=feature_names).
    ↪ sort_values(ascending=False)

        plt.figure(figsize=(10, 6))
        feature_importance.plot(kind="barh")
        plt.title("Feature Importance")
        plt.ylabel("Coefficient Value")
        plt.xlabel("Feature")
        plt.tight_layout()

```



```

plt.show()

class Lasso(LinearRegression):
    def __init__(
        self,
        l: Scalar,
        momentum: float,
        train_method: TrainMethod,
        weight_init_method: WeightInitMethod,
        lr: Scalar,
        num_epochs: int,
        batch_size: int | None,
        params_for_logging: dict,
    ) -> None:
        super().__init__(
            regularization=LassoPenalty(l),
            momentum=momentum,
            train_method=train_method,
            weight_init_method=weight_init_method,
            lr=lr,
            num_epochs=num_epochs,
            batch_size=batch_size,
            params_for_logging=params_for_logging,
        )

class Ridge(LinearRegression):
    def __init__(
        self,
        l: Scalar,
        momentum: float,
        train_method: TrainMethod,
        weight_init_method: WeightInitMethod,
        lr: Scalar,
        num_epochs: int,
        batch_size: int | None,
        params_for_logging: dict,
    ) -> None:
        super().__init__(
            regularization=RidgePenalty(l),
            momentum=momentum,
            train_method=train_method,
            weight_init_method=weight_init_method,
            lr=lr,
            num_epochs=num_epochs,
            batch_size=batch_size,

```

```

        params_for_logging=params_for_logging,
    )

```

```
[ ]: # This is not used in the assignment but I implemented it anyway
```

```

# class ElasticNet(LinearRegression):
#     def __init__(
#         self,
#         l: Scalar,
#         l_ratio: Scalar,
#         momentum: float,
#         train_method: TrainMethod,
#         weight_init_method: WeightInitMethod,
#         lr: Scalar,
#         num_epochs: int,
#         batch_size: int | None,
#         params_for_logging: dict,
#     ) -> None:
#         super().__init__(
#             regularization=ElasticPenalty(l, l_ratio),
#             momentum=momentum,
#             train_method=train_method,
#             weight_init_method=weight_init_method,
#             lr=lr,
#             num_epochs=num_epochs,
#             batch_size=batch_size,
#             params_for_logging=params_for_logging,
#         )

```

## 1.5 Preparing the data

### 1.5.1 Data Cleaning

- I just use the old function from Assignment 1 so the data stays the same

```

[49]: def data_cleaning(input_df: pd.DataFrame) -> pd.DataFrame:
    df = input_df.copy()

    # Drop unused columns first to reduce size
    df = df.drop(columns=["torque"])

    # Drop rows that will definitely not be used
    df = df[~df["fuel"].isin(["CNG", "LPG"])]
    df = df[df["owner"] != "Test Drive Car"]

    # Remove Units
    df["mileage"] = df["mileage"].str.split(" ").str[0]
    df["engine"] = df["engine"].str.split(" ").str[0]

```

```

df["max_power"] = df["max_power"].str.split(" ").str[0]

# Numeric Type conversion
df["mileage"] = df["mileage"].astype(float)
df["engine"] = df["engine"].astype(float)
df["max_power"] = df["max_power"].astype(float)

# Fix Brand column
df = df.rename(columns={"name": "brand"})
df["brand"] = df["brand"].str.split(" ").str[0]

return df

# Load, Clean, Split, Preprocess Data
df = pd.read_csv("Cars.csv")
df_train, df_test = train_test_split(df, random_state=42, test_size=0.2)
df_train = data_cleaning(df_train)
df_test = data_cleaning(df_test)
X_train = df_train.drop(columns=["selling_price"])
X_test = df_test.drop(columns=["selling_price"])
y_train = df_train["selling_price"]
y_test = df_test["selling_price"]

```

```
[8]: X_train.shape
```

```
[8]: (6421, 11)
```

```

[9]: print("X_train shape:", X_train.shape, type(X_train))
      print("X_test shape:", X_test.shape, type(X_test))
      print("y_train shape:", y_train.shape, type(y_train))
      print("y_test shape:", y_test.shape, type(y_test))

```

```

X_train shape: (6421, 11) <class 'pandas.core.frame.DataFrame'>
X_test shape: (1607, 11) <class 'pandas.core.frame.DataFrame'>
y_train shape: (6421,) <class 'pandas.core.series.Series'>
y_test shape: (1607,) <class 'pandas.core.series.Series'>

```

### 1.5.2 Preprocessor

- Preprocessor is mostly the same as Assignment 1 with some tweak for polynomial features for numerical values, I will explain in the report at the end why numerical features only.

```

[ ]: # This function was added to utils folder for easy workflow during the
      ↪assignment period
      # But I paste it here so that it shows up in the file for submission

```

```

def get_preprocessor(polynomial_degree: int | None = None) -> ColumnTransformer:
    # Define transformers for each feature group
    brand_le = (
        "brand_le",
        Pipeline(
            [
                # Pre-imputer for handling missing values before encoding
                ("pre_imputer", SimpleImputer(strategy="most_frequent")),
                ("encoder", OrdinalEncoder(handle_unknown="use_encoded_value",
↪unknown_value=np.nan)),
                ("post_imputer", SimpleImputer(strategy="most_frequent")),
                # Post-imputer for handling any NaN introduced by unknown
↪categories during encoding
                # So the pipeline is robust to unseen categories in production
↪data
                # This won't happen if frontend limits the choices to existing
↪categories only
                # But it's better to be safe than sorry
            ],
        ),
        ["brand"],
    )

    fuel_le = (
        "fuel_le",
        Pipeline(
            [
                ("pre_imputer", SimpleImputer(strategy="most_frequent")),
                ("encoder", OrdinalEncoder(handle_unknown="use_encoded_value",
↪unknown_value=np.nan)),
                ("post_imputer", SimpleImputer(strategy="most_frequent")),
            ],
        ),
        ["fuel"],
    )

    transmission_le = (
        "transmission_le",
        Pipeline(
            [
                ("pre_imputer", SimpleImputer(strategy="most_frequent")),
                ("encoder", OrdinalEncoder(handle_unknown="use_encoded_value",
↪unknown_value=np.nan)),
                ("post_imputer", SimpleImputer(strategy="most_frequent")),
            ],
        ),
    )

```

```

        ["transmission"],
    )

    owner_le = (
        "owner_le",
        Pipeline(
            [
                ("pre_imputer", SimpleImputer(strategy="most_frequent")),
                (
                    "encoder",
                    OrdinalEncoder(
                        categories=[
                            "First Owner",
                            "Second Owner",
                            "Third Owner",
                            "Fourth & Above Owner",
                        ],
                    ),
                    handle_unknown="use_encoded_value",
                    unknown_value=np.nan,
                ),
                ("post_imputer", SimpleImputer(strategy="most_frequent")),
            ],
        ),
        ["owner"],
    )

    seller_type_ohe = (
        "seller_type_ohe",
        Pipeline(
            [
                ("imputer", SimpleImputer(strategy="most_frequent")),
                ("encoder", OneHotEncoder(sparse_output=False, drop="first",
↪handle_unknown="ignore")),
            ],
        ),
        ["seller_type"],
    )

    pipeline_steps = [
        (
            "imputer",
            ColumnTransformer(
                transformers=[

```

```

        ("year_imputer", SimpleImputer(strategy="median"),
↪["year"]),
        ("km_imputer", SimpleImputer(strategy="median"),
↪["km_driven"]),
        ("mileage_imputer", SimpleImputer(strategy="mean"),
↪["mileage"]),
        ("engine_imputer", SimpleImputer(strategy="median"),
↪["engine"]),
        ("max_power_imputer", SimpleImputer(strategy="median"),
↪["max_power"]),
        ("seats_imputer", SimpleImputer(strategy="most_frequent"),
↪["seats"]),
    ],
    remainder="passthrough",
),
("scaler", StandardScaler()),
]

# This is the added polynomial feature step
if polynomial_degree is not None:
    pipeline_steps.insert(-1, ("poly",
↪PolynomialFeatures(degree=polynomial_degree, include_bias=False)))

num_scaler = (
    "num_scaler",
    Pipeline(pipeline_steps),
    ["year", "km_driven", "mileage", "engine", "max_power", "seats"],
)

# Combine all transformers into the ColumnTransformer
return ColumnTransformer(
    transformers=[
        brand_le,
        fuel_le,
        transmission_le,
        owner_le,
        seller_type_ohe,
        num_scaler,
    ],
    remainder="passthrough",
    verbose_feature_names_out=True,
)

```

## 1.6 Preparing the experiments

- Here I use itertools to create all combinations of the parameter sets and convert them into list of dictionaries, each dictionary has 5 keys associated with each parameters. The reason I use this data structure so i can loop through list of them and pass the param dict inside to populate the experiment and for logging inside each cv for easy reference as well.

```
[10]: model_types = ["Normal", "Ridge", "Lasso", "Polynomial"]
momentum = [0, 0.9] # 0 means no momentum
train_methods = ["batch", "stochastic", "mini-batch"]
weight_init_methods = ["zero", "xavier"]
learning_rates = [0.01, 0.001, 0.0001]

all_combinations = list(
    itertools.product(
        model_types,
        momentum,
        train_methods,
        weight_init_methods,
        learning_rates,
    ),
)
keys = ["model_type", "momentum", "train_method", "weight_init_method",
        ↪ "learning_rate"]
params_combinations = [dict(zip(keys, comb, strict=False)) for comb in
        ↪ all_combinations]
len(params_combinations)
```

[10]: 144

```
[ ]: def start_experiment(
    X_train_original: pd.DataFrame,
    y_train_original: pd.Series,
    params_combinations: list[dict],
    num_epochs: int = 500,
    batch_size: int = 32,
) -> None:
    for i, params in enumerate(params_combinations):
        # Populate all params with num_epochs, batch_size, and lambda (if
        ↪ applicable)
        params["num_epochs"] = num_epochs
        if params["train_method"] == "mini-batch":
            params["batch_size"] = batch_size
            train_method_log_string =
            ↪ f"{params['train_method']}_BatchSize-{params['batch_size']}"
        else:
            params["batch_size"] = None
```

```

        train_method_log_string = f"{params['train_method']}"
    if params["model_type"] in {"Ridge", "Lasso":
        params["lambda"] = 0.1

    log_string = (
        f"Reg-{params['model_type']}"
        f"_TrainMethod-{train_method_log_string}"
        f"_Momentum-{params['momentum']}"
        f"_WeightInit-{params['weight_init_method']}"
        f"_LR-{params['learning_rate']}"
    )
    mlflow.start_run(run_name=log_string)
    mlflow.log_params(params)

    print("==== Experiment: ", i + 1, " / ",
    len(params_combinations), "====")
    print(log_string)

    if params["model_type"] in ["Normal", "Polynomial"]:
        model = LinearRegression(
            regularization=None,
            momentum=params["momentum"],
            train_method=params["train_method"],
            weight_init_method=params["weight_init_method"],
            lr=params["learning_rate"],
            num_epochs=params["num_epochs"],
            batch_size=params["batch_size"],
            params_for_logging=params,
        )
    elif params["model_type"] == "Ridge":
        model = Ridge(
            l=params["lambda"],
            momentum=params["momentum"],
            train_method=params["train_method"],
            weight_init_method=params["weight_init_method"],
            lr=params["learning_rate"],
            num_epochs=params["num_epochs"],
            batch_size=params["batch_size"],
            params_for_logging=params,
        )
    elif params["model_type"] == "Lasso":
        model = Lasso(
            l=params["lambda"],
            momentum=params["momentum"],
            train_method=params["train_method"],
            weight_init_method=params["weight_init_method"],
            lr=params["learning_rate"],

```



```

        num_epochs=params["num_epochs"],
        batch_size=params["batch_size"],
        params_for_logging=params,
    )
else:
    print("Error: Please select correct model type.")
    break

# For polynomial model, add polynomial features step in the preprocessor
if params["model_type"] == "Polynomial":
    pipeline = Pipeline(steps=[("preprocess",
    ↪get_preprocessor(polynomial_degree=2))])
else:
    pipeline = Pipeline(steps=[("preprocess", get_preprocessor())])
    pipeline.steps.append(("model", model))

# After adding all the steps, fit the pipeline
# The reason to fit the whole pipeline here instead of just the model
# is to ensure that the shape of X after preprocessing will match the
↪input of model
# so that during prediction, I can just call pipeline.predict(X)
↪directly
# using the original X_train without any preprocessing
# which will be useful during inference in production where input data
↪is raw but human-readable
    pipeline.fit(X_train_original, y_train_original)

# Get model from pipeline
# (last step so index = -1)
# and it's a tuple of (name, model) so get the model with [1]
    fitted_model: LinearRegression = pipeline.steps[-1][1]

# After fitting the model, get average kfold scores of all folds,
# Because each fold has its own theta after fitting, which the theta
↪comes from
# randomized kfold split, so I cannot just use the theta from the last
↪fold
# to evaluate the model performance.
    avg_kfold_scores: KFoldScore = fitted_model.get_average_kfold_scores()

    print("Average training loss (MSE): ", avg_kfold_scores.train_loss)
    print("Average validation loss (MSE): ", avg_kfold_scores.val_loss)
    print("Average validation R2: ", avg_kfold_scores.val_r2)

    mlflow.log_metric(key="avg_train_mse", value=avg_kfold_scores.
    ↪train_loss)

```

```

mlflow.log_metric(key="avg_val_mse", value=avg_kfold_scores.val_loss)
mlflow.log_metric(key="avg_val_r2", value=avg_kfold_scores.val_r2)

# No logging the model for now as i need to fit model again
# based on the the best average val loss

# Below this is the old code provided in Regularization.ipynb
# which saves only the last fold model which is not correct
# I will refit the model with the best params and save the model later
# signature = mlflow.models.infer_signature(X_train_original, pipeline.
↪predict(X_train_original))
# mlflow.sklearn.log_model(pipeline, name="skpipeline",
↪signature=signature)

mlflow.end_run()

```

```
[13]: params_combinations[107]
```

```
[13]: {'model_type': 'Lasso',
      'momentum': 0.9,
      'train_method': 'mini-batch',
      'weight_init_method': 'xavier',
      'learning_rate': 0.0001}
```

## 2 Task 2

### 2.1 Run the experiments

```
[ ]: mlflow.end_run() # just in case there is some run stuck
```

```
[18]: # suppress overflow warnings for log
import warnings

with warnings.catch_warnings():
    warnings.simplefilter("ignore", RuntimeWarning)
    start_experiment(
        X_train,
        np.log(y_train.to_numpy()),
        params_combinations=params_combinations,
        num_epochs=500,
        batch_size=50, # only used for mini-batch training, will be ignored
↪otherwise
    )

```

```

===== Experiment: 1 / 144 =====
Reg-Normal_TrainMethod-batch-Momentum-0-WeightInit-zero_LR-0.01
Fold 0 Train Loss: inf

```

```

Fold 0 Val Loss:    inf
Fold 0 Val R2:      -inf
  View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/b6c7ef49dc884734a69e0f4b74348e20
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: inf
Fold 1 Val Loss:    inf
Fold 1 Val R2:      -inf
  View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/c883802a3e3347dc9ad940257e696bf4
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: inf
Fold 2 Val Loss:    inf
Fold 2 Val R2:      -inf
  View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/5721ff64e7404aac913cf9a557f82093
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE):  inf
Average validation loss (MSE):  inf
Average validation R2:  -inf
  View run Reg-Normal_TrainMethod-batch_Momentum-0_WeightInit-zero_LR-0.01 at: http://localhost:5000/#/experiments/769694705593400528/runs/13e082e0fa874dbc87948bb105bc8813
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment:  2 / 144 =====
Reg-Normal_TrainMethod-batch_Momentum-0_WeightInit-zero_LR-0.001
Fold 0 Train Loss: 11.975079568466512
Fold 0 Val Loss:   12.097783612252575
Fold 0 Val R2:     -16.365669948314217
  View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/e5f0d7f3470841458830386140b34a3d
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: 12.083536035356094
Fold 1 Val Loss:   11.477878254543468
Fold 1 Val R2:     -14.967174799852245
  View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/38c04585fe3f443f95d1fb1c7589e0a7
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 12.019689702116025
Fold 2 Val Loss:   12.651457301612359
Fold 2 Val R2:     -16.951608005682036
  View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/40bbeab4e2f342fab264e5120241bb6a
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE):  12.026101768646209
Average validation loss (MSE): 12.075706389469467
Average validation R2:  -16.094817584616166
  View run Reg-Normal_TrainMethod-batch_Momentum-0_WeightInit-zero_LR-0.001 at:

```

<http://localhost:5000/#/experiments/769694705593400528/runs/25bafefc63a24461bb488104d674e437>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 3 / 144 =====

Reg-Normal\_TrainMethod-batch\_Momentum-0\_WeightInit-zero\_LR-0.0001

Fold 0 Train Loss: 22.949542079913442

Fold 0 Val Loss: 22.726925296053363

Fold 0 Val R2: -31.623189195708377

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/de5897dc25c846f18848668fbd9ff13d>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 23.271790469490657

Fold 1 Val Loss: 22.044336099195423

Fold 1 Val R2: -29.66644897572548

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e8dbc11c30d046618a3aca1f05f99216>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 22.445168378410298

Fold 2 Val Loss: 23.91038964956118

Fold 2 Val R2: -32.92731224705128

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/fabcab0d806c4494a9a78e2aede621f6>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 22.888833642604798

Average validation loss (MSE): 22.893883681603324

Average validation R2: -31.405650139495048

View run Reg-Normal\_TrainMethod-batch\_Momentum-0\_WeightInit-zero\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/017b6875ee1441f5bfb158f1afd963b7>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 4 / 144 =====

Reg-Normal\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.01

Fold 0 Train Loss: inf

Fold 0 Val Loss: inf

Fold 0 Val R2: -inf

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/bdcdb25181b144278ec61d4de2a6fbb4>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: inf

Fold 1 Val Loss: inf

Fold 1 Val R2: -inf

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/533d07e45bbbc49febee0b18da717b7e1>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: inf

Fold 2 Val Loss: inf

Fold 2 Val R2: -inf

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/run>

s/83ad1f502be343ab9065c4350e968f8e  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): inf  
Average validation loss (MSE): inf  
Average validation R2: -inf  
View run Reg-Normal\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.01 at:  
<http://localhost:5000/#/experiments/769694705593400528/runs/b3ec879409e7431d88a4577fc38128a7>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 5 / 144 =====  
Reg-Normal\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.001  
Fold 0 Train Loss: 11.97869905941238  
Fold 0 Val Loss: 12.10856182299838  
Fold 0 Val R2: -16.38114144759409  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/d51106c3ec2d40968c9edc6f9c9fa912>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 12.062005473440042  
Fold 1 Val Loss: 11.458995918824902  
Fold 1 Val R2: -14.94090709180015  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f17405b92fd64735a376cb483bb7a7ad>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 12.004431883776684  
Fold 2 Val Loss: 12.630677794692552  
Fold 2 Val R2: -16.922123215599537  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e9df18c85eab4d1bb6689db5ac238a31>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 12.015045472209701  
Average validation loss (MSE): 12.066078512171947  
Average validation R2: -16.081390584997923  
View run Reg-Normal\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.001  
at: <http://localhost:5000/#/experiments/769694705593400528/runs/3fc02ea2897a4cd1873358642a9bf4e2>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 6 / 144 =====  
Reg-Normal\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.0001  
Fold 0 Train Loss: 22.89363028907115  
Fold 0 Val Loss: 22.67162885865396  
Fold 0 Val R2: -31.543814352185542  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/85c81ea61e564af883e3c966888d8acf>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 23.309975464534215  
Fold 1 Val Loss: 22.080448298405102  
Fold 1 Val R2: -29.716685594759124  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs>

```

s/a3af3ec5b5f14211b273c8ef7c25bfcc
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 22.42904371367591
Fold 2 Val Loss: 23.893357636067382
Fold 2 Val R2: -32.903144910237906
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/run
s/4d088d928cc849719dcb2ebbe5806396
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE): 22.877549822427095
Average validation loss (MSE): 22.881811597708815
Average validation R2: -31.387881619060853
View run Reg-Normal_TrainMethod-batch_Momentum-0_WeightInit-xavier_LR-0.0001
at: http://localhost:5000/#/experiments/769694705593400528/runs/ab4eee6fa3864d44
aa698e4f63bc8abf
View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment: 7 / 144 =====
Reg-Normal_TrainMethod-stochastic_Momentum-0_WeightInit-zero_LR-0.01
Fold 0 Train Loss: nan
Fold 0 Val Loss: nan
Fold 0 Val R2: nan
View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/run
s/a99f0f02444741e7905f7952c2b7985b
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: nan
Fold 1 Val Loss: nan
Fold 1 Val R2: nan
View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/run
s/7d79996e2263435aa9359ece80daae81
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: nan
Fold 2 Val Loss: nan
Fold 2 Val R2: nan
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/run
s/1e887cbd426c46848de2d34425e04e2b
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE): nan
Average validation loss (MSE): nan
Average validation R2: nan
View run Reg-Normal_TrainMethod-stochastic_Momentum-0_WeightInit-zero_LR-0.01
at: http://localhost:5000/#/experiments/769694705593400528/runs/e9da2bfb68044974
b6f98ef1a2004e54
View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment: 8 / 144 =====
Reg-Normal_TrainMethod-stochastic_Momentum-0_WeightInit-zero_LR-0.001
Fold 0 Train Loss: 0.0014044362078914584
Fold 0 Val Loss: 0.11678922261116916
Fold 0 Val R2: 0.832355813396126
View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/run

```

s/7338fdef43ee44c593e6e960d657d3a7

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 0.14368656865584564

Fold 1 Val Loss: 0.21061894715375606

Fold 1 Val R2: 0.7070025077122908

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/dc055eff5ee64453998809beebf28583>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 0.05253812396083354

Fold 2 Val Loss: 0.10021683402664351

Fold 2 Val R2: 0.8577987280732061

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/8b7d4d4f0bec4dbb986a54ca95923f5f>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 0.06587637627485689

Average validation loss (MSE): 0.1425416679305229

Average validation R2: 0.7990523497272076

View run Reg-Normal\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/5d2a41ab1deb4bad a37c962bdedaf951>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 9 / 144 =====

Reg-Normal\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.0001

Fold 0 Train Loss: 0.11890949879332118

Fold 0 Val Loss: 0.1583164081941564

Fold 0 Val R2: 0.7727459359317801

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/05ba699242a94363a9301ce3f6941533>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 0.0540730256969829

Fold 1 Val Loss: 0.08728469206400838

Fold 1 Val R2: 0.878575995961134

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/65135b683afd43e2ae09e37ca57aebb8>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 0.05941368598382313

Fold 2 Val Loss: 0.09385270801932899

Fold 2 Val R2: 0.8668290154668581

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e6ad72dc622c438ebf76e23b14ddeec>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 0.07746540349137573

Average validation loss (MSE): 0.11315126942583126

Average validation R2: 0.8393836491199241

View run Reg-Normal\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/50556fdb025a43baaff295a3625a929e>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

```

===== Experiment: 10 / 144 =====
Reg-Normal_TrainMethod-stochastic_Momentum-0_WeightInit-xavier_LR-0.01
Fold 0 Train Loss: nan
Fold 0 Val Loss: nan
Fold 0 Val R2: nan
View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/188917c91cf3442a9efbdbae9dc6c092
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: nan
Fold 1 Val Loss: nan
Fold 1 Val R2: nan
View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/71eed8f2e4064fe4be0181c6a4bbe450
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: nan
Fold 2 Val Loss: nan
Fold 2 Val R2: nan
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/080dc07dd58541928ba9db14eb3f1a84
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE): nan
Average validation loss (MSE): nan
Average validation R2: nan
View run Reg-Normal_TrainMethod-stochastic_Momentum-0_WeightInit-xavier_LR-0.01 at: http://localhost:5000/#/experiments/769694705593400528/runs/e3ef02d299b24d3a91786d96d8dabaf1
View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment: 11 / 144 =====
Reg-Normal_TrainMethod-stochastic_Momentum-0_WeightInit-xavier_LR-0.001
Fold 0 Train Loss: 0.11561084235953517
Fold 0 Val Loss: 0.12676559447581498
Fold 0 Val R2: 0.8180353075385387
View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/7f3fb1fcdclc4d8d94ff31d962893807
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: 0.07947322535385834
Fold 1 Val Loss: 0.1370673920509266
Fold 1 Val R2: 0.8093219879405251
View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/ad1a699bbdf948d6a01c06ee669f0b31
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 0.1704859577246799
Fold 2 Val Loss: 0.10103552195617924
Fold 2 Val R2: 0.8566370623109436
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/99ac82c24b0744fca1ed967906e8667e
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE): 0.12185667514602445

```



Average validation loss (MSE): 0.12162283616097362  
Average validation R2: 0.8279981192633358  
View run Reg-Normal\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/c86f9c7e7ee9427e80c8df7821e14369>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 12 / 144 =====  
Reg-Normal\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.0001  
Fold 0 Train Loss: 0.022422054920936674  
Fold 0 Val Loss: 0.10823542033449725  
Fold 0 Val R2: 0.8446343027377092  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/56ace7cbc40244429c02875b67ef8cbf>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 0.07880601199762921  
Fold 1 Val Loss: 0.08665149455043153  
Fold 1 Val R2: 0.879456853481827  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/fc4d76db44a44ebbbb1bea8b1508aafb>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 0.09917496380822957  
Fold 2 Val Loss: 0.09320248193558624  
Fold 2 Val R2: 0.8677516446543219  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/02e02d93f4384436bd9ba31603bfa322>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 0.06680101024226515  
Average validation loss (MSE): 0.09602979894017166  
Average validation R2: 0.863947600291286  
View run Reg-Normal\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/c3165b4a21494db4b8c7f786702dc455>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 13 / 144 =====  
Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.01  
Fold 0 Train Loss: inf  
Fold 0 Val Loss: inf  
Fold 0 Val R2: -inf  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f5a8bdba8a7e41c28920ebc22f2ea274>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: inf  
Fold 1 Val Loss: inf  
Fold 1 Val R2: -inf  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/8d5aaa81dddb446a98571c02ad907856>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: inf  
 Fold 2 Val Loss: inf  
 Fold 2 Val R2: -inf  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/9e18db6e3b7c41b0b8c989b45a187249>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): inf  
 Average validation loss (MSE): inf  
 Average validation R2: -inf  
 View run Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/c577ff73fba140a9a611450424856521>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 14 / 144 =====  
 Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.001  
 Fold 0 Train Loss: 0.04624634905846925  
 Fold 0 Val Loss: 0.13605373407653737  
 Fold 0 Val R2: 0.8047027193629117  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/38702664f7e14616bb92dc8b8b679c37>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: 0.23700230872882194  
 Fold 1 Val Loss: 0.16654803008413555  
 Fold 1 Val R2: 0.7683107060425762  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/cf69dd52667942b980abfaedbc709e5b>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 0.10077394910988821  
 Fold 2 Val Loss: 0.1192829552225985  
 Fold 2 Val R2: 0.8307451226474509  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/0784802a7c3c4a088f7138a207687d18>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 0.12800753563239312  
 Average validation loss (MSE): 0.14062823979442382  
 Average validation R2: 0.8012528493509796  
 View run Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/b26479a88c194be8a947276a7b4acced>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 15 / 144 =====  
 Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.0001  
 Fold 0 Train Loss: 2.7761767005820728  
 Fold 0 Val Loss: 2.562515377632426  
 Fold 0 Val R2: -2.678342885913028  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs>

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s/48f6d8eebc584b8bb2d403e022da19d4
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: 3.056640060997847
Fold 1 Val Loss: 2.648048795518529
Fold 1 Val R2: -2.683769513746625
View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/a5b5c75bd07e40ffb7ec85336b8486d5
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 1.9760626186226327
Fold 2 Val Loss: 2.4665008416905683
Fold 2 Val R2: -2.4998067969664604
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/4417d25dbad44e7ca17c2999e204f566
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE): 2.602959793400851
Average validation loss (MSE): 2.559021671613841
Average validation R2: -2.6206397322087045
View run Reg-Normal_TrainMethod-mini-batch_BatchSize-50_Momentum-0_WeightInit-zero_LR-0.0001 at: http://localhost:5000/#/experiments/769694705593400528/runs/0f0e7e72169846368c8d79192324b676
View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment: 16 / 144 =====
Reg-Normal_TrainMethod-mini-batch_BatchSize-50_Momentum-0_WeightInit-xavier_LR-0.01
Fold 0 Train Loss: inf
Fold 0 Val Loss: inf
Fold 0 Val R2: -inf
View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/1d0bc8fa47a34b0180561ccac8199a85
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: inf
Fold 1 Val Loss: inf
Fold 1 Val R2: -inf
View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/0e71e9ca87234a958a548791abb022ed
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: inf
Fold 2 Val Loss: inf
Fold 2 Val R2: -inf
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/c8395a26e4554beeba0c36fc030535d8
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE): inf
Average validation loss (MSE): inf
Average validation R2: -inf
View run Reg-Normal_TrainMethod-mini-batch_BatchSize-50_Momentum-0_WeightInit-xavier_LR-0.01 at: http://localhost:5000/#/experiments/769694705593400528/runs/7dbaa2582e284ad1806db95792174608

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View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 17 / 144 =====  
 Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-xavier\_LR-0.001  
 Fold 0 Train Loss: 0.11196066240425889  
 Fold 0 Val Loss: 0.1394221177494175  
 Fold 0 Val R2: 0.7998675990634145  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/b89e3bfa55ca4bfbbc93b178eca14cb0>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: 0.10568934058443047  
 Fold 1 Val Loss: 0.12127892147605031  
 Fold 1 Val R2: 0.8312857397682274  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/43f714600c7744e7890a6441e8e9638c>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 0.20646041948335683  
 Fold 2 Val Loss: 0.12248728811612089  
 Fold 2 Val R2: 0.8261983794025525  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/53262a486977488186f8ec2a5e2c9461>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 0.14137014082401542  
 Average validation loss (MSE): 0.12772944244719622  
 Average validation R2: 0.8191172394113981  
 View run Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-xavier\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e35509de5851486fa0c9fdff3fbd7b86>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 18 / 144 =====  
 Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-xavier\_LR-0.0001  
 Fold 0 Train Loss: 1.7638146768134904  
 Fold 0 Val Loss: 2.6023988328863106  
 Fold 0 Val R2: -2.735593283385492  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/0e1a1a35bfc646d7b444866c8d299bbb>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: 2.6665169697248747  
 Fold 1 Val Loss: 2.6456216828938257  
 Fold 1 Val R2: -2.680393094283191  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f4820c716e6c4c6d8ac423b1ee0f1c55>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 2.716920096619829  
 Fold 2 Val Loss: 2.4666032224695735  
 Fold 2 Val R2: -2.4999520687377825  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs>

s/d228e939d25a43988c703d3dd078b54a

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 2.3824172477193977

Average validation loss (MSE): 2.5715412460832368

Average validation R2: -2.6386461488021555

View run Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/3392425f6b1d4399a0e20cac23878ee6>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 19 / 144 =====

Reg-Normal\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.01

Fold 0 Train Loss: inf

Fold 0 Val Loss: inf

Fold 0 Val R2: -inf

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/5045f7bd1db947d09c704a4f39689538>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: inf

Fold 1 Val Loss: inf

Fold 1 Val R2: -inf

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/0ef3de686ec940e5ac089da302992932>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: inf

Fold 2 Val Loss: inf

Fold 2 Val R2: -inf

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e6449a4702504bd08f4675051cf3cf6a>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): inf

Average validation loss (MSE): inf

Average validation R2: -inf

View run Reg-Normal\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/a3237d968ab54d6ca6f2499f19377f08>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 20 / 144 =====

Reg-Normal\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.001

Fold 0 Train Loss: 22.916976041146338

Fold 0 Val Loss: 22.695223772591724

Fold 0 Val R2: -31.57768348896586

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/37326654246c47e19cb6323520cae0fc>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 23.23877493609242

Fold 1 Val Loss: 22.01294373728522

Fold 1 Val R2: -29.62277824504823

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/37326654246c47e19cb6323520cae0fc>

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s/9e47e4fde28b48c6a171bd8c9f5b47c5
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 22.414505960065295
Fold 2 Val Loss: 23.877590409200383
Fold 2 Val R2: -32.88077222468041
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/run
s/a73640ff82004b2a98407886e6676f42
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE): 22.856752312434683
Average validation loss (MSE): 22.86191930635911
Average validation R2: -31.360411319564832
View run Reg-Normal_TrainMethod-batch_Momentum-0.9_WeightInit-zero_LR-0.001
at: http://localhost:5000/#/experiments/769694705593400528/runs/f87275c838134bfe
a2d9c42629ff3414
View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment: 21 / 144 =====
Reg-Normal_TrainMethod-batch_Momentum-0.9_WeightInit-zero_LR-0.0001
Fold 0 Train Loss: 24.877838458861238
Fold 0 Val Loss: 24.558376370804346
Fold 0 Val R2: -34.252131480508346
View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/run
s/8932418cb7154b92889d812b368822ba
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: 25.22440933822383
Fold 1 Val Loss: 23.935243899156614
Fold 1 Val R2: -32.29693996916608
View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/run
s/ceb4ab23fd984c09b0bc4e657fef57a4
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 24.2618630888794
Fold 2 Val Loss: 25.873075398001408
Fold 2 Val R2: -35.71223767930543
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/run
s/4c86489ca5c94e0cbd06fb2a0f1ca8f5
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE): 24.788036961988155
Average validation loss (MSE): 24.788898555987455
Average validation R2: -34.087103042993284
View run Reg-Normal_TrainMethod-batch_Momentum-0.9_WeightInit-zero_LR-0.0001
at: http://localhost:5000/#/experiments/769694705593400528/runs/a9c0436560354ab8
bdfa3013c5b824dc
View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment: 22 / 144 =====
Reg-Normal_TrainMethod-batch_Momentum-0.9_WeightInit-xavier_LR-0.01
Fold 0 Train Loss: inf
Fold 0 Val Loss: inf
Fold 0 Val R2: -inf
View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/run

```

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s/de4d59d6ca9b42e78c5ebc81290bc4d6
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: inf
Fold 1 Val Loss:    inf
Fold 1 Val R2:      -inf
View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/6c851be60df84bb592d3a0878322ced5
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: inf
Fold 2 Val Loss:    inf
Fold 2 Val R2:      -inf
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/be955b3f2b78479dbb7e6025a2f2812a
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE):  inf
Average validation loss (MSE):  inf
Average validation R2:  -inf
View run Reg-Normal_TrainMethod-batch_Momentum-0.9_WeightInit-xavier_LR-0.01
at: http://localhost:5000/#/experiments/769694705593400528/runs/1a4f0712119e46b89581e5d7e505580c
View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment:  23  /  144 =====
Reg-Normal_TrainMethod-batch_Momentum-0.9_WeightInit-xavier_LR-0.001
Fold 0 Train Loss: 22.87865944034137
Fold 0 Val Loss:   22.65640047639492
Fold 0 Val R2:     -31.52195488861493
View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/a2c44b419ea241449c87a87820b08830
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: 23.170634381748847
Fold 1 Val Loss:   21.95925104742021
Fold 1 Val R2:     -29.548084948470574
View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/94da2612fbb6447e9333147f9884c117
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 22.44424101412565
Fold 2 Val Loss:   23.905434934763417
Fold 2 Val R2:     -32.92028182393804
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/0e9cecdb93bd4b2e949bccf4c4012e0e
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE):  22.831178278738623
Average validation loss (MSE): 22.840362152859516
Average validation R2:  -31.33010722034118
View run Reg-Normal_TrainMethod-batch_Momentum-0.9_WeightInit-xavier_LR-0.001
at: http://localhost:5000/#/experiments/769694705593400528/runs/6cca0a7453bd4eb99acf82c58d53025e
View experiment at: http://localhost:5000/#/experiments/769694705593400528

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===== Experiment: 24 / 144 =====
Reg-Normal_TrainMethod-batch_Momentum-0.9_WeightInit-xavier_LR-0.0001
Fold 0 Train Loss: 24.877359424946164
Fold 0 Val Loss: 24.557593255711833
Fold 0 Val R2: -34.251007364003705
View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/4fa2f58ab8af477bb3975651d43e69b9
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: 25.242986789891766
Fold 1 Val Loss: 23.95154029546884
Fold 1 Val R2: -32.31961030969021
View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/4a393e1997994fe2ada38ac1c7f9c6fe
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 24.30897341716839
Fold 2 Val Loss: 25.921438241916785
Fold 2 Val R2: -35.78086145878877
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/e60b17eba15446228355b0bfef0ae1c9
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE): 24.809773210668776
Average validation loss (MSE): 24.81019059769915
Average validation R2: -34.117159710827565
View run Reg-Normal_TrainMethod-batch_Momentum-0.9_WeightInit-xavier_LR-0.0001
at: http://localhost:5000/#/experiments/769694705593400528/runs/b4bfae4ec40e4c73b52dc9a4abd1b1f6
View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment: 25 / 144 =====
Reg-Normal_TrainMethod-stochastic_Momentum-0.9_WeightInit-zero_LR-0.01
Fold 0 Train Loss: nan
Fold 0 Val Loss: nan
Fold 0 Val R2: nan
View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/274a0de92dfa4ad7bb1378593626396c
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: nan
Fold 1 Val Loss: nan
Fold 1 Val R2: nan
View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/6ac56d6d4aab4e4b8c220c55dfffb1bd2
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: nan
Fold 2 Val Loss: nan
Fold 2 Val R2: nan
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/14a2957112bf4048b4235369a508864b
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE): nan

```



Average validation loss (MSE): nan  
Average validation R2: nan  
View run Reg-Normal\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e4d5ac9873594a89b79c0c14aeabaafd>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 26 / 144 =====  
Reg-Normal\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.001  
Fold 0 Train Loss: 0.048024512518600765  
Fold 0 Val Loss: 0.1101555917173  
Fold 0 Val R2: 0.8418780075726858  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/b03fd17521654f91976b7bc5aebf22cd>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 0.04704695191511015  
Fold 1 Val Loss: 0.14088999985459022  
Fold 1 Val R2: 0.8040042588586528  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/8f2e1e1d6f484eeea93e9f0ea1db87c7>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 0.1524762249091759  
Fold 2 Val Loss: 0.09894979970191442  
Fold 2 Val R2: 0.8595965686685644  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/01047fadbb9d401d9c09c31026ce156e>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 0.08251589644762894  
Average validation loss (MSE): 0.11666513042460154  
Average validation R2: 0.8351596116999677  
View run Reg-Normal\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f9c4f26b463a40daaac30ffd91e647db>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 27 / 144 =====  
Reg-Normal\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.0001  
Fold 0 Train Loss: 0.12643363322148568  
Fold 0 Val Loss: 0.4939209080273612  
Fold 0 Val R2: 0.29100505148002465  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/d90cfe97250c4a85823ba66010d5fa06>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 0.008773529931420483  
Fold 1 Val Loss: 0.42455781518657626  
Fold 1 Val R2: 0.409386587190538  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/8e898861fcd8413cb948c6b1098d4319>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 0.2596632930428621

Fold 2 Val Loss: 0.38347255282363696  
 Fold 2 Val R2: 0.4558769962136451  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/1232ee298e604d8483e2a2e52cf463ff>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 0.13162348539858942  
 Average validation loss (MSE): 0.4339837586791915  
 Average validation R2: 0.3854228782947359  
 View run Reg-Normal\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e32e4c1fb05b47e2a1823481eba2d9c0>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 28 / 144 =====  
 Reg-Normal\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.01  
 Fold 0 Train Loss: nan  
 Fold 0 Val Loss: nan  
 Fold 0 Val R2: nan  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f97d4049d224433a8895e71364ec14e5>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: nan  
 Fold 1 Val Loss: nan  
 Fold 1 Val R2: nan  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/a307dc51bf3c47a8be49effcc1d10f0e>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: nan  
 Fold 2 Val Loss: nan  
 Fold 2 Val R2: nan  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/515bcd2489349458e13dba7921b3f64>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): nan  
 Average validation loss (MSE): nan  
 Average validation R2: nan  
 View run Reg-Normal\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/2e931622c2e54e4ab33a27000700ca19>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 29 / 144 =====  
 Reg-Normal\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.001  
 Fold 0 Train Loss: 0.6722341336161765  
 Fold 0 Val Loss: 0.8410301949191865  
 Fold 0 Val R2: -0.20725029060207678  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/173f1b5c7cfa4e688eff90d0db2ff964>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: 0.22521966455020156

Fold 1 Val Loss: 0.3012628536446142  
 Fold 1 Val R2: 0.5809054131636466  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/8461e8590a424c90b5b8a41c3542fc59>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 0.40702050962235325  
 Fold 2 Val Loss: 0.468222657933307  
 Fold 2 Val R2: 0.33562202249016326  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/dd638325ba3f4cb6839187b2b2831817>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 0.4348247692629104  
 Average validation loss (MSE): 0.5368385688323692  
 Average validation R2: 0.23642571501724438  
 View run Reg-Normal\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/76a933185af24932a0c5c0c25b20da3a>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 30 / 144 =====  
 Reg-Normal\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.0001  
 Fold 0 Train Loss: 0.3871165677610804  
 Fold 0 Val Loss: 0.654410864263285  
 Fold 0 Val R2: 0.06063098468073336  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/2d4bf21373a541f49ada91cd035265b6>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: 0.33189638955560535  
 Fold 1 Val Loss: 0.5894480163011155  
 Fold 1 Val R2: 0.18000354220690218  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/2f73fc98c81d430dbbab7597b5d30745>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 0.05045980237516761  
 Fold 2 Val Loss: 0.3826515252240201  
 Fold 2 Val R2: 0.45704198181797473  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/9d71048445544b7890af3ced3282b56c>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 0.2564909198972844  
 Average validation loss (MSE): 0.5421701352628069  
 Average validation R2: 0.23255883623520343  
 View run Reg-Normal\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/d90b23a4fd1d4f0b8d7e5a4a3a9a84c0>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 31 / 144 =====  
 Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-zero\_LR-0.01

Fold 0 Train Loss: inf  
 Fold 0 Val Loss: inf  
 Fold 0 Val R2: -inf  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/5fd47866106d41ce8c865acf9232bdfe>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: inf  
 Fold 1 Val Loss: inf  
 Fold 1 Val R2: -inf  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/899992bf2a9644aab155728d1220ffa5>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: inf  
 Fold 2 Val Loss: inf  
 Fold 2 Val R2: -inf  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/609d0fd2c4cd475fbe35d49872161854>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): inf  
 Average validation loss (MSE): inf  
 Average validation R2: -inf  
 View run Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-zero\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/549172320ac249a0aab30f2cda8676d3>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 32 / 144 =====  
 Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-zero\_LR-0.001  
 Fold 0 Train Loss: 2.829376403704916  
 Fold 0 Val Loss: 2.5978623914565167  
 Fold 0 Val R2: -2.729081483610046  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/97ecb9390479494096ef3b88e375d469>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: 2.481257508863766  
 Fold 1 Val Loss: 2.6491980491480973  
 Fold 1 Val R2: -2.6853682703448154  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e9f021b380b74bf0a151bf2116261680>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 1.7363327707061618  
 Fold 2 Val Loss: 2.5425869807270476  
 Fold 2 Val R2: -2.607768157471121  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/9a321aec5b414ceb922523dca961680c>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 2.348988894424948  
 Average validation loss (MSE): 2.596549140443887

Average validation R2: -2.6740726371419945

View run Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-zero\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/7c24a3ee1eba464984c83e71a6ddd4a8>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 33 / 144 =====

Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-zero\_LR-0.0001

Fold 0 Train Loss: 14.970559947716914

Fold 0 Val Loss: 13.176410695071091

Fold 0 Val R2: -17.91397685459485

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/618f4b4810c5437b947757e21fc726eb>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 14.153743812976915

Fold 1 Val Loss: 12.548820398368754

Fold 1 Val R2: -16.456990254569924

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/023e4a27c6c44338b72b4423f1324f89>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 13.334742301989765

Fold 2 Val Loss: 13.816620745511422

Fold 2 Val R2: -18.60490034258624

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/3f501b41f30b451894ea0334f8be8847>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 14.153015354227863

Average validation loss (MSE): 13.180617279650422

Average validation R2: -17.658622483917004

View run Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-zero\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/5f085fb958434c50b225927a34a7b5e9>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 34 / 144 =====

Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-xavier\_LR-0.01

Fold 0 Train Loss: inf

Fold 0 Val Loss: inf

Fold 0 Val R2: -inf

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f1aa4aaf4c084d6b9654be2b8e7d1f14>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: inf

Fold 1 Val Loss: inf

Fold 1 Val R2: -inf

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/29e7656646a84619b31ed707640a11db>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: inf  
 Fold 2 Val Loss: inf  
 Fold 2 Val R2: -inf  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/25097a7ab93048298eb4872fbe4932c8>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): inf  
 Average validation loss (MSE): inf  
 Average validation R2: -inf  
 View run Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-xavier\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/2c8cb07042e3460db3d1cbbcd601e2eb>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 35 / 144 =====  
 Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-xavier\_LR-0.001  
 Fold 0 Train Loss: 1.6454240934040065  
 Fold 0 Val Loss: 2.595379568968816  
 Fold 0 Val R2: -2.725517535266814  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/a386e4e98b0b486487d4da5a84a6a86a>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: 5.317148896043423  
 Fold 1 Val Loss: 7.1912575279179665  
 Fold 1 Val R2: -9.0039452791344  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/30a6c38f264a41f0bebd490eec70f5f>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 2.8588539745597883  
 Fold 2 Val Loss: 2.685467715556075  
 Fold 2 Val R2: -2.8105067734317988  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/bd793420671a4672a35ebf8777b889ff>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 3.273808988002406  
 Average validation loss (MSE): 4.157368270814286  
 Average validation R2: -4.846656529277671  
 View run Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-xavier\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/c5bb738c6db642cab1e50197629a05ed>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 36 / 144 =====  
 Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-xavier\_LR-0.0001  
 Fold 0 Train Loss: 18.88269744957764  
 Fold 0 Val Loss: 13.179171208058998  
 Fold 0 Val R2: -17.91793941161955  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/>

s/9bcf660a85ed4582bcabf609822b243e  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 9.113265109874183  
Fold 1 Val Loss: 12.557052587521952  
Fold 1 Val R2: -16.468442266891323  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/c358e14fefaf4f40b61f40cea50c5858>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 18.33305189075723  
Fold 2 Val Loss: 13.848641810259572  
Fold 2 Val R2: -18.65033618357905  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/b8d562065be0455cba3d81fef8c4cbe7>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 15.443004816736353  
Average validation loss (MSE): 13.19495520194684  
Average validation R2: -17.678905954029975  
View run Reg-Normal\_TrainMethod-mini-  
batch\_BatchSize-50\_Momentum-0.9\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/47564d67a4334072bd2550b272340713>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 37 / 144 =====  
Reg-Ridge\_TrainMethod-batch\_Momentum-0\_WeightInit-zero\_LR-0.01  
Fold 0 Train Loss: inf  
Fold 0 Val Loss: inf  
Fold 0 Val R2: -inf  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e62d4d23ee9443d2b1e3d9583ede8b87>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: inf  
Fold 1 Val Loss: inf  
Fold 1 Val R2: -inf  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f3231310533b424e9ff5077d2cb81fb8>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: inf  
Fold 2 Val Loss: inf  
Fold 2 Val R2: -inf  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/d98473e4f6e9433fbbb624286d4de178>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): inf  
Average validation loss (MSE): inf  
Average validation R2: -inf  
View run Reg-Ridge\_TrainMethod-batch\_Momentum-0\_WeightInit-zero\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/99d4242811ac4c95ba5532db3e944e75>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

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===== Experiment: 38 / 144 =====
Reg-Ridge_TrainMethod-batch_Momentum-0_WeightInit-zero_LR-0.001
Fold 0 Train Loss: 11.97513846605496
Fold 0 Val Loss: 12.097838272977345
Fold 0 Val R2: -16.36574841062896
View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/7e56a2103d7544a687452118a7232e78
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: 12.083596610115396
Fold 1 Val Loss: 11.477934204325463
Fold 1 Val R2: -14.967252633048355
View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/6a8c4d85f5a24683bbca091a3d777550
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 12.019745720346238
Fold 2 Val Loss: 12.651519831711251
Fold 2 Val R2: -16.95169673188939
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/060df3528d7f440c8a31fd3742a8ec97
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE): 12.026160265505531
Average validation loss (MSE): 12.075764103004687
Average validation R2: -16.094899258522233
View run Reg-Ridge_TrainMethod-batch_Momentum-0_WeightInit-zero_LR-0.001 at: http://localhost:5000/#/experiments/769694705593400528/runs/7ce72ee5f8ea4d5398ceca23920ea51
View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment: 39 / 144 =====
Reg-Ridge_TrainMethod-batch_Momentum-0_WeightInit-zero_LR-0.0001
Fold 0 Train Loss: 22.949543386823173
Fold 0 Val Loss: 22.726926438632585
Fold 0 Val R2: -31.62319083581488
View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/9311e48b775444e9978317c7360b990b
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: 23.27179178564198
Fold 1 Val Loss: 22.044337407644058
Fold 1 Val R2: -29.66645079594239
View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/0ff1876a5e234f99874cf27b36878ae2
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 22.445169592071515
Fold 2 Val Loss: 23.91039102819888
Fold 2 Val R2: -32.92731420324993
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/3f751e7ef6884d4db630455dcc6aa250
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE): 22.888834921512224

```



Average validation loss (MSE): 22.89388495815851  
 Average validation R2: -31.4056519450024  
 View run Reg-Ridge\_TrainMethod-batch\_Momentum-0\_WeightInit-zero\_LR-0.0001 at:  
<http://localhost:5000/#/experiments/769694705593400528/runs/5392921b91084d7997c7bcd7e05d397>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 40 / 144 =====  
 Reg-Ridge\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.01  
 Fold 0 Train Loss: inf  
 Fold 0 Val Loss: inf  
 Fold 0 Val R2: -inf  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/9d1124d89bca43b28eb37b14e30cf955>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: inf  
 Fold 1 Val Loss: inf  
 Fold 1 Val R2: -inf  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/210898691f2f4b508e5a2911a10fcf3d>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: inf  
 Fold 2 Val Loss: inf  
 Fold 2 Val R2: -inf  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/7bbb46f04a5540f296063d476be3403e>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): inf  
 Average validation loss (MSE): inf  
 Average validation R2: -inf  
 View run Reg-Ridge\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.01 at:  
<http://localhost:5000/#/experiments/769694705593400528/runs/13d2c9d5c4e746fe9ae3e43c99e16d49>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 41 / 144 =====  
 Reg-Ridge\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.001  
 Fold 0 Train Loss: 11.9782849164734  
 Fold 0 Val Loss: 12.094312828611846  
 Fold 0 Val R2: -16.36068783877261  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/549e4daf1117439f85ae1f7abb040792>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: 12.100162608092845  
 Fold 1 Val Loss: 11.494801288662538  
 Fold 1 Val R2: -14.990716872518515  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/10ea430592ff40f1b727a61b402e1f8b>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 12.00618193944799

Fold 2 Val Loss: 12.638101842686769  
Fold 2 Val R2: -16.93265746443983  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/209ae0a5ec584704b344b5ba79196506>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 12.028209821338079  
Average validation loss (MSE): 12.075738653320386  
Average validation R2: -16.094687391910316  
View run Reg-Ridge\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e73248f92e2d433a93d943ab459e3cdc>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 42 / 144 =====  
Reg-Ridge\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.0001  
Fold 0 Train Loss: 22.897987860968524  
Fold 0 Val Loss: 22.677941969377397  
Fold 0 Val R2: -31.552876458161663  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/6b79e2854d7640309cb6a67b4f113b71>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 23.17472474703187  
Fold 1 Val Loss: 21.953452160095708  
Fold 1 Val R2: -29.540017965575167  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/ebd49cdc90df438ba91c221eac9ca258>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 22.450472045512676  
Fold 2 Val Loss: 23.913041705016617  
Fold 2 Val R2: -32.93107534396655  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/1ca7cac350e24dc193aa1801b4e87682>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 22.841061551171023  
Average validation loss (MSE): 22.84814527816324  
Average validation R2: -31.341323255901127  
View run Reg-Ridge\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e02ed45edc6b461ab96e3c4fe8de963f>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 43 / 144 =====  
Reg-Ridge\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.01  
Fold 0 Train Loss: nan  
Fold 0 Val Loss: nan  
Fold 0 Val R2: nan  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/156ffdc6b71342fc8659f247d3828eb7>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: nan

Fold 1 Val Loss: nan  
 Fold 1 Val R2: nan  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/35956d01612c4ec595e73280528f72ea>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: nan  
 Fold 2 Val Loss: nan  
 Fold 2 Val R2: nan  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/6d6f1296b26e42de939ada7132fd5c6c>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): nan  
 Average validation loss (MSE): nan  
 Average validation R2: nan  
 View run Reg-Ridge\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/19f6c1635de840ffbfe6764ce98c90d1>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 44 / 144 =====  
 Reg-Ridge\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.001  
 Fold 0 Train Loss: 0.36257771330939115  
 Fold 0 Val Loss: 0.10926883815035544  
 Fold 0 Val R2: 0.8431508911241368  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/65a1853d4422a08bf3b491>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: 0.03139633477052898  
 Fold 1 Val Loss: 0.16707278092277136  
 Fold 1 Val R2: 0.7675807115103943  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/5c9816ccc52c4806b3d24f20e0dab5bf>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 0.20627238783764668  
 Fold 2 Val Loss: 0.21077217857909558  
 Fold 2 Val R2: 0.700927771548237  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/470eddb3d40a4e0c83acc66ac3d36c13>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 0.20008214530585558  
 Average validation loss (MSE): 0.16237126588407413  
 Average validation R2: 0.7705531247275893  
 View run Reg-Ridge\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/d8eb4fba18874d41a2c661e7b5313f22>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 45 / 144 =====  
 Reg-Ridge\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.0001  
 Fold 0 Train Loss: 0.16534662985337445

Fold 0 Val Loss: 0.1095510826786754  
 Fold 0 Val R2: 0.842745745398224  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/7974ce74c0be4968a25e4644350acadb>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: 0.10754841163300903  
 Fold 1 Val Loss: 0.09230630966003943  
 Fold 1 Val R2: 0.871590293189622  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/877dfca397024dd3986f77000d48202f>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 0.007097321444748584  
 Fold 2 Val Loss: 0.15766884016153684  
 Fold 2 Val R2: 0.7762780092590815  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/612381c2c62d44b3b4d7b7033fb2eaae>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 0.09333078764371068  
 Average validation loss (MSE): 0.1198420775000839  
 Average validation R2: 0.8302046826156425  
 View run Reg-Ridge\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/a084ce24ef8248bea803b1a9fd36ee69>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 46 / 144 =====  
 Reg-Ridge\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.01  
 Fold 0 Train Loss: nan  
 Fold 0 Val Loss: nan  
 Fold 0 Val R2: nan  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/544fbf1a5eff41d8b7adee6c0ad89a4e>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: nan  
 Fold 1 Val Loss: nan  
 Fold 1 Val R2: nan  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/4cc2b1fcbdaa48e589a8f5eb31587572>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: nan  
 Fold 2 Val Loss: nan  
 Fold 2 Val R2: nan  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/220c62f0620e415e800c8bb27576f0ab>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): nan  
 Average validation loss (MSE): nan  
 Average validation R2: nan  
 View run Reg-Ridge\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.01

at: [http://localhost:5000/#/experiments/769694705593400528/runs/40e160cdcc7e4aed  
a1bee209145f456a](http://localhost:5000/#/experiments/769694705593400528/runs/40e160cdcc7e4aed<br/>a1bee209145f456a)

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 47 / 144 =====

Reg-Ridge\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.001

Fold 0 Train Loss: 0.10851461863271075

Fold 0 Val Loss: 0.22017202537327338

Fold 0 Val R2: 0.6839557685085578

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/902a0342eae94331a1eb47fd25c9481f>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 0.3364362534398015

Fold 1 Val Loss: 0.3687181251853357

Fold 1 Val R2: 0.48706663146757445

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/a61b56a6bb32414f9a83ec9df6e7a2d9>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 0.2758692338561935

Fold 2 Val Loss: 0.23942043964589843

Fold 2 Val R2: 0.6602777230632972

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/484b59c9527c42b0bb34af8082289bec>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 0.24027336864290194

Average validation loss (MSE): 0.2761035300681692

Average validation R2: 0.6104333743464765

View run Reg-Ridge\_TrainMethod-stochastic\_Momentum-0\_WeightInit-

xavier\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/1c91b5a329e740ee8f164cca9947ffd3>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 48 / 144 =====

Reg-Ridge\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.0001

Fold 0 Train Loss: 0.05249794177152423

Fold 0 Val Loss: 0.11659517368690438

Fold 0 Val R2: 0.8326343594241103

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/359df0dad3bc47cc9f68ac2ff99c3199>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 0.4941269893163301

Fold 1 Val Loss: 0.09124157412479329

Fold 1 Val R2: 0.8730714744698085

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/9b3a0e0cd0454d089079b6abd9911994>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 0.006675921815509741

Fold 2 Val Loss: 0.0978329629850346

Fold 2 Val R2: 0.8611812884735488

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs>

s/869f32cf39cf4547883fdca5abc39864  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 0.18443361763445468  
Average validation loss (MSE): 0.10188990359891076  
Average validation R2: 0.8556290407891559  
View run Reg-Ridge\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/68dddac15a064a53b7aa1c6c33f93b39>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 49 / 144 =====  
Reg-Ridge\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.01  
Fold 0 Train Loss: inf  
Fold 0 Val Loss: inf  
Fold 0 Val R2: -inf  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/ef611e20a029413a97877ab7cc8be08f>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: inf  
Fold 1 Val Loss: inf  
Fold 1 Val R2: -inf  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/c338800b436c4a3db712eda0119413a9>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: inf  
Fold 2 Val Loss: inf  
Fold 2 Val R2: -inf  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/cc7f9df877b74ea0a2edcc84063710a0>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): inf  
Average validation loss (MSE): inf  
Average validation R2: -inf  
View run Reg-Ridge\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/8910d7bb3e6747e6a75facd527a5e0ec>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 50 / 144 =====  
Reg-Ridge\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.001  
Fold 0 Train Loss: 0.06509457329391866  
Fold 0 Val Loss: 0.13639064306542334  
Fold 0 Val R2: 0.8042191059597352  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/075e7c16f1774675b4100b20f7f60295>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 0.04448068070406577  
Fold 1 Val Loss: 0.11774041810664246  
Fold 1 Val R2: 0.8362082437864973

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/113da6e5218941a791d3846e54e308dc>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 0.10350165465020254

Fold 2 Val Loss: 0.1171907332240247

Fold 2 Val R2: 0.8337138517261535

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/b475a963f4044786a38dc1cccb3ea15d>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 0.07102563621606232

Average validation loss (MSE): 0.12377393146536349

Average validation R2: 0.8247137338241286

View run Reg-Ridge\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/01692705d7974e2ea59a7b7140db9dce>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 51 / 144 =====

Reg-Ridge\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.0001

Fold 0 Train Loss: 3.6802319075800463

Fold 0 Val Loss: 2.549515162465646

Fold 0 Val R2: -2.6596818275672063

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/9c139bd616df42168e92c97bb91eb663>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 1.8155333384453376

Fold 1 Val Loss: 2.636006564741032

Fold 1 Val R2: -2.6670172534821157

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/dafd61056f4e4f359c79be1d724e6419>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 0.9302182943160152

Fold 2 Val Loss: 2.4574262859213825

Fold 2 Val R2: -2.486930583253644

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/65b37d304c1f465e90eb28a02b819d3d>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 2.1419945134471328

Average validation loss (MSE): 2.547649337709353

Average validation R2: -2.6045432214343216

View run Reg-Ridge\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/7589ba22b2a5463abad246ad2f71919f>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 52 / 144 =====

Reg-Ridge\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-xavier\_LR-0.01

Fold 0 Train Loss: inf

```

Fold 0 Val Loss:    inf
Fold 0 Val R2:      -inf
  View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/790f811d37c74eb4bf218fac3c125e16
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: inf
Fold 1 Val Loss:    inf
Fold 1 Val R2:      -inf
  View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/b6190e31995b4d16ac41980a5ebf6d5e
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: inf
Fold 2 Val Loss:    inf
Fold 2 Val R2:      -inf
  View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/391a3553573e49f890ed652e8c3d42c3
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE):  inf
Average validation loss (MSE):  inf
Average validation R2:  -inf
  View run Reg-Ridge_TrainMethod-mini-batch_BatchSize-50_Momentum-0_WeightInit-xavier_LR-0.01 at: http://localhost:5000/#/experiments/769694705593400528/runs/3a5ab6d96a884d3f826720b322292756
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment:  53 / 144 =====
Reg-Ridge_TrainMethod-mini-batch_BatchSize-50_Momentum-0_WeightInit-xavier_LR-0.001
Fold 0 Train Loss: 0.08737752581116162
Fold 0 Val Loss:   0.1355828010587688
Fold 0 Val R2:     0.8053787165220979
  View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/950848ebff7048858dcc234313df5726
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: 0.16997264808705953
Fold 1 Val Loss:   0.1198989319193272
Fold 1 Val R2:     0.8332054791125132
  View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/6d6685e4cb1c46c88961152cf2880496
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 0.08255018534669009
Fold 2 Val Loss:   0.12219119791041706
Fold 2 Val R2:     0.8266185124497106
  View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/86ec1821fcc04def9b4b2c76075433cf
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE):  0.11330011974830374
Average validation loss (MSE): 0.1258909769628377
Average validation R2:  0.8217342360281071

```



View run Reg-Ridge\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-xavier\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/c6df0bc311c14f96b272db2035bc7e71>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 54 / 144 =====

Reg-Ridge\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-xavier\_LR-0.0001

Fold 0 Train Loss: 3.352533127977296

Fold 0 Val Loss: 3.3380899672569764

Fold 0 Val R2: -3.791635433985923

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/9dc1d0ed574d4d4c8e88f5d177720b11>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 2.577768145861001

Fold 1 Val Loss: 2.6338717002563583

Fold 1 Val R2: -2.6640473880030773

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/34839925446c460cb5d41942216aceb8>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 2.508255773673837

Fold 2 Val Loss: 2.4599177168558297

Fold 2 Val R2: -2.490465764256289

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/dfe0501ca6594a5d96e9cb0d9ae57592>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 2.8128523491707114

Average validation loss (MSE): 2.8106264614563883

Average validation R2: -2.98204952874843

View run Reg-Ridge\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/cf25c8eac8f24e72852c09a3b48ae66b>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 55 / 144 =====

Reg-Ridge\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.01

Fold 0 Train Loss: inf

Fold 0 Val Loss: inf

Fold 0 Val R2: -inf

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/b40af70d8cfb4cfdb9c1cdd887f8c551>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: inf

Fold 1 Val Loss: inf

Fold 1 Val R2: -inf

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/0d1aa99976fe419ab71902a943330f72>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: inf

Fold 2 Val Loss: inf

Fold 2 Val R2: -inf

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/4feb705a9bf84452b8c56368861b192e>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): inf

Average validation loss (MSE): inf

Average validation R2: -inf

View run Reg-Ridge\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/c0b8d5bcc7be4359a97f1f9ddcc81e7a>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 56 / 144 =====

Reg-Ridge\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.001

Fold 0 Train Loss: 22.91697744766722

Fold 0 Val Loss: 22.69522501128352

Fold 0 Val R2: -31.577685267036422

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/806ae00ed9cb4ab4960b6380cdad83ac>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 23.238776352509124

Fold 1 Val Loss: 22.0129451413353

Fold 1 Val R2: -29.6227801982588

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/39f0506798df48b5bc8c0f84f3734424>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 22.414507266570936

Fold 2 Val Loss: 23.877591887829592

Fold 2 Val R2: -32.8807743227606

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/895eb9c5591540228ab3a3181772e9df>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 22.85675368891576

Average validation loss (MSE): 22.86192068014947

Average validation R2: -31.360413262685274

View run Reg-Ridge\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/41d592e6e8104218af0c2beac0d1fe61>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 57 / 144 =====

Reg-Ridge\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.0001

Fold 0 Train Loss: 24.87783887159844

Fold 0 Val Loss: 24.558376674273887

Fold 0 Val R2: -34.25213191612135

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/a1d5987edba4f1ab9e8f92abef373cf>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 25.224409743331577

Fold 1 Val Loss: 23.935244351296145

Fold 1 Val R2: -32.29694059814914  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/060bf5a847e84f9eaa5ac5b5640f5f7b>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 24.2618634844201  
 Fold 2 Val Loss: 25.873075846518763  
 Fold 2 Val R2: -35.71223831572284  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/916d2eff044f4e498ad05106017ab20b>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 24.78803736645004  
 Average validation loss (MSE): 24.78889895736293  
 Average validation R2: -34.087103609997776  
 View run Reg-Ridge\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/303ca6fbc78c4614a9d75e46f830db3e>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 58 / 144 =====  
 Reg-Ridge\_TrainMethod-batch\_Momentum-0.9\_WeightInit-xavier\_LR-0.01  
 Fold 0 Train Loss: inf  
 Fold 0 Val Loss: inf  
 Fold 0 Val R2: -inf  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/42e35cad865e4accae8fa0f870557f8a>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: inf  
 Fold 1 Val Loss: inf  
 Fold 1 Val R2: -inf  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/a64dc9275a444c46992ff5f333bc97b7>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: inf  
 Fold 2 Val Loss: inf  
 Fold 2 Val R2: -inf  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/8cb3d6bfb5b6417ea81b8f1f01dbf314>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): inf  
 Average validation loss (MSE): inf  
 Average validation R2: -inf  
 View run Reg-Ridge\_TrainMethod-batch\_Momentum-0.9\_WeightInit-xavier\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/588fbc568cb54e8f96baea293c73b2ce>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 59 / 144 =====  
 Reg-Ridge\_TrainMethod-batch\_Momentum-0.9\_WeightInit-xavier\_LR-0.001  
 Fold 0 Train Loss: 22.914346182088103  
 Fold 0 Val Loss: 22.696442043147194

Fold 0 Val R2: -31.579432246015266  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/9e105a8101b340178729b09b47fb8fcc>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 23.28390218179826  
Fold 1 Val Loss: 22.06053894567495  
Fold 1 Val R2: -29.68898917664557  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/70ab0d1ff48245b2a79fed75376c266b>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 22.388366075809937  
Fold 2 Val Loss: 23.848427259241305  
Fold 2 Val R2: -32.83939158181885  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/5a5c8fa7b3a14228a0202cea59fcdfa9>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 22.8622048132321  
Average validation loss (MSE): 22.868469416021146  
Average validation R2: -31.369271001493228  
View run Reg-Ridge\_TrainMethod-batch\_Momentum-0.9\_WeightInit-xavier\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/9b53e2c6c6a142bb8027dbc81e43256e>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 60 / 144 =====  
Reg-Ridge\_TrainMethod-batch\_Momentum-0.9\_WeightInit-xavier\_LR-0.0001  
Fold 0 Train Loss: 24.861970243619364  
Fold 0 Val Loss: 24.543332551369215  
Fold 0 Val R2: -34.2305369462163  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/453661bef71c48b4991610aceae557b8>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 25.133927401943485  
Fold 1 Val Loss: 23.846144641479988  
Fold 1 Val R2: -32.172991675735055  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/fdf806213eef4ac2bc65e30bde902635>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 24.27242447975459  
Fold 2 Val Loss: 25.885835183041223  
Fold 2 Val R2: -35.73034299743683  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e0f81d8b53e8449284c04eb9a86d18bd>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 24.756107375105813  
Average validation loss (MSE): 24.75843745863014  
Average validation R2: -34.04462387312939  
View run Reg-Ridge\_TrainMethod-batch\_Momentum-0.9\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/febf5d2262274416>

8bf0383b0fe9a909

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 61 / 144 =====

Reg-Ridge\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.01

Fold 0 Train Loss: nan

Fold 0 Val Loss: nan

Fold 0 Val R2: nan

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/41987cd0c9704033924d47102c0d3442>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: nan

Fold 1 Val Loss: nan

Fold 1 Val R2: nan

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f8001f8681884a25a8818520d5688a8f>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: nan

Fold 2 Val Loss: nan

Fold 2 Val R2: nan

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/aa00a3f771ec4d539e7d7d1e718de731>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): nan

Average validation loss (MSE): nan

Average validation R2: nan

View run Reg-Ridge\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.01  
at: <http://localhost:5000/#/experiments/769694705593400528/runs/aaa35e63e4ff481eb850c19a8afcd026>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 62 / 144 =====

Reg-Ridge\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.001

Fold 0 Train Loss: 0.23629896477124118

Fold 0 Val Loss: 0.21331482775000085

Fold 0 Val R2: 0.6937988798182618

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/4e9a9d60ee8c4ffba3d85ce0c8d02b0f>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 0.1344029541618459

Fold 1 Val Loss: 0.1473518159415629

Fold 1 Val R2: 0.7950150585293717

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/5427a894b1064d8ca06a68df8687c658>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 0.35168864600027755

Fold 2 Val Loss: 0.4844362137674142

Fold 2 Val R2: 0.312616024700878

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/de977cb691e44d798ed3a57b412d3312>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 0.24079685497778822  
Average validation loss (MSE): 0.281700952486326  
Average validation R2: 0.6004766543495038  
View run Reg-Ridge\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/07792e405bde4a5fb34242b76eae1f0d>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 63 / 144 =====  
Reg-Ridge\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.0001  
Fold 0 Train Loss: 4.466474699228479e-05  
Fold 0 Val Loss: 0.1761568716992415  
Fold 0 Val R2: 0.7471369805326706  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/71ce586b0155443abc67be4dd900d87e>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 0.39570250429358045  
Fold 1 Val Loss: 0.15931866094449593  
Fold 1 Val R2: 0.7783676693754632  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/638886acd08a435cb9c967518a14662c>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 0.024105255927820028  
Fold 2 Val Loss: 0.16886037308620172  
Fold 2 Val R2: 0.7603979404846595  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/6a581750c5da4053a31e66c57e5f221f>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 0.13995080832279758  
Average validation loss (MSE): 0.16811196857664637  
Average validation R2: 0.7619675301309311  
View run Reg-Ridge\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/9a32660849944a9eb5114999c7dce2c1>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 64 / 144 =====  
Reg-Ridge\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.01  
Fold 0 Train Loss: nan  
Fold 0 Val Loss: nan  
Fold 0 Val R2: nan  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/1286e255d18741cd8b0ee23b60cbe0e6>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: nan  
Fold 1 Val Loss: nan  
Fold 1 Val R2: nan  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/d7b7b8d14b9c4e86a1b915f9536ce612>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: nan  
Fold 2 Val Loss: nan  
Fold 2 Val R2: nan  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/7027d17becb343cbb882ba8b9b9fdcc7>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): nan  
Average validation loss (MSE): nan  
Average validation R2: nan  
View run Reg-Ridge\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/9f577bf965c647ab92ea4884f4a552ba>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 65 / 144 =====  
Reg-Ridge\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.001  
Fold 0 Train Loss: 0.030186863675414878  
Fold 0 Val Loss: 0.13543162996788405  
Fold 0 Val R2: 0.8055957138956807  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/01a6f19833464dad80bbfe2aa69eebd8>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 0.02496545660388075  
Fold 1 Val Loss: 0.09114434927555479  
Fold 1 Val R2: 0.8732067264848807  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/025e6aa51abb4ebe998d37dd541becbf>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 0.000456664355990655  
Fold 2 Val Loss: 0.11377811119797976  
Fold 2 Val R2: 0.8385561439161061  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/db08647249fe41cb9941855bd08753e2>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 0.018536328211762095  
Average validation loss (MSE): 0.11345136348047287  
Average validation R2: 0.8391195280988892  
View run Reg-Ridge\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/72d3df254a9c4230900b9a922732157d>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 66 / 144 =====  
Reg-Ridge\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.0001  
Fold 0 Train Loss: 0.06062037303470461  
Fold 0 Val Loss: 0.17600006224615122  
Fold 0 Val R2: 0.7473620714496865  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/7fd10070a2e94f84863370b38c4f7e2c>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 0.8299983895016555  
Fold 1 Val Loss: 0.1516111714826356  
Fold 1 Val R2: 0.7890897583170171  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/8fa0387b7d43424aa12e070672cb87ca>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 0.1445091060349683  
Fold 2 Val Loss: 0.1579579909162867  
Fold 2 Val R2: 0.775867722848586  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/a560e0e8c13c445e98574d7318b75207>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 0.34504262285710946  
Average validation loss (MSE): 0.1618564082150245  
Average validation R2: 0.7707731842050966  
View run Reg-Ridge\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/bd5abd9be41c4282bcd368d3c69b6f79>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 67 / 144 =====  
Reg-Ridge\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-zero\_LR-0.01  
Fold 0 Train Loss: inf  
Fold 0 Val Loss: inf  
Fold 0 Val R2: -inf  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/82fbbad2bb044e8e806e3bfa668e0e30>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: inf  
Fold 1 Val Loss: inf  
Fold 1 Val R2: -inf  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/9bd2587169424ad08642ae9ce5f31af1>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: inf  
Fold 2 Val Loss: inf  
Fold 2 Val R2: -inf  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/3b8c9587dbb64ee59f6d65ada15d061f>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): inf  
Average validation loss (MSE): inf  
Average validation R2: -inf  
View run Reg-Ridge\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-zero\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/4596f11ff44740e6ac612d634c3ef90a>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>



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===== Experiment: 68 / 144 =====
Reg-Ridge_TrainMethod-mini-batch_BatchSize-50_Momentum-0.9_WeightInit-
zero_LR-0.001
Fold 0 Train Loss: 0.9617504078195552
Fold 0 Val Loss: 2.5948651884868728
Fold 0 Val R2: -2.7247791717811065
View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/run
s/dba351ee466a42e7ab280b1dac30cb6a
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: 4.486413374888476
Fold 1 Val Loss: 2.6361964702188256
Fold 1 Val R2: -2.667281435928744
View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/run
s/4b1491ce8af14832b38cc71079229860
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 1.6178413952909438
Fold 2 Val Loss: 2.466096540574034
Fold 2 Val R2: -2.499233119564145
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/run
s/b8eb65ca77b544218367680722006465
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE): 2.3553350593329916
Average validation loss (MSE): 2.5657193997599106
Average validation R2: -2.630431242424665
View run Reg-Ridge_TrainMethod-mini-
batch_BatchSize-50_Momentum-0.9_WeightInit-zero_LR-0.001 at: http://localhost:50
00/#/experiments/769694705593400528/runs/32aa3e071feb42be8767a9ca8519cd5f
View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment: 69 / 144 =====
Reg-Ridge_TrainMethod-mini-batch_BatchSize-50_Momentum-0.9_WeightInit-
zero_LR-0.0001
Fold 0 Train Loss: 9.88628312430723
Fold 0 Val Loss: 13.18140764554966
Fold 0 Val R2: -17.921149688523787
View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/run
s/11b375935ca943a1a9f31a34b0f50fea
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: 19.604027717279894
Fold 1 Val Loss: 12.552370677795189
Fold 1 Val R2: -16.461929140567236
View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/run
s/645c962e3aaf4b16abdf4195c44b3371
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 17.87503687531024
Fold 2 Val Loss: 13.820696348744534
Fold 2 Val R2: -18.610683362667114
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/run
s/acfcc1237c6442d48ee77590f3dd7f1e

```

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 15.788449238965788  
Average validation loss (MSE): 13.184824890696461  
Average validation R2: -17.664587397252713  
View run Reg-Ridge\_TrainMethod-mini-  
batch\_BatchSize-50\_Momentum-0.9\_WeightInit-zero\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/fa8ac5a0596540469fa2a85288acf5fc>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 70 / 144 =====  
Reg-Ridge\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-  
xavier\_LR-0.01  
Fold 0 Train Loss: inf  
Fold 0 Val Loss: inf  
Fold 0 Val R2: -inf  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/617bbb0d93414cb6a4594ac376511306>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: inf  
Fold 1 Val Loss: inf  
Fold 1 Val R2: -inf  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/d72d44c868ab48d38dc01e879f03da99>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: inf  
Fold 2 Val Loss: inf  
Fold 2 Val R2: -inf  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/ef590290effb401993af297f8a35423b>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): inf  
Average validation loss (MSE): inf  
Average validation R2: -inf  
View run Reg-Ridge\_TrainMethod-mini-  
batch\_BatchSize-50\_Momentum-0.9\_WeightInit-xavier\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/0e4a81e6376f4b9ba601050cd42b8530>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 71 / 144 =====  
Reg-Ridge\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-  
xavier\_LR-0.001  
Fold 0 Train Loss: 1.7908848155229442  
Fold 0 Val Loss: 2.8049443038941253  
Fold 0 Val R2: -3.026335613698394  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/0200d9169fcc4ca786e86b8646ab3888>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 1.0170946029106107  
Fold 1 Val Loss: 2.657649159233039  
Fold 1 Val R2: -2.697124829265071

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/030796a3d02845eda5eb89f47690adf8>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 2.4730462346397615

Fold 2 Val Loss: 2.468325341823278

Fold 2 Val R2: -2.5023956458561876

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/ddb24b99eafc46caa2183d62d50e448f>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 1.760341884357772

Average validation loss (MSE): 2.6436396016501473

Average validation R2: -2.741952029606551

View run Reg-Ridge\_TrainMethod-mini-

batch\_BatchSize-50\_Momentum-0.9\_WeightInit-xavier\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/166b055d3bb24621853e72278517f031>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 72 / 144 =====

Reg-Ridge\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-xavier\_LR-0.0001

Fold 0 Train Loss: 10.20857643703496

Fold 0 Val Loss: 13.20778111120153

Fold 0 Val R2: -17.959007275878854

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/272dfceadb724f35b59c65cf47b07e51>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 16.48470829256757

Fold 1 Val Loss: 12.561402138620936

Fold 1 Val R2: -16.474493040488866

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/80cdf98809a4ac8bd797780c2d656a3>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 16.68572299064514

Fold 2 Val Loss: 13.847956591329757

Fold 2 Val R2: -18.649363901783136

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/97c9ced1a6144ea7bb3f260052ff20df>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 14.459669240082556

Average validation loss (MSE): 13.205713280384074

Average validation R2: -17.694288072716954

View run Reg-Ridge\_TrainMethod-mini-

batch\_BatchSize-50\_Momentum-0.9\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/9b4f0d83c41e4230bce52f79258e60e6>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 73 / 144 =====

Reg-Lasso\_TrainMethod-batch\_Momentum-0\_WeightInit-zero\_LR-0.01

Fold 0 Train Loss: inf

Fold 0 Val Loss: inf

```

Fold 0 Val R2:      -inf
  View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/0d22c661c030483d9a305abd3b303766
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: inf
Fold 1 Val Loss:    inf
Fold 1 Val R2:      -inf
  View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/f80e804d8b34461e9f456360c7a97ef6
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: inf
Fold 2 Val Loss:    inf
Fold 2 Val R2:      -inf
  View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/fc21115a5542400e89dde7206ab66e49
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE):  inf
Average validation loss (MSE):  inf
Average validation R2:  -inf
  View run Reg-Lasso_TrainMethod-batch_Momentum-0_WeightInit-zero_LR-0.01 at: http://localhost:5000/#/experiments/769694705593400528/runs/05d6929659a0479cb7403534665829a6
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment:  74 / 144 =====
Reg-Lasso_TrainMethod-batch_Momentum-0_WeightInit-zero_LR-0.001
Fold 0 Train Loss: 11.975138255920982
Fold 0 Val Loss:   12.097834607262227
Fold 0 Val R2:     -16.365743148706535
  View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/8088ff3099bc4926b58ad0012c83d404
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: 12.083593916592005
Fold 1 Val Loss:   11.47793252741929
Fold 1 Val R2:     -14.967250300260588
  View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/2f98f2638cc3425c8431c5c67e2339a7
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 12.019745759070286
Fold 2 Val Loss:   12.65152156837612
Fold 2 Val R2:     -16.95169919610566
  View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/87c1b1e902a44884b2c1bbee44b53c90
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE):  12.026159310527758
Average validation loss (MSE): 12.075762901019212
Average validation R2:  -16.094897548357594
  View run Reg-Lasso_TrainMethod-batch_Momentum-0_WeightInit-zero_LR-0.001 at: http://localhost:5000/#/experiments/769694705593400528/runs/79bbc3194f2f4dca8af07

```

7098299cb11

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 75 / 144 =====

Reg-Lasso\_TrainMethod-batch\_Momentum-0\_WeightInit-zero\_LR-0.0001

Fold 0 Train Loss: 22.949557093976136

Fold 0 Val Loss: 22.726939751624393

Fold 0 Val R2: -31.623209945846043

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/6314dd2d982f4737a256fe2b8ad17f70>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 23.271805517263

Fold 1 Val Loss: 22.044350595933004

Fold 1 Val R2: -29.666469142513964

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/9f61c6ca20874d0492345bc21948a181>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 22.445182949354866

Fold 2 Val Loss: 23.91040534870682

Fold 2 Val R2: -32.927334523133936

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/2c9f4791aa8043b6998ca995631ee186>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 22.888848520198

Average validation loss (MSE): 22.893898565421406

Average validation R2: -31.405671203831314

View run Reg-Lasso\_TrainMethod-batch\_Momentum-0\_WeightInit-zero\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e578332fdf7244b6b9512c2cad6df44>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 76 / 144 =====

Reg-Lasso\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.01

Fold 0 Train Loss: inf

Fold 0 Val Loss: inf

Fold 0 Val R2: -inf

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/47fddcff5cfc431b89c7bef4f640d597>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: inf

Fold 1 Val Loss: inf

Fold 1 Val R2: -inf

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f2707bef99954efc8229b022290cec94>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: inf

Fold 2 Val Loss: inf

Fold 2 Val R2: -inf

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/75c40ffdee664c64bd1265aa5d7d7745>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): inf  
Average validation loss (MSE): inf  
Average validation R2: -inf  
View run Reg-Lasso\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.01 at:  
<http://localhost:5000/#/experiments/769694705593400528/runs/fe95f909b4c6405798ed749cd36093b2>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 77 / 144 =====  
Reg-Lasso\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.001  
Fold 0 Train Loss: 11.966792787529572  
Fold 0 Val Loss: 12.09307437984384  
Fold 0 Val R2: -16.35891011706404  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/2a790a5aa71049108d519370e72ef6ad>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 12.070959224107687  
Fold 1 Val Loss: 11.468600726804599  
Fold 1 Val R2: -14.954268589851425  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/6adbed38b551495ba294c61c0e64515e>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 12.007152818649821  
Fold 2 Val Loss: 12.636294338474448  
Fold 2 Val R2: -16.930092731673113  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/b4615dc9b6472eb27f2666b4836ae3>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 12.01496827676236  
Average validation loss (MSE): 12.065989815040963  
Average validation R2: -16.081090479529525  
View run Reg-Lasso\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.001 at:  
<http://localhost:5000/#/experiments/769694705593400528/runs/a9e531dae64b4e88b8a3338d7743c888>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 78 / 144 =====  
Reg-Lasso\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.0001  
Fold 0 Train Loss: 23.00319734922717  
Fold 0 Val Loss: 22.780304741970838  
Fold 0 Val R2: -31.699812308630307  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/d4f35d8821464355aacf82b2599093cb>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 23.2747947996554  
Fold 1 Val Loss: 22.042002984196706  
Fold 1 Val R2: -29.663203318802964  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/22f4cb880b5c4723a88572b0d375dee9>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 22.37700727431926  
Fold 2 Val Loss: 23.843979834580857  
Fold 2 Val R2: -32.83308097093539  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/798613ec7ca541fe96ae156e8f5f7213>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 22.884999807733948  
Average validation loss (MSE): 22.88876252024947  
Average validation R2: -31.398698866122885  
View run Reg-Lasso\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/6acfa4975a674d44bec67dc87cec8e4a>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 79 / 144 =====  
Reg-Lasso\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.01  
Fold 0 Train Loss: nan  
Fold 0 Val Loss: nan  
Fold 0 Val R2: nan  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/a99df768a53b4457a6a9ec5e70aaec8c>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: nan  
Fold 1 Val Loss: nan  
Fold 1 Val R2: nan  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/28cb47eb5696420bbb4736d237482f89>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: nan  
Fold 2 Val Loss: nan  
Fold 2 Val R2: nan  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/1f23b19c31f042cd998476bb2959372e>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): nan  
Average validation loss (MSE): nan  
Average validation R2: nan  
View run Reg-Lasso\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/da91b7fbb8bc406db07e5f253f5d30cb>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 80 / 144 =====  
Reg-Lasso\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.001  
Fold 0 Train Loss: 0.010810714284027549  
Fold 0 Val Loss: 0.14377905291695456  
Fold 0 Val R2: 0.793613470164215  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/64e6160f43a14b9e9633aa4be839d35b>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 1.0868037559064736  
Fold 1 Val Loss: 0.3924982925642776  
Fold 1 Val R2: 0.4539854224767921  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/b8034c831f594126924323b6b27f68af>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 0.5819127452333706  
Fold 2 Val Loss: 0.17082410271158158  
Fold 2 Val R2: 0.7576115338578536  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/62449cb2234d495da38e3d4831d77822>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 0.5598424051412906  
Average validation loss (MSE): 0.2357004827309379  
Average validation R2: 0.6684034754996202  
View run Reg-Lasso\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/aaa08bf7dd02448f9cbfab048d9aaebb>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 81 / 144 =====  
Reg-Lasso\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.0001  
Fold 0 Train Loss: 0.021674058166811217  
Fold 0 Val Loss: 0.12865260029527234  
Fold 0 Val R2: 0.8153266196249892  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f0ca4776cf044f789cf76c3c63359d2b>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 0.003271331991527933  
Fold 1 Val Loss: 0.10837877009004648  
Fold 1 Val R2: 0.8492314756923163  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/bf347cad8ba54ecd845725f9a2c20749>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 0.04559879010668058  
Fold 2 Val Loss: 0.1128687460882536  
Fold 2 Val R2: 0.8398464747921995  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/77c384c5e9c0409a9a277c6df115e0dd>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 0.023514726755006574  
Average validation loss (MSE): 0.11663337215785748  
Average validation R2: 0.834801523369835  
View run Reg-Lasso\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/142db873b19f46dd9418a0231271c22f>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 82 / 144 =====



Reg-Lasso\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.01

Fold 0 Train Loss: nan

Fold 0 Val Loss: nan

Fold 0 Val R2: nan

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/7d9e024c473449ebb3b1a81194b2a4e1>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: nan

Fold 1 Val Loss: nan

Fold 1 Val R2: nan

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/52a793d4efd54600a29f6b52c4f20533>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: nan

Fold 2 Val Loss: nan

Fold 2 Val R2: nan

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/02cfe78977e345e787a83e5ce9e43ede>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): nan

Average validation loss (MSE): nan

Average validation R2: nan

View run Reg-Lasso\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f0eb240d5fe544b384afad4747bb82bc>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 83 / 144 =====

Reg-Lasso\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.001

Fold 0 Train Loss: 0.0010283007183249353

Fold 0 Val Loss: 0.15875874422754838

Fold 0 Val R2: 0.7721109880927104

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/51520a70663449c883afaede1fa96eaf>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 0.037125133069561216

Fold 1 Val Loss: 0.112808357879542

Fold 1 Val R2: 0.8430693609741045

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/aa7547a2d3014ba3acd793f1b8d5806b>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 0.032844735698511915

Fold 2 Val Loss: 0.17471438443320356

Fold 2 Val R2: 0.7520914731381034

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/ef8f8c100faa4f04a0b4182674b4094e>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 0.02366605649546602

Average validation loss (MSE): 0.14876049551343132

Average validation R2: 0.7890906074016394

View run Reg-Lasso\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/a67cdb906bca467392f5ce2c4f6b8572>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 84 / 144 =====

Reg-Lasso\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.0001

Fold 0 Train Loss: 0.25703999415667445

Fold 0 Val Loss: 0.12092033541095723

Fold 0 Val R2: 0.8264258394686936

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/db862b5e179742a38017ece88efd9954>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 0.03071393190854613

Fold 1 Val Loss: 0.1062116709824806

Fold 1 Val R2: 0.8522461835931787

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/7b7d156a61284b16b8b6bb9b9088f7bd>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 0.004568419888612389

Fold 2 Val Loss: 0.11555468225798604

Fold 2 Val R2: 0.8360353033122804

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/a4186bcd9f0148e2a916137b73d712a2>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 0.097440781984611

Average validation loss (MSE): 0.11422889621714129

Average validation R2: 0.8382357754580508

View run Reg-Lasso\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/7c298ced05f74f85840acfd9a59898ec>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 85 / 144 =====

Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.01

Fold 0 Train Loss: inf

Fold 0 Val Loss: inf

Fold 0 Val R2: -inf

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/3ed3abd6027d46979471c9e1af06f5f8>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: inf

Fold 1 Val Loss: inf

Fold 1 Val R2: -inf

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/ff79d6c747e2454198000088e23b3b8c>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: inf

Fold 2 Val Loss: inf

Fold 2 Val R2: -inf

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/cf37f71f249540ba98dd21434cc80877>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): inf

Average validation loss (MSE): inf

Average validation R2: -inf

View run Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/374d333d85664db59572d8c72cecb9f1>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 86 / 144 =====

Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.001

Fold 0 Train Loss: 0.15559560213903403

Fold 0 Val Loss: 0.15162512934149017

Fold 0 Val R2: 0.7823508804250714

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/6d79537cc86d42978b5fdcccb8907fe7>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 0.17143297930437193

Fold 1 Val Loss: 0.1227441435238495

Fold 1 Val R2: 0.8292474312900429

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/057a88852dc4494690501b550039c99a>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 0.06456866192144788

Fold 2 Val Loss: 0.11697247390505609

Fold 2 Val R2: 0.8340235477275156

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/99a5d1a298d84931861da530e62e325e>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 0.13053241445495128

Average validation loss (MSE): 0.13044724892346526

Average validation R2: 0.8152072864808767

View run Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/ed51af183c2402cbb5e8a23522b466b>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 87 / 144 =====

Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.0001

Fold 0 Train Loss: 2.6456452109697564

Fold 0 Val Loss: 2.564647836870154

Fold 0 Val R2: -2.6814039080380323

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/bfee68766b6147eb9d28b48a719bc9af>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 2.264009737202899  
 Fold 1 Val Loss: 3.0094059046858286  
 Fold 1 Val R2: -3.186462781551551  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/af7f0e64a0354d609b7ae78b146ac87a>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 2.161576933858678  
 Fold 2 Val Loss: 2.4645475209900063  
 Fold 2 Val R2: -2.497035159937625  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e459ad5edb0c4ad6af55ac6056823d65>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 2.3570772940104443  
 Average validation loss (MSE): 2.679533754181996  
 Average validation R2: -2.788300616509069  
 View run Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/291f59855ad249729f96149aed8e7b27>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 88 / 144 =====  
 Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-xavier\_LR-0.01  
 Fold 0 Train Loss: inf  
 Fold 0 Val Loss: inf  
 Fold 0 Val R2: -inf  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/17353e9f672349cfb3f78748aca8a089>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: inf  
 Fold 1 Val Loss: inf  
 Fold 1 Val R2: -inf  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/8998696f47ae48db9026046022daac72>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: inf  
 Fold 2 Val Loss: inf  
 Fold 2 Val R2: -inf  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f2c9eec506a14bd0bfc4e6f5e63cd5e6>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): inf  
 Average validation loss (MSE): inf  
 Average validation R2: -inf  
 View run Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-xavier\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/5639d4d569584798a60d48c90ff804c1>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 89 / 144 =====

Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-xavier\_LR-0.001

Fold 0 Train Loss: 0.07753942698255621

Fold 0 Val Loss: 0.13429244488051814

Fold 0 Val R2: 0.8072309483213652

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/2ac097541fd64d8ab2a28622de35aa75>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 0.09914094933057022

Fold 1 Val Loss: 0.1207361060897532

Fold 1 Val R2: 0.8320408643622372

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/38e98ebc1a2c4ea7aacec017c4a4fa5b>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 0.07431708166927281

Fold 2 Val Loss: 0.12140507805826985

Fold 2 Val R2: 0.8277339661950616

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/814f635d7cb64d3ca5a5e6b5bbb61c08>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 0.08366581932746642

Average validation loss (MSE): 0.12547787634284704

Average validation R2: 0.8223352596262213

View run Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-xavier\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/b4da161c84e3401192cee8d967be15de>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 90 / 144 =====

Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-xavier\_LR-0.0001

Fold 0 Train Loss: 6.865887779317839

Fold 0 Val Loss: 3.9659543929913452

Fold 0 Val R2: -4.6928985693711605

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/0b333b02e30a43b1887d36ade261cf89>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 4.22940267484937

Fold 1 Val Loss: 3.0585145535945037

Fold 1 Val R2: -3.254779099595698

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/8707d2b76c654a329e372d167cd42a06>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 1.4340345832210624

Fold 2 Val Loss: 2.579213245377021

Fold 2 Val R2: -2.6597384823146397

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/090b36b0762d4192b174a4aeb545628d>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 4.176441679129424  
 Average validation loss (MSE): 3.2012273973209564  
 Average validation R2: -3.5358053837604992  
 View run Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/afe1de3c5abf49d2bf213a98471dc209>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 91 / 144 =====  
 Reg-Lasso\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.01  
 Fold 0 Train Loss: inf  
 Fold 0 Val Loss: inf  
 Fold 0 Val R2: -inf  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f778c6f5d7df44a1adcb821caa1fae82>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: inf  
 Fold 1 Val Loss: inf  
 Fold 1 Val R2: -inf  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/17cd3d64bbaf4184a39cddb172489d37>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: inf  
 Fold 2 Val Loss: inf  
 Fold 2 Val R2: -inf  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/74232c2211d042d4a8d1c338419fc093>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): inf  
 Average validation loss (MSE): inf  
 Average validation R2: -inf  
 View run Reg-Lasso\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/7dfe099473a743398f4f2004eb90c55e>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 92 / 144 =====  
 Reg-Lasso\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.001  
 Fold 0 Train Loss: 22.91699131383728  
 Fold 0 Val Loss: 22.695238490945055  
 Fold 0 Val R2: -31.577704616312353  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/9fca9c11310e419cb2b868a593ec17ae>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: 23.23879023342278  
 Fold 1 Val Loss: 22.012958469009416  
 Fold 1 Val R2: -29.6227987387327  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/1394f10c885645cab38ab11123619e26>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 22.414520742139235  
 Fold 2 Val Loss: 23.877606331089076  
 Fold 2 Val R2: -32.8807948168212  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/19c83b197c914a7f8f962830e315351f>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 22.856767429799763  
 Average validation loss (MSE): 22.861934430347848  
 Average validation R2: -31.360432723955416  
 View run Reg-Lasso\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/43cc76db882941c894fe25491366e8ba>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 93 / 144 =====  
 Reg-Lasso\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.0001  
 Fold 0 Train Loss: 24.877840508564102  
 Fold 0 Val Loss: 24.558378226039675  
 Fold 0 Val R2: -34.252134143591555  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/375cdf6083a24051b252d60d10a597f1>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: 25.224411390510053  
 Fold 1 Val Loss: 23.93524595740331  
 Fold 1 Val R2: -32.296942832446575  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/3c05500916d049dcbfdc295b16cc3975>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 24.261865129700592  
 Fold 2 Val Loss: 25.873077613116514  
 Fold 2 Val R2: -35.71224082241196  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/3deb8f99d3464446952224ffcd3654bc>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 24.788039009591582  
 Average validation loss (MSE): 24.78890059885317  
 Average validation R2: -34.08710593281669  
 View run Reg-Lasso\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/61ef35a10048416cb877b3dd1723b370>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 94 / 144 =====  
 Reg-Lasso\_TrainMethod-batch\_Momentum-0.9\_WeightInit-xavier\_LR-0.01  
 Fold 0 Train Loss: inf  
 Fold 0 Val Loss: inf  
 Fold 0 Val R2: -inf  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/c199d08d50c0436384bdabe78fcb990a>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

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Fold 1 Train Loss: inf
Fold 1 Val Loss:    inf
Fold 1 Val R2:      -inf
  View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/run
s/ed0935b9fbcb4284a3f2b52a4e5f6149
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: inf
Fold 2 Val Loss:    inf
Fold 2 Val R2:      -inf
  View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/run
s/4979bafa4c7d479d9db81b2bb389f303
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE):  inf
Average validation loss (MSE):  inf
Average validation R2:  -inf
  View run Reg-Lasso_TrainMethod-batch_Momentum-0.9_WeightInit-xavier_LR-0.01
at: http://localhost:5000/#/experiments/769694705593400528/runs/e7cb6325dd5444c3
b0153187885d102a
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment:  95 / 144 =====
Reg-Lasso_TrainMethod-batch_Momentum-0.9_WeightInit-xavier_LR-0.001
Fold 0 Train Loss: 22.975353537989538
Fold 0 Val Loss:   22.753062085495554
Fold 0 Val R2:     -31.6607070480282
  View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/run
s/ab9ee1d2de91424da72f9dddc978d6e
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: 23.20278538804801
Fold 1 Val Loss:   21.979360314477002
Fold 1 Val R2:     -29.576059472600555
  View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/run
s/b5f039a31e694c949b8d117ffef8dach
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 22.38812565725719
Fold 2 Val Loss:   23.84796420880254
Fold 2 Val R2:     -32.838734542889235
  View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/run
s/48fc1f068926445199f140c877a7e9c9
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE):  22.855421527764914
Average validation loss (MSE): 22.8601288695917
Average validation R2:  -31.358500354505995
  View run Reg-Lasso_TrainMethod-batch_Momentum-0.9_WeightInit-xavier_LR-0.001
at: http://localhost:5000/#/experiments/769694705593400528/runs/576f7bf98f78497c
9bceaf64af92d749
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment:  96 / 144 =====
Reg-Lasso_TrainMethod-batch_Momentum-0.9_WeightInit-xavier_LR-0.0001

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Fold 0 Train Loss: 24.9063153113509  
 Fold 0 Val Loss: 24.58873310860997  
 Fold 0 Val R2: -34.29570682507857  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/0969844b0382455b98b0d9e53b61d461>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: 25.25394006757529  
 Fold 1 Val Loss: 23.95732528384228  
 Fold 1 Val R2: -32.327657957393505  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/4d36fbe3900d4a1981674b06d471fc01>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 24.26116930644439  
 Fold 2 Val Loss: 25.869555272754088  
 Fold 2 Val R2: -35.70724284692649  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/3db94d6b4a1749c5923e88c6631f925e>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 24.807141561790193  
 Average validation loss (MSE): 24.80520455506878  
 Average validation R2: -34.11020254313285  
 View run Reg-Lasso\_TrainMethod-batch\_Momentum-0.9\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/0a0e23138a5b44958b49afb2b59d5f7b>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 97 / 144 =====  
 Reg-Lasso\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.01  
 Fold 0 Train Loss: nan  
 Fold 0 Val Loss: nan  
 Fold 0 Val R2: nan  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f4e430d414334736aefe6176dbf702ab>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: nan  
 Fold 1 Val Loss: nan  
 Fold 1 Val R2: nan  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/5bfc2449e2ec406495be4a4e05de2f59>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: nan  
 Fold 2 Val Loss: nan  
 Fold 2 Val R2: nan  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/37d0311bab7c492eb66d0cd05047b5c1>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): nan  
 Average validation loss (MSE): nan  
 Average validation R2: nan

View run Reg-Lasso\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.01  
at: <http://localhost:5000/#/experiments/769694705593400528/runs/fe3ba0f76ca345068acc422ca605e4a6>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 98 / 144 =====

Reg-Lasso\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.001

Fold 0 Train Loss: 0.029547395753365256

Fold 0 Val Loss: 0.15345451982435052

Fold 0 Val R2: 0.7797248960009687

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/2edc86cdb0244e74baf14063ce38765c>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 0.039887350821688115

Fold 1 Val Loss: 0.1462376683609615

Fold 1 Val R2: 0.7965649781902858

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/727fbd4d29bb412eb85c337c87bddaa2>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 0.21686096680876832

Fold 2 Val Loss: 0.2438603634103179

Fold 2 Val R2: 0.6539777554711202

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/8b631ed9e4ce464cad328d860b7f403b>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 0.0954319044612739

Average validation loss (MSE): 0.18118418386520996

Average validation R2: 0.7434225432207916

View run Reg-Lasso\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/fceb8e81fefa4269b2c3e077956f9416>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 99 / 144 =====

Reg-Lasso\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.0001

Fold 0 Train Loss: 0.08159183034928821

Fold 0 Val Loss: 0.22778225135458258

Fold 0 Val R2: 0.6730317284645904

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/c206f9d959c8476aa2ff1b1982e895c6>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 0.08907041869420027

Fold 1 Val Loss: 0.22584984765017402

Fold 1 Val R2: 0.6858144061144023

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/1226b181752049c8bb440b9d67f272b3>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 0.11788625388654268

Fold 2 Val Loss: 0.22405165101611174

Fold 2 Val R2: 0.6820850502689113

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e0c9551ccbb646f58fbff953f5ace304>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 0.09618283431001039

Average validation loss (MSE): 0.22589458334028945

Average validation R2: 0.6803103949493012

View run Reg-Lasso\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/6ac73434188d4c32ba2285d6a86b6f67>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 100 / 144 =====

Reg-Lasso\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.01

Fold 0 Train Loss: nan

Fold 0 Val Loss: nan

Fold 0 Val R2: nan

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/d5e250e6faf14fa5bbff2af1a1290120>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: nan

Fold 1 Val Loss: nan

Fold 1 Val R2: nan

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/caad5620cbb84dfe8be2e7fd9f6626e4>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: nan

Fold 2 Val Loss: nan

Fold 2 Val R2: nan

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/ecc71b925f1547999943a77bf7a61c0a>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): nan

Average validation loss (MSE): nan

Average validation R2: nan

View run Reg-Lasso\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/6f056424339149aba0755b6721703721>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 101 / 144 =====

Reg-Lasso\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.001

Fold 0 Train Loss: 0.09107375116158253

Fold 0 Val Loss: 0.1372734886805267

Fold 0 Val R2: 0.8029518320475392

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e4a8aec7d8274690af657208c2e76f20>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 0.04410997518779536

Fold 1 Val Loss: 0.11707068078765784

Fold 1 Val R2: 0.8371399327803214

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/68727ae0638d49ceab25ebf028df14dd>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 0.2850201481049599

Fold 2 Val Loss: 0.2575195041999499

Fold 2 Val R2: 0.6345963091045703

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f4d74c39a1c34585bb80fe05d20a3607>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 0.14006795815144593

Average validation loss (MSE): 0.17062122455604478

Average validation R2: 0.758229357977477

View run Reg-Lasso\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e3c8ecb26643455b9aef30ec8f69971d>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 102 / 144 =====

Reg-Lasso\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.0001

Fold 0 Train Loss: 0.005361283359540786

Fold 0 Val Loss: 0.22881080278011912

Fold 0 Val R2: 0.6715553022733791

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/5b86eeec97f0403882ee66c54db7fb8d>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 0.7070000248408637

Fold 1 Val Loss: 0.22582137135262187

Fold 1 Val R2: 0.6858540202321498

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/0c40f9db52fa4737b8e675a40eae7690>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 0.1852151314276307

Fold 2 Val Loss: 0.22071137407462002

Fold 2 Val R2: 0.6868246894151806

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/69e16e6b2dcd4aedb03243fdec610fad>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 0.2991921465426784

Average validation loss (MSE): 0.22511451606912034

Average validation R2: 0.6814113373069032

View run Reg-Lasso\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/8fa73d428a4a4216a2c196f897bf766f>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 103 / 144 =====

Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-zero\_LR-0.01

Fold 0 Train Loss: inf

Fold 0 Val Loss: inf

```

Fold 0 Val R2:      -inf
  View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/d6a5f573eb8b456b85a993157fd81357
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: inf
Fold 1 Val Loss:    inf
Fold 1 Val R2:      -inf
  View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/103253ea3a924c4f8ec8bfe89c265704
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: inf
Fold 2 Val Loss:    inf
Fold 2 Val R2:      -inf
  View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/1b4ad0f4603642569e6e4ab72de0535a
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE):  inf
Average validation loss (MSE):  inf
Average validation R2:  -inf
  View run Reg-Lasso_TrainMethod-mini-
batch_BatchSize-50_Momentum-0.9_WeightInit-zero_LR-0.01 at: http://localhost:5000/#/experiments/769694705593400528/runs/cfb33c54ea394164b2a464ff4a17bde
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment:  104 /  144 =====
Reg-Lasso_TrainMethod-mini-batch_BatchSize-50_Momentum-0.9_WeightInit-
zero_LR-0.001
Fold 0 Train Loss: 2.5238604955901898
Fold 0 Val Loss:   2.6767458752970676
Fold 0 Val R2:     -2.8423141705760115
  View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/458ed3b71c8d4364a55a6c815145a7e3
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: 3.9391645705495693
Fold 1 Val Loss:   2.8920652124113344
Fold 1 Val R2:     -3.0232270943337625
  View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/c213f5710788490395ad1d5e8870c613
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 2.0062202993895517
Fold 2 Val Loss:   2.5304683884164456
Fold 2 Val R2:     -2.5905726507753926
  View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/1f0f150083104880abcfa5a64caf1fe0
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE):  2.82308178850977
Average validation loss (MSE): 2.6997598253749495
Average validation R2:  -2.8187046385617225
  View run Reg-Lasso_TrainMethod-mini-

```

batch\_BatchSize-50\_Momentum-0.9\_WeightInit-zero\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/689dec9b5b284c309cf1944e3a23d03f>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 105 / 144 =====  
Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-zero\_LR-0.0001  
Fold 0 Train Loss: 14.275685548686313  
Fold 0 Val Loss: 13.181759688287867  
Fold 0 Val R2: -17.921655025550415  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/687cda5ea1bd4d78acee35e9c557a216>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 17.734984249312713  
Fold 1 Val Loss: 12.56219536086585  
Fold 1 Val R2: -16.47559651257306  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/b66fac80574f40d586e52d5eb9f99c63>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 15.97670617203005  
Fold 2 Val Loss: 17.20806296204007  
Fold 2 Val R2: -23.41713973869792  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/fe9d8b08e6294220a53528f806bc7afc>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 15.995791990009693  
Average validation loss (MSE): 14.317339337064595  
Average validation R2: -19.271463758940463  
View run Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-zero\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/5b0087f7557d4dba96c912088e0d6200>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 106 / 144 =====  
Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-xavier\_LR-0.01  
Fold 0 Train Loss: inf  
Fold 0 Val Loss: inf  
Fold 0 Val R2: -inf  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/37b935534a7e4eb59f1e7c4766388efe>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: inf  
Fold 1 Val Loss: inf  
Fold 1 Val R2: -inf  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/b7e4739fb897472fad2a7a45c9f0e784>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: inf  
Fold 2 Val Loss: inf

Fold 2 Val R2: -inf

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e23d5e1a47084317b14ec03cc5f571c8>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): inf

Average validation loss (MSE): inf

Average validation R2: -inf

View run Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-xavier\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/3a46183daf244537b4e83006e75fa5cb>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 107 / 144 =====

Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-xavier\_LR-0.001

Fold 0 Train Loss: 4.1892123340329315

Fold 0 Val Loss: 2.7568531553403304

Fold 0 Val R2: -2.957303546338987

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f985c20c701b4ebcb4aa026cb8fd0c0d>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 1.8751824803819572

Fold 1 Val Loss: 2.7089165606737895

Fold 1 Val R2: -2.7684442440719517

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/7c51c0a415794c05be49f701249dde32>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 2.713891023017557

Fold 2 Val Loss: 2.4670234108208478

Fold 2 Val R2: -2.5005482891091444

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/8129e4758c384589a6602f54a6cacc0>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 2.9260952791441484

Average validation loss (MSE): 2.6442643756116557

Average validation R2: -2.742098693173361

View run Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-xavier\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e4f304e458874b33b552644e5c85163e>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 108 / 144 =====

Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-xavier\_LR-0.0001

Fold 0 Train Loss: 13.097219615869967

Fold 0 Val Loss: 13.161919018776382

Fold 0 Val R2: -17.893174889904653

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/8b0d10450aed455aa68487911ae1661e>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 9.81872334623705  
 Fold 1 Val Loss: 12.51806450805749  
 Fold 1 Val R2: -16.414204936078622  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/1c35b02738b44a8f9b84c01ba0b23c07>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 11.324919469533747  
 Fold 2 Val Loss: 13.85301532593404  
 Fold 2 Val R2: -18.656541922342814  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/424ecb9a71a840c8ac85e1d69d2f47af>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 11.413620810546922  
 Average validation loss (MSE): 13.17766628425597  
 Average validation R2: -17.65464058277536  
 View run Reg-Lasso\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/480cdce88e7644b1b0d6ab85ec29e182>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 109 / 144 =====  
 Reg-Polynomial\_TrainMethod-batch\_Momentum-0\_WeightInit-zero\_LR-0.01  
 Fold 0 Train Loss: inf  
 Fold 0 Val Loss: inf  
 Fold 0 Val R2: -inf  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/49c7c092f06b4958b5e2f8087afcd29b>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: inf  
 Fold 1 Val Loss: inf  
 Fold 1 Val R2: -inf  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/d546bf58522f4af7b25acaaba5524989>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: inf  
 Fold 2 Val Loss: inf  
 Fold 2 Val R2: -inf  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/129d475f338e4c578a47c203cdbd437b>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): inf  
 Average validation loss (MSE): inf  
 Average validation R2: -inf  
 View run Reg-Polynomial\_TrainMethod-batch\_Momentum-0\_WeightInit-zero\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/c9015e0eb43d481fadbfa86ade35220f>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 110 / 144 =====  
 Reg-Polynomial\_TrainMethod-batch\_Momentum-0\_WeightInit-zero\_LR-0.001



Fold 0 Train Loss: 11.52532542852829  
 Fold 0 Val Loss: 11.929641243257235  
 Fold 0 Val R2: -16.124311284786504  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/13406fb450d14de3b4fca536b61d7efe>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: 11.421407118212802  
 Fold 1 Val Loss: 11.239233973206689  
 Fold 1 Val R2: -14.635190536682256  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/7d4ba495f62147ab9d55d8b66338e89e>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 11.505410786440653  
 Fold 2 Val Loss: 11.991257804852431  
 Fold 2 Val R2: -16.014827183612443  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f969f04641de443886732db8186bdf04>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 11.484047777727248  
 Average validation loss (MSE): 11.720044340438784  
 Average validation R2: -15.591443001693733  
 View run Reg-Polynomial\_TrainMethod-batch\_Momentum-0\_WeightInit-zero\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/537ed55d07fe49359de1fca9dc51e980>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 111 / 144 =====  
 Reg-Polynomial\_TrainMethod-batch\_Momentum-0\_WeightInit-zero\_LR-0.0001  
 Fold 0 Train Loss: 22.08255400077714  
 Fold 0 Val Loss: 22.048277641218352  
 Fold 0 Val R2: -30.649029666757446  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/d9a843c9369a4413a10f98a78d4006e9>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: 22.465885236833557  
 Fold 1 Val Loss: 21.266378253503007  
 Fold 1 Val R2: -28.584211594076116  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/0892968756aa42b0aee43963610ac5bb>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 21.716148187288248  
 Fold 2 Val Loss: 23.071326067588682  
 Fold 2 Val R2: -31.736734738364277  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/fec8787a39e742129661e6fb6d8e9f86>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 22.088195808299645  
 Average validation loss (MSE): 22.128660654103346  
 Average validation R2: -30.32332533306595

View run Reg-Polynomial\_TrainMethod-batch\_Momentum-0\_WeightInit-zero\_LR-0.0001  
at: <http://localhost:5000/#/experiments/769694705593400528/runs/e05c0f912187492fb363516a22c2fcab>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 112 / 144 =====

Reg-Polynomial\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.01  
Fold 0 Train Loss: inf  
Fold 0 Val Loss: inf  
Fold 0 Val R2: -inf

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/b893f20189f44da09268102213f9ed85>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: inf  
Fold 1 Val Loss: inf  
Fold 1 Val R2: -inf

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/1f57c6e237af4101bfb09a1727969e60>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: inf  
Fold 2 Val Loss: inf  
Fold 2 Val R2: -inf

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/0750b6bfab3d4ec39622842d88540154>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): inf  
Average validation loss (MSE): inf  
Average validation R2: -inf

View run Reg-Polynomial\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.01  
at: <http://localhost:5000/#/experiments/769694705593400528/runs/d058ebf00a8e46d3b5bb1e91306e58e5>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 113 / 144 =====

Reg-Polynomial\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.001  
Fold 0 Train Loss: 11.555659027133679  
Fold 0 Val Loss: 11.990578911072207  
Fold 0 Val R2: -16.211783788892387

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/3f9d6f8732ad41a4a1cfa34297e32db2>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 11.4286612821207  
Fold 1 Val Loss: 11.248375357262027  
Fold 1 Val R2: -14.64790735366595

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/cfed4b9ec37c4ca39479b32800afb5e7>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 11.509514594147817  
Fold 2 Val Loss: 11.993451265998768  
Fold 2 Val R2: -16.01793956456082

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/689673f7eb06455aab3a8963029f85f5>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 11.497944967800732

Average validation loss (MSE): 11.744135178111

Average validation R2: -15.62587690237305

View run Reg-Polynomial\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/16b46aba42414e5e8fbcf145d7696c1e>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 114 / 144 =====

Reg-Polynomial\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.0001

Fold 0 Train Loss: 22.04500122704224

Fold 0 Val Loss: 21.997362709573864

Fold 0 Val R2: -30.575944221793485

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/205215884b9e4317b40dd7a9aea3c229>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 22.377992708332513

Fold 1 Val Loss: 21.191601561226953

Fold 1 Val R2: -28.48018779367957

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/358aacbc86df4eacebaee2b54c6897ba9>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 21.679788548573782

Fold 2 Val Loss: 23.035593011328878

Fold 2 Val R2: -31.6860318190461

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/6ab624ff67dc4ed3857ea8f98d85c505>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 22.034260827982845

Average validation loss (MSE): 22.074852427376566

Average validation R2: -30.247387944839716

View run Reg-Polynomial\_TrainMethod-batch\_Momentum-0\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/b4613fa7adb9409bb9fac405e8066f86>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 115 / 144 =====

Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.01

Fold 0 Train Loss: nan

Fold 0 Val Loss: nan

Fold 0 Val R2: nan

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/a06a296b1d2946318cedd690016bf753>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: nan

Fold 1 Val Loss: nan

Fold 1 Val R2: nan

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/11e51f004bcf46c1a9587f9b9a806a87>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: nan

Fold 2 Val Loss: nan

Fold 2 Val R2: nan

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/66299cda00174e71a7e3d31e3ff8e51f>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): nan

Average validation loss (MSE): nan

Average validation R2: nan

View run Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/580c471780494c6186bf6e8f0f018f50>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 116 / 144 =====

Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.001

Fold 0 Train Loss: 0.07253427323903051

Fold 0 Val Loss: 0.1645669144888579

Fold 0 Val R2: 0.7637736949988447

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/8e0f2f899c6c407d85529566d8b7c2ab>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 6.200714668040327e+305

Fold 1 Val Loss: inf

Fold 1 Val R2: -inf

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/256ff90678cb43718959c948607e932f>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 7.579955309928816e+305

Fold 2 Val Loss: inf

Fold 2 Val R2: -inf

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/724c443a938947c09c459995c3d40e8f>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 4.593556659323048e+305

Average validation loss (MSE): inf

Average validation R2: -inf

View run Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/5360d5a5102d4a6ea4ac92f85d2a3ac9>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 117 / 144 =====

Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.0001

Fold 0 Train Loss: 0.002994001493536139

Fold 0 Val Loss: 0.13509670541894134

Fold 0 Val R2: 0.806076478749884

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/96823d646d004b3e876e17b5b0be996a>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 0.1866995292544271

Fold 1 Val Loss: 0.09479294121660728

Fold 1 Val R2: 0.8681310753929155

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/c2982c32455d49d2a312dfbc2234b743>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 0.48641496222435066

Fold 2 Val Loss: 0.08422600439324802

Fold 2 Val R2: 0.8804886916418913

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/7c6a5988d15848c389b2dc187a270fdb>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 0.22536949765743797

Average validation loss (MSE): 0.10470521700959888

Average validation R2: 0.8515654152615636

View run Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/a66111434fac401ea98861f464a1b8cf>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 118 / 144 =====

Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.01

Fold 0 Train Loss: nan

Fold 0 Val Loss: nan

Fold 0 Val R2: nan

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f4ac1b8f77b44ab7a4b0b7c5ddf69c6c>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: nan

Fold 1 Val Loss: nan

Fold 1 Val R2: nan

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/4660e93d55f64254b00cfd68c86b16e2>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: nan

Fold 2 Val Loss: nan

Fold 2 Val R2: nan

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/880708c4aea141c9aabe44d39b21c871>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): nan

Average validation loss (MSE): nan

Average validation R2: nan

View run Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/0ca7701b6fb54bef8dc9747c8b501986>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 119 / 144 =====  
Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.001  
Fold 0 Train Loss: 0.1435733085986241  
Fold 0 Val Loss: 0.18376818586181967  
Fold 0 Val R2: 0.7362113784673142  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/56a71c06a17949f29fca927a1d564832>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 5.272683055669698e+305  
Fold 1 Val Loss: inf  
Fold 1 Val R2: -inf  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/108265c116414f50bd95ab3079635466>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 8.726080241758783e+305  
Fold 2 Val Loss: inf  
Fold 2 Val R2: -inf  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/86bd7ef0c56a48ad96b40749ee98d8a9>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 4.66625443247616e+305  
Average validation loss (MSE): inf  
Average validation R2: -inf  
View run Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/a2442fcea4c41a19b75b864292aa18c>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 120 / 144 =====  
Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.0001  
Fold 0 Train Loss: 0.09569696290228029  
Fold 0 Val Loss: 0.13444898082544102  
Fold 0 Val R2: 0.8070062500095335  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/d2fe393fe3174ab7aa1fd55776705aa0>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 0.016739285424723638  
Fold 1 Val Loss: 0.09485436633842106  
Fold 1 Val R2: 0.8680456253092543  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/344c94a0613342f6be6703d6dc715105>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 4.810750405722156e-05  
Fold 2 Val Loss: 0.09851851009355939  
Fold 2 Val R2: 0.8602085410130568  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/d333f020c1064754895d7b66d1642790>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 0.03749478527702038  
 Average validation loss (MSE): 0.1092739524191405  
 Average validation R2: 0.8450868054439482  
 View run Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/2cadb43e250e4e9c9892841f015454b5>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 121 / 144 =====  
 Reg-Polynomial\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.01  
 Fold 0 Train Loss: inf  
 Fold 0 Val Loss: inf  
 Fold 0 Val R2: -inf  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/c4fde240b7be4bb3afda3ad856b82b8d>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: inf  
 Fold 1 Val Loss: inf  
 Fold 1 Val R2: -inf  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/cf2737b675ee48bda7caf9ad7658ae2c>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: inf  
 Fold 2 Val Loss: inf  
 Fold 2 Val R2: -inf  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e8f8dfa0df004482b0d066c009aa21f4>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): inf  
 Average validation loss (MSE): inf  
 Average validation R2: -inf  
 View run Reg-Polynomial\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/459b4ff93eba45b3a2cd6aa5b176cef4>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 122 / 144 =====  
 Reg-Polynomial\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-0.001  
 Fold 0 Train Loss: 0.13844314975417163  
 Fold 0 Val Loss: 0.1378284734923298  
 Fold 0 Val R2: 0.8021551833904794  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/a508472fb5cb4df289eb8f0b50519827>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: 0.08999068672892056  
 Fold 1 Val Loss: 0.17187000450318024  
 Fold 1 Val R2: 0.7609071690869904  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/a508472fb5cb4df289eb8f0b50519827>

```

s/ef52c0494e20408792ae11c207c79d2a
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 0.1344171718180993
Fold 2 Val Loss: 0.11934454920392941
Fold 2 Val R2: 0.8306577247309895
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/run
s/3099f456cd494622bf23653f45117a2b
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE): 0.12095033610039717
Average validation loss (MSE): 0.14301434239981314
Average validation R2: 0.7979066924028198
View run Reg-Polynomial_TrainMethod-mini-
batch_BatchSize-50_Momentum-0_WeightInit-zero_LR-0.001 at: http://localhost:5000
/#/experiments/769694705593400528/runs/ab46d87537384db1b2263d58a2eb5849
View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment: 123 / 144 =====
Reg-Polynomial_TrainMethod-mini-batch_BatchSize-50_Momentum-0_WeightInit-
zero_LR-0.0001
Fold 0 Train Loss: 2.6655488304726918
Fold 0 Val Loss: 2.5481629643360315
Fold 0 Val R2: -2.6577408252170045
View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/run
s/dc1fb7cdb88d4088a1949b6ee13ffe87
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: 1.8673814371146356
Fold 1 Val Loss: 2.820488279608069
Fold 1 Val R2: -2.9236545625154697
View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/run
s/fb2b81fd5ba144a19a4d70b23e7aae7c
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 3.0163661795036436
Fold 2 Val Loss: 2.809288626759996
Fold 2 Val R2: -2.9862007198166056
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/run
s/37a4f4f16bd6441980e92caec50a0b0b
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE): 2.5164321490303236
Average validation loss (MSE): 2.7259799569013654
Average validation R2: -2.8558653691830265
View run Reg-Polynomial_TrainMethod-mini-
batch_BatchSize-50_Momentum-0_WeightInit-zero_LR-0.0001 at: http://localhost:500
0/#/experiments/769694705593400528/runs/d6399f95705c48a18eb0a0b71c96bea1
View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment: 124 / 144 =====
Reg-Polynomial_TrainMethod-mini-batch_BatchSize-50_Momentum-0_WeightInit-
xavier_LR-0.01
Fold 0 Train Loss: inf
Fold 0 Val Loss: inf

```



```

Fold 0 Val R2:      -inf
  View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/40b1b69352314462bca0c1a6191a158c
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: inf
Fold 1 Val Loss:    inf
Fold 1 Val R2:      -inf
  View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/9ff405f238a54f319277bc90af2ab39c
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: inf
Fold 2 Val Loss:    inf
Fold 2 Val R2:      -inf
  View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/68e626e7f413442781d08e4e60da2a8f
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE):  inf
Average validation loss (MSE):  inf
Average validation R2:  -inf
  View run Reg-Polynomial_TrainMethod-mini-
batch_BatchSize-50_Momentum-0_WeightInit-xavier_LR-0.01 at: http://localhost:5000/#/experiments/769694705593400528/runs/aa4788be3a9e424b95aca3601d2c11e6
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment:  125 /  144 =====
Reg-Polynomial_TrainMethod-mini-batch_BatchSize-50_Momentum-0_WeightInit-
xavier_LR-0.001
Fold 0 Train Loss: 0.07291934794538132
Fold 0 Val Loss:   0.1357203513677043
Fold 0 Val R2:     0.8051812709946509
  View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/93e81ede80e143e6b2b99b44fe6d266a
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: 0.09234811963838835
Fold 1 Val Loss:   0.17122888467033903
Fold 1 Val R2:     0.7617990475519476
  View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/1b3c2bfdb4114f2e8513200741b8a1f5
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 0.07377069451078763
Fold 2 Val Loss:   0.1175767486686497
Fold 2 Val R2:     0.8331661205216888
  View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/2efe7e4808dd47619814f1082e5a066d
  View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE):  0.07967938736485243
Average validation loss (MSE): 0.1415086615688977
Average validation R2:  0.8000488130227624
  View run Reg-Polynomial_TrainMethod-mini-

```

batch\_BatchSize-50\_Momentum-0\_WeightInit-xavier\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/3ba3721a1fe8485dabaedfb88bd2f01d>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 126 / 144 =====  
Reg-Polynomial\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-xavier\_LR-0.0001  
Fold 0 Train Loss: 1.506286276431693  
Fold 0 Val Loss: 2.5411007997939756  
Fold 0 Val R2: -2.6476034957285006  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/4c4c197323ae4293a10996dfd8170716>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: 2.9849645717176236  
Fold 1 Val Loss: 2.8729359629183207  
Fold 1 Val R2: -2.9966158981116706  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/90b9fe606c4b45f0aedd4a960affaa97>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: 2.6515864536366234  
Fold 2 Val Loss: 3.5550837395785373  
Fold 2 Val R2: -4.044436241519359  
View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/6eb9859d726c4095ac26b839beff89e5>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Average training loss (MSE): 2.3809457672619803  
Average validation loss (MSE): 2.9897068340969444  
Average validation R2: -3.229551878453177  
View run Reg-Polynomial\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/d9f8b217e0394567bd6b0db443133e42>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
===== Experiment: 127 / 144 =====  
Reg-Polynomial\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.01  
Fold 0 Train Loss: inf  
Fold 0 Val Loss: inf  
Fold 0 Val R2: -inf  
View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/5160a8eca78646968ae8e0951a146cd9>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 1 Train Loss: inf  
Fold 1 Val Loss: inf  
Fold 1 Val R2: -inf  
View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/68361f5a19034a8dbb11bad5623d73ff>  
View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
Fold 2 Train Loss: inf  
Fold 2 Val Loss: inf  
Fold 2 Val R2: -inf

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f07188071d43494b962ff72239d090a2>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): inf

Average validation loss (MSE): inf

Average validation R2: -inf

View run Reg-Polynomial\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/c0cfed54fb8c416a9adda7aa8e33fd82>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 128 / 144 =====

Reg-Polynomial\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.001

Fold 0 Train Loss: 22.04794547952821

Fold 0 Val Loss: 22.015696767597507

Fold 0 Val R2: -30.602261703627878

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e5c6d5bb051b41338e714f58d5703026>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 22.430054724919483

Fold 1 Val Loss: 21.2336344256494

Fold 1 Val R2: -28.53866080399452

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/97935967e2b4426d9c32bba075cd5b76>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 21.683780689393725

Fold 2 Val Loss: 23.036486576431464

Fold 2 Val R2: -31.687299730723524

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/5ec32a2d36144d7ba1a56c0d868a4af7>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 22.053926964613808

Average validation loss (MSE): 22.09527258989279

Average validation R2: -30.276074079448637

View run Reg-Polynomial\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/83eb656c8cb04c0ab707b8b134a4472f>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 129 / 144 =====

Reg-Polynomial\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.0001

Fold 0 Train Loss: 24.749061321753338

Fold 0 Val Loss: 24.447027489745146

Fold 0 Val R2: -34.0922965901215

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/6e4e3e629f574adf9a107fe70d51e5af>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 25.11253795925317

Fold 1 Val Loss: 23.824009651435627

Fold 1 Val R2: -32.142199115699725

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/6843022cb65b4926a3623c8db2194480>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 24.154146384604353

Fold 2 Val Loss: 25.754679661766694

Fold 2 Val R2: -35.5442417088998

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/84974003c7db444f86f21d4b9dff3f35>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 24.67191522187029

Average validation loss (MSE): 24.675238934315825

Average validation R2: -33.926245804907005

View run Reg-Polynomial\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/36642820e8d7490d8bba7d0ce197a622>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 130 / 144 =====

Reg-Polynomial\_TrainMethod-batch\_Momentum-0.9\_WeightInit-xavier\_LR-0.01

Fold 0 Train Loss: inf

Fold 0 Val Loss: inf

Fold 0 Val R2: -inf

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/2c17704f57714e11b283d670e799ebb0>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: inf

Fold 1 Val Loss: inf

Fold 1 Val R2: -inf

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/482645e93f8d4270bfb5cf3da704e373>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: inf

Fold 2 Val Loss: inf

Fold 2 Val R2: -inf

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/ada9e6a491d34f61ac0f9dd8669a05ba>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): inf

Average validation loss (MSE): inf

Average validation R2: -inf

View run Reg-Polynomial\_TrainMethod-batch\_Momentum-0.9\_WeightInit-xavier\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/54d121f0e3a54b74937bad58b663942c>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 131 / 144 =====

Reg-Polynomial\_TrainMethod-batch\_Momentum-0.9\_WeightInit-xavier\_LR-0.001

Fold 0 Train Loss: 21.994821431663162

Fold 0 Val Loss: 21.952105837911056

Fold 0 Val R2: -30.510980595283126

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/aac3e8406c674e848db994ec0410815f>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 22.45871854532938

Fold 1 Val Loss: 21.26772682432326

Fold 1 Val R2: -28.586087625999387

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/0a9464c289974fac935496737809c303>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 21.61870188710477

Fold 2 Val Loss: 22.976840400278267

Fold 2 Val R2: -31.602665625117083

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/3dc57897e520439292407e4961436238>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 22.024080621365773

Average validation loss (MSE): 22.065557687504192

Average validation R2: -30.233244615466532

View run Reg-Polynomial\_TrainMethod-batch\_Momentum-0.9\_WeightInit-

xavier\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/77b6e30550594b3ba31315f6ca14c49e>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 132 / 144 =====

Reg-Polynomial\_TrainMethod-batch\_Momentum-0.9\_WeightInit-xavier\_LR-0.0001

Fold 0 Train Loss: 24.773580713819506

Fold 0 Val Loss: 24.472402194196135

Fold 0 Val R2: -34.12872051343301

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/faa005d5435d47729dc228043253eb38>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 25.12722286049949

Fold 1 Val Loss: 23.83096509297847

Fold 1 Val R2: -32.15187501123214

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/454ec20710b04b8f8e4ffb5bc7439697>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 24.229697978046033

Fold 2 Val Loss: 25.836841571180884

Fold 2 Val R2: -35.6608241986193

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f85398a0ae724c1daf3cf3c2f352a2fb>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 24.710167184121676

Average validation loss (MSE): 24.713402952785163

Average validation R2: -33.98047324109482

View run Reg-Polynomial\_TrainMethod-batch\_Momentum-0.9\_WeightInit-

xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/c9186cb40f5d4bdcbbe02ba2c94e6520>

```

View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment: 133 / 144 =====
Reg-Polynomial_TrainMethod-stochastic_Momentum-0.9_WeightInit-zero_LR-0.01
Fold 0 Train Loss: nan
Fold 0 Val Loss: nan
Fold 0 Val R2: nan
View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/4955188062d54119b86952ab0e811ce3
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: nan
Fold 1 Val Loss: nan
Fold 1 Val R2: nan
View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/9ea9d1f424f6432ca510c45d93a65f4b
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: nan
Fold 2 Val Loss: nan
Fold 2 Val R2: nan
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/1d2a2d2d9eb94ae1aa2f4d7f330f49a9
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE): nan
Average validation loss (MSE): nan
Average validation R2: nan
View run Reg-Polynomial_TrainMethod-stochastic_Momentum-0.9_WeightInit-zero_LR-0.01 at: http://localhost:5000/#/experiments/769694705593400528/runs/8bc323c021ad4038851629486eb80d91
View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment: 134 / 144 =====
Reg-Polynomial_TrainMethod-stochastic_Momentum-0.9_WeightInit-zero_LR-0.001
Fold 0 Train Loss: 0.12632074566409826
Fold 0 Val Loss: 0.2380027259680438
Fold 0 Val R2: 0.6583608272035746
View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/e3e5e6431bb84c81b7c1c955b948a880
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: 0.02514839318060109
Fold 1 Val Loss: 0.09705956785699758
Fold 1 Val R2: 0.8649779121539878
View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/f189380f03324a048f2ba2b0efa44e64
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 0.0612297268640001
Fold 2 Val Loss: 0.08607838830947741
Fold 2 Val R2: 0.8778602774483784
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/714f0bde0e274b15ae95096ead9cc081
View experiment at: http://localhost:5000/#/experiments/769694705593400528

```

Average training loss (MSE): 0.07089962190289982  
 Average validation loss (MSE): 0.14038022737817293  
 Average validation R2: 0.800399672268647  
 View run Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/bf6cd6913c84724942c4d6f6e6f9291>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 135 / 144 =====  
 Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.0001  
 Fold 0 Train Loss: 0.004301911965695168  
 Fold 0 Val Loss: 0.40825267325300263  
 Fold 0 Val R2: 0.4139768567154468  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/a420b87622bc400597a12ab7c7fc02fa>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: 0.10017603712070257  
 Fold 1 Val Loss: 2.5655076620505852  
 Fold 1 Val R2: -2.5689443973764554  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/6fc15ae389b846a08cead380da6a3894>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 0.4818745163172106  
 Fold 2 Val Loss: 0.3638932232703893  
 Fold 2 Val R2: 0.4836588114444623  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/c46a7a497eb84bf7887464a0e33bc044>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 0.19545082180120277  
 Average validation loss (MSE): 1.1125511861913258  
 Average validation R2: -0.5571029097388487  
 View run Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-zero\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/bfc66335ef1a412e8c16ae0e31c52f45>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 136 / 144 =====  
 Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.01  
 Fold 0 Train Loss: nan  
 Fold 0 Val Loss: nan  
 Fold 0 Val R2: nan  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f8099edc17d24cc49f112b8c3cb9808f>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: nan  
 Fold 1 Val Loss: nan  
 Fold 1 Val R2: nan  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/fb3d80d8853e4cc2a82648e4ca92c101>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: nan  
 Fold 2 Val Loss: nan  
 Fold 2 Val R2: nan  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/6b48872f7f38499e81ef958ce4d0e28c>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): nan  
 Average validation loss (MSE): nan  
 Average validation R2: nan  
 View run Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/b1746c789a5d4271ac97fd1d4694635e>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 137 / 144 =====  
 Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.001  
 Fold 0 Train Loss: 2.6113854928134193  
 Fold 0 Val Loss: 2.46598249163885  
 Fold 0 Val R2: -2.5397755010885152  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/35e94c47d077454eb5a19967bcf1838d>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: 0.08327879669429075  
 Fold 1 Val Loss: 0.1569350686208626  
 Fold 1 Val R2: 0.7816835466167968  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/1333e42e93794311bb7c11e4afba5345>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 0.006390489831644427  
 Fold 2 Val Loss: 0.0863136178792051  
 Fold 2 Val R2: 0.8775265017475696  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f56f31019bd0484a9bd767afb9617f50>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 0.9003515931131182  
 Average validation loss (MSE): 0.9030770593796392  
 Average validation R2: -0.29352181757471624  
 View run Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/6f86568e70d248c9a0ccf7e40aeda60c>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 138 / 144 =====  
 Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.0001  
 Fold 0 Train Loss: 0.8965890624772241  
 Fold 0 Val Loss: 0.42082337580038365  
 Fold 0 Val R2: 0.39593233893822677  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/aeb2234721e34af7a03e4ae588c98816>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>



Fold 1 Train Loss: 0.7438480521037493  
 Fold 1 Val Loss: 2.5911744247760344  
 Fold 1 Val R2: -2.6046500981945786  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/509d92ff5811485fa8f3746735a6f44d>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: 0.014373195363338295  
 Fold 2 Val Loss: 0.36401467041256524  
 Fold 2 Val R2: 0.48348648572436714  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/fe7f8c8c622143c0ba641e1e32d9711b>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): 0.551603436648104  
 Average validation loss (MSE): 1.1253374903296611  
 Average validation R2: -0.5750770911773283  
 View run Reg-Polynomial\_TrainMethod-stochastic\_Momentum-0.9\_WeightInit-xavier\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/d5e57d23213a4003b779443308c5a75b>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 139 / 144 =====  
 Reg-Polynomial\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-zero\_LR-0.01  
 Fold 0 Train Loss: inf  
 Fold 0 Val Loss: inf  
 Fold 0 Val R2: -inf  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/37c7a1c5bf414e0eb125d7800e8c8bf4>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: inf  
 Fold 1 Val Loss: inf  
 Fold 1 Val R2: -inf  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/4140136c05db4acba2e89e4b11b97f37>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: inf  
 Fold 2 Val Loss: inf  
 Fold 2 Val R2: -inf  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/6b5657a6d00247a58d8a61c8c510c84f>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): inf  
 Average validation loss (MSE): inf  
 Average validation R2: -inf  
 View run Reg-Polynomial\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-zero\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/3ad2f539a90d4e0d9e34840d880aba38>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 140 / 144 =====

Reg-Polynomial\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-zero\_LR-0.001

Fold 0 Train Loss: 3.39199902293523

Fold 0 Val Loss: 2.563441365925883

Fold 0 Val R2: -2.679672088649296

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/5f9593b2e8be4fb9a23e5f4d3826d7>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 2.611167313489138

Fold 1 Val Loss: 2.8488290757918637

Fold 1 Val R2: -2.9630801807872533

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/f02241cd26754e3abdc5946948f32b6a>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 2.4099224818032416

Fold 2 Val Loss: 2.5021187008802963

Fold 2 Val R2: -2.5503462590167403

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/5aa743df502f4da4b98bb6b50855e183>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 2.8043629394092036

Average validation loss (MSE): 2.6381297141993474

Average validation R2: -2.731032842817763

View run Reg-Polynomial\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-zero\_LR-0.001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/e8e32659147d4432b4310480005c113f>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

===== Experiment: 141 / 144 =====

Reg-Polynomial\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-zero\_LR-0.0001

Fold 0 Train Loss: 26.5648713344905

Fold 0 Val Loss: 12.98149198526221

Fold 0 Val R2: -17.634182299676205

View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/090a15660d094564b527577f77c1831c>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 1 Train Loss: 13.946545936581547

Fold 1 Val Loss: 12.209689428106769

Fold 1 Val R2: -15.98521634635016

View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/73971fd231dc4367a65a8e7d30d87de7>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Fold 2 Train Loss: 8.484230412395735

Fold 2 Val Loss: 13.073422708052094

Fold 2 Val R2: -17.550349904561863

View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/24e29230bd3e4404a3eb38fdf08c2187>

View experiment at: <http://localhost:5000/#/experiments/769694705593400528>

Average training loss (MSE): 16.331882561155926  
 Average validation loss (MSE): 12.75486804047369  
 Average validation R2: -17.056582850196076  
 View run Reg-Polynomial\_TrainMethod-mini-  
 batch\_BatchSize-50\_Momentum-0.9\_WeightInit-zero\_LR-0.0001 at: <http://localhost:5000/#/experiments/769694705593400528/runs/c62120a95e5947afb87cd09c71ce1f33>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 142 / 144 =====  
 Reg-Polynomial\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-xavier\_LR-0.01  
 Fold 0 Train Loss: inf  
 Fold 0 Val Loss: inf  
 Fold 0 Val R2: -inf  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/6a1c4a4721bc4e2d809d09035b172f9e>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: inf  
 Fold 1 Val Loss: inf  
 Fold 1 Val R2: -inf  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/4247f80461fa4352a57514b88ad76f2e>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 2 Train Loss: inf  
 Fold 2 Val Loss: inf  
 Fold 2 Val R2: -inf  
 View run Fold-2 at: <http://localhost:5000/#/experiments/769694705593400528/runs/9639d96eca9a4809a3dbe7fa629a9963>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Average training loss (MSE): inf  
 Average validation loss (MSE): inf  
 Average validation R2: -inf  
 View run Reg-Polynomial\_TrainMethod-mini-  
 batch\_BatchSize-50\_Momentum-0.9\_WeightInit-xavier\_LR-0.01 at: <http://localhost:5000/#/experiments/769694705593400528/runs/0718dd6320fc49a4915940752ffec4a8>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 ===== Experiment: 143 / 144 =====  
 Reg-Polynomial\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0.9\_WeightInit-xavier\_LR-0.001  
 Fold 0 Train Loss: 3.4983590605624477  
 Fold 0 Val Loss: 2.6773338246209515  
 Fold 0 Val R2: -2.8431581378870625  
 View run Fold-0 at: <http://localhost:5000/#/experiments/769694705593400528/runs/7ebc726078e744249d0c2cf2f58dce3d>  
 View experiment at: <http://localhost:5000/#/experiments/769694705593400528>  
 Fold 1 Train Loss: 2.327555229089308  
 Fold 1 Val Loss: 2.8900485482369964  
 Fold 1 Val R2: -3.020421660378971  
 View run Fold-1 at: <http://localhost:5000/#/experiments/769694705593400528/runs/7ebc726078e744249d0c2cf2f58dce3d>

```

s/cdaacd5bccfe4126b9021f26ca180e2b
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 1.3152727734842162
Fold 2 Val Loss: 2.4830972194805514
Fold 2 Val R2: -2.523355994604112
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/273614647c184160ae1ff089749e3d6c
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE): 2.380395687711991
Average validation loss (MSE): 2.6834931974461664
Average validation R2: -2.7956452642900484
View run Reg-Polynomial_TrainMethod-mini-batch_BatchSize-50_Momentum-0.9_WeightInit-xavier_LR-0.001 at: http://localhost:5000/#/experiments/769694705593400528/runs/d0e1c94b2067451e9561e5e0ddce109e
View experiment at: http://localhost:5000/#/experiments/769694705593400528
===== Experiment: 144 / 144 =====
Reg-Polynomial_TrainMethod-mini-batch_BatchSize-50_Momentum-0.9_WeightInit-xavier_LR-0.0001
Fold 0 Train Loss: 11.439569580654688
Fold 0 Val Loss: 12.968843896628497
Fold 0 Val R2: -17.616026698639697
View run Fold-0 at: http://localhost:5000/#/experiments/769694705593400528/runs/970232f6316e42058b34e6e8cc53b372
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 1 Train Loss: 10.831028672297178
Fold 1 Val Loss: 12.169874512256792
Fold 1 Val R2: -15.929828781948498
View run Fold-1 at: http://localhost:5000/#/experiments/769694705593400528/runs/b43dc3b4c63f4cd1ac974a973c39363e
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Fold 2 Train Loss: 11.719401017544671
Fold 2 Val Loss: 13.077980232602007
Fold 2 Val R2: -17.55681674014023
View run Fold-2 at: http://localhost:5000/#/experiments/769694705593400528/runs/1ad85eaa459048d68b07875036290022
View experiment at: http://localhost:5000/#/experiments/769694705593400528
Average training loss (MSE): 11.329999756832178
Average validation loss (MSE): 12.738899547162433
Average validation R2: -17.03422407357614
View run Reg-Polynomial_TrainMethod-mini-batch_BatchSize-50_Momentum-0.9_WeightInit-xavier_LR-0.0001 at: http://localhost:5000/#/experiments/769694705593400528/runs/d7416a70e47f43cc88f46b47c1d4a0e9
View experiment at: http://localhost:5000/#/experiments/769694705593400528

```

## 2.2 Convert the output from MLFlow to be normal csv

- All the training logs are inside `mlruns` folder which I cannot let Git tracks them because they create a lot of git diff lines, instead I load all of them and save them into CSV file so I can

push it to GitHub for backup just in case I somehow lost my laptop and the re-training will cost me another 2+ hours :(

```
[19]: runs = mlflow.search_runs(search_all_experiments=True, output_format="pandas")
      runs.to_csv("mlflow_runs.csv", index=False)
```

```
[20]: runs[runs["tags.mlflow.parentRunId"].isna()]
```

```
[20]:
```

|     | run_id                           | experiment_id      | status \ |
|-----|----------------------------------|--------------------|----------|
| 3   | d7416a70e47f43cc88f46b47c1d4a0e9 | 769694705593400528 | FINISHED |
| 7   | d0e1c94b2067451e9561e5e0ddce109e | 769694705593400528 | FINISHED |
| 11  | 0718dd6320fc49a4915940752ffec4a8 | 769694705593400528 | FINISHED |
| 15  | c62120a95e5947afb87cd09c71ce1f33 | 769694705593400528 | FINISHED |
| 19  | e8e32659147d4432b4310480005c113f | 769694705593400528 | FINISHED |
| ..  | ...                              | ...                | ...      |
| 559 | 3fc02ea2897a4cd1873358642a9bf4e2 | 769694705593400528 | FINISHED |
| 563 | b3ec879409e7431d88a4577fc38128a7 | 769694705593400528 | FINISHED |
| 567 | 017b6875ee1441f5bfb158f1afd963b7 | 769694705593400528 | FINISHED |
| 571 | 25bafefc63a24461bb488104d674e437 | 769694705593400528 | FINISHED |
| 575 | 13e082e0fa874dbc87948bb105bc8813 | 769694705593400528 | FINISHED |

|     | artifact_uri \                                    |
|-----|---------------------------------------------------|
| 3   | mlflow-artifacts:/769694705593400528/d7416a70e... |
| 7   | mlflow-artifacts:/769694705593400528/d0e1c94b2... |
| 11  | mlflow-artifacts:/769694705593400528/0718dd632... |
| 15  | mlflow-artifacts:/769694705593400528/c62120a95... |
| 19  | mlflow-artifacts:/769694705593400528/e8e326591... |
| ..  | ...                                               |
| 559 | mlflow-artifacts:/769694705593400528/3fc02ea28... |
| 563 | mlflow-artifacts:/769694705593400528/b3ec87940... |
| 567 | mlflow-artifacts:/769694705593400528/017b6875e... |
| 571 | mlflow-artifacts:/769694705593400528/25bafefc6... |
| 575 | mlflow-artifacts:/769694705593400528/13e082e0f... |

|     | start_time                       | end_time \                       |
|-----|----------------------------------|----------------------------------|
| 3   | 2025-09-10 13:46:11.582000+00:00 | 2025-09-10 13:47:10.619000+00:00 |
| 7   | 2025-09-10 13:45:09.971000+00:00 | 2025-09-10 13:46:11.553000+00:00 |
| 11  | 2025-09-10 13:45:09.095000+00:00 | 2025-09-10 13:45:09.946000+00:00 |
| 15  | 2025-09-10 13:44:08.302000+00:00 | 2025-09-10 13:45:09.071000+00:00 |
| 19  | 2025-09-10 13:43:08.525000+00:00 | 2025-09-10 13:44:08.275000+00:00 |
| ..  | ...                              | ...                              |
| 559 | 2025-09-10 09:52:14.606000+00:00 | 2025-09-10 09:53:14.126000+00:00 |
| 563 | 2025-09-10 09:51:44.636000+00:00 | 2025-09-10 09:52:14.578000+00:00 |
| 567 | 2025-09-10 09:50:20.742000+00:00 | 2025-09-10 09:51:44.588000+00:00 |
| 571 | 2025-09-10 09:49:03.496000+00:00 | 2025-09-10 09:50:20.706000+00:00 |
| 575 | 2025-09-10 09:48:40.555000+00:00 | 2025-09-10 09:49:03.466000+00:00 |

|     | metrics.train_loss | metrics.val_loss | metrics.val_r2 | \ |
|-----|--------------------|------------------|----------------|---|
| 3   | NaN                | NaN              | NaN            |   |
| 7   | NaN                | NaN              | NaN            |   |
| 11  | NaN                | NaN              | NaN            |   |
| 15  | NaN                | NaN              | NaN            |   |
| 19  | NaN                | NaN              | NaN            |   |
| ..  | ...                | ...              | ...            |   |
| 559 | NaN                | NaN              | NaN            |   |
| 563 | NaN                | NaN              | NaN            |   |
| 567 | NaN                | NaN              | NaN            |   |
| 571 | NaN                | NaN              | NaN            |   |
| 575 | NaN                | NaN              | NaN            |   |

|     | metrics.avg_train_mse | ... | params.weight_init_method | params.num_epochs | \ |
|-----|-----------------------|-----|---------------------------|-------------------|---|
| 3   | 11.330000             | ... | xavier                    | 500               |   |
| 7   | 2.380396              | ... | xavier                    | 500               |   |
| 11  | inf                   | ... | xavier                    | 500               |   |
| 15  | 16.331883             | ... | zero                      | 500               |   |
| 19  | 2.804363              | ... | zero                      | 500               |   |
| ..  | ...                   | ... | ...                       | ...               |   |
| 559 | 12.015045             | ... | xavier                    | 500               |   |
| 563 | inf                   | ... | xavier                    | 500               |   |
| 567 | 22.888834             | ... | zero                      | 500               |   |
| 571 | 12.026102             | ... | zero                      | 500               |   |
| 575 | inf                   | ... | zero                      | 500               |   |

|     | params.learning_rate | params.batch_size | params.lambda | \ |
|-----|----------------------|-------------------|---------------|---|
| 3   | 0.0001               | 50                | None          |   |
| 7   | 0.001                | 50                | None          |   |
| 11  | 0.01                 | 50                | None          |   |
| 15  | 0.0001               | 50                | None          |   |
| 19  | 0.001                | 50                | None          |   |
| ..  | ...                  | ...               | ...           |   |
| 559 | 0.001                | None              | None          |   |
| 563 | 0.01                 | None              | None          |   |
| 567 | 0.0001               | None              | None          |   |
| 571 | 0.001                | None              | None          |   |
| 575 | 0.01                 | None              | None          |   |

|     | tags.mlflow.parentRunId | tags.mlflow.source.type | \ |
|-----|-------------------------|-------------------------|---|
| 3   | None                    | LOCAL                   |   |
| 7   | None                    | LOCAL                   |   |
| 11  | None                    | LOCAL                   |   |
| 15  | None                    | LOCAL                   |   |
| 19  | None                    | LOCAL                   |   |
| ..  | ...                     | ...                     |   |
| 559 | None                    | LOCAL                   |   |

|     |      |       |
|-----|------|-------|
| 563 | None | LOCAL |
| 567 | None | LOCAL |
| 571 | None | LOCAL |
| 575 | None | LOCAL |

|     | tags.mlflow.source.name \                         |
|-----|---------------------------------------------------|
| 3   | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |
| 7   | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |
| 11  | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |
| 15  | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |
| 19  | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |
| ..  | ...                                               |
| 559 | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |
| 563 | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |
| 567 | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |
| 571 | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |
| 575 | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |

|     | tags.mlflow.runName                               | tags.mlflow.user |
|-----|---------------------------------------------------|------------------|
| 3   | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... | promb            |
| 7   | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... | promb            |
| 11  | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... | promb            |
| 15  | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... | promb            |
| 19  | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... | promb            |
| ..  | ...                                               | ...              |
| 559 | Reg-Normal_TrainMethod-batch_Momentum-0_Weight... | promb            |
| 563 | Reg-Normal_TrainMethod-batch_Momentum-0_Weight... | promb            |
| 567 | Reg-Normal_TrainMethod-batch_Momentum-0_Weight... | promb            |
| 571 | Reg-Normal_TrainMethod-batch_Momentum-0_Weight... | promb            |
| 575 | Reg-Normal_TrainMethod-batch_Momentum-0_Weight... | promb            |

[144 rows x 25 columns]

- Loading the mlflow runs from csv looks the same here

```
[38]: run_df = pd.read_csv("mlflow_runs.csv")
run_df[run_df["tags.mlflow.parentRunId"].isna()]
```

| [38]: | run_id                           | experiment_id      | status \ |
|-------|----------------------------------|--------------------|----------|
| 3     | d7416a70e47f43cc88f46b47c1d4a0e9 | 769694705593400528 | FINISHED |
| 7     | d0e1c94b2067451e9561e5e0ddce109e | 769694705593400528 | FINISHED |
| 11    | 0718dd6320fc49a4915940752ffec4a8 | 769694705593400528 | FINISHED |
| 15    | c62120a95e5947afb87cd09c71ce1f33 | 769694705593400528 | FINISHED |
| 19    | e8e32659147d4432b4310480005c113f | 769694705593400528 | FINISHED |
| ..    | ...                              | ...                | ...      |
| 559   | 3fc02ea2897a4cd1873358642a9bf4e2 | 769694705593400528 | FINISHED |
| 563   | b3ec879409e7431d88a4577fc38128a7 | 769694705593400528 | FINISHED |
| 567   | 017b6875ee1441f5bfb158f1afd963b7 | 769694705593400528 | FINISHED |

|     |                                  |                    |          |
|-----|----------------------------------|--------------------|----------|
| 571 | 25bafefc63a24461bb488104d674e437 | 769694705593400528 | FINISHED |
| 575 | 13e082e0fa874dbc87948bb105bc8813 | 769694705593400528 | FINISHED |

|     | artifact_uri \                                    |
|-----|---------------------------------------------------|
| 3   | mlflow-artifacts:/769694705593400528/d7416a70e... |
| 7   | mlflow-artifacts:/769694705593400528/d0e1c94b2... |
| 11  | mlflow-artifacts:/769694705593400528/0718dd632... |
| 15  | mlflow-artifacts:/769694705593400528/c62120a95... |
| 19  | mlflow-artifacts:/769694705593400528/e8e326591... |
| ..  | ...                                               |
| 559 | mlflow-artifacts:/769694705593400528/3fc02ea28... |
| 563 | mlflow-artifacts:/769694705593400528/b3ec87940... |
| 567 | mlflow-artifacts:/769694705593400528/017b6875e... |
| 571 | mlflow-artifacts:/769694705593400528/25bafefc6... |
| 575 | mlflow-artifacts:/769694705593400528/13e082e0f... |

|     | start_time                       | end_time \                       |
|-----|----------------------------------|----------------------------------|
| 3   | 2025-09-10 13:46:11.582000+00:00 | 2025-09-10 13:47:10.619000+00:00 |
| 7   | 2025-09-10 13:45:09.971000+00:00 | 2025-09-10 13:46:11.553000+00:00 |
| 11  | 2025-09-10 13:45:09.095000+00:00 | 2025-09-10 13:45:09.946000+00:00 |
| 15  | 2025-09-10 13:44:08.302000+00:00 | 2025-09-10 13:45:09.071000+00:00 |
| 19  | 2025-09-10 13:43:08.525000+00:00 | 2025-09-10 13:44:08.275000+00:00 |
| ..  | ...                              | ...                              |
| 559 | 2025-09-10 09:52:14.606000+00:00 | 2025-09-10 09:53:14.126000+00:00 |
| 563 | 2025-09-10 09:51:44.636000+00:00 | 2025-09-10 09:52:14.578000+00:00 |
| 567 | 2025-09-10 09:50:20.742000+00:00 | 2025-09-10 09:51:44.588000+00:00 |
| 571 | 2025-09-10 09:49:03.496000+00:00 | 2025-09-10 09:50:20.706000+00:00 |
| 575 | 2025-09-10 09:48:40.555000+00:00 | 2025-09-10 09:49:03.466000+00:00 |

|     | metrics.train_loss | metrics.val_loss | metrics.val_r2 \ |
|-----|--------------------|------------------|------------------|
| 3   | NaN                | NaN              | NaN              |
| 7   | NaN                | NaN              | NaN              |
| 11  | NaN                | NaN              | NaN              |
| 15  | NaN                | NaN              | NaN              |
| 19  | NaN                | NaN              | NaN              |
| ..  | ...                | ...              | ...              |
| 559 | NaN                | NaN              | NaN              |
| 563 | NaN                | NaN              | NaN              |
| 567 | NaN                | NaN              | NaN              |
| 571 | NaN                | NaN              | NaN              |
| 575 | NaN                | NaN              | NaN              |

|    | metrics.avg_train_mse ... | params.weight_init_method | params.num_epochs \ |
|----|---------------------------|---------------------------|---------------------|
| 3  | 11.330000 ...             | xavier                    | 500                 |
| 7  | 2.380396 ...              | xavier                    | 500                 |
| 11 | inf ...                   | xavier                    | 500                 |
| 15 | 16.331883 ...             | zero                      | 500                 |



|     |           |     |        |     |
|-----|-----------|-----|--------|-----|
| 19  | 2.804363  | ... | zero   | 500 |
| ..  | ...       | ... | ...    | ... |
| 559 | 12.015045 | ... | xavier | 500 |
| 563 | inf       | ... | xavier | 500 |
| 567 | 22.888834 | ... | zero   | 500 |
| 571 | 12.026102 | ... | zero   | 500 |
| 575 | inf       | ... | zero   | 500 |

|     | params.learning_rate | params.batch_size | params.lambda | \ |
|-----|----------------------|-------------------|---------------|---|
| 3   | 0.0001               | 50.0              | NaN           |   |
| 7   | 0.0010               | 50.0              | NaN           |   |
| 11  | 0.0100               | 50.0              | NaN           |   |
| 15  | 0.0001               | 50.0              | NaN           |   |
| 19  | 0.0010               | 50.0              | NaN           |   |
| ..  | ...                  | ...               | ...           |   |
| 559 | 0.0010               | NaN               | NaN           |   |
| 563 | 0.0100               | NaN               | NaN           |   |
| 567 | 0.0001               | NaN               | NaN           |   |
| 571 | 0.0010               | NaN               | NaN           |   |
| 575 | 0.0100               | NaN               | NaN           |   |

|     | tags.mlflow.parentRunId | tags.mlflow.source.type | \ |
|-----|-------------------------|-------------------------|---|
| 3   | NaN                     | LOCAL                   |   |
| 7   | NaN                     | LOCAL                   |   |
| 11  | NaN                     | LOCAL                   |   |
| 15  | NaN                     | LOCAL                   |   |
| 19  | NaN                     | LOCAL                   |   |
| ..  | ...                     | ...                     |   |
| 559 | NaN                     | LOCAL                   |   |
| 563 | NaN                     | LOCAL                   |   |
| 567 | NaN                     | LOCAL                   |   |
| 571 | NaN                     | LOCAL                   |   |
| 575 | NaN                     | LOCAL                   |   |

|     | tags.mlflow.source.name                           | \ |
|-----|---------------------------------------------------|---|
| 3   | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |   |
| 7   | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |   |
| 11  | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |   |
| 15  | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |   |
| 19  | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |   |
| ..  | ...                                               |   |
| 559 | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |   |
| 563 | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |   |
| 567 | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |   |
| 571 | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |   |
| 575 | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |   |

|     | tags.mlflow.runName                               | tags.mlflow.user |
|-----|---------------------------------------------------|------------------|
| 3   | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... | promb            |
| 7   | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... | promb            |
| 11  | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... | promb            |
| 15  | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... | promb            |
| 19  | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... | promb            |
| ..  | ...                                               | ...              |
| 559 | Reg-Normal_TrainMethod-batch_Momentum-0_Weight... | promb            |
| 563 | Reg-Normal_TrainMethod-batch_Momentum-0_Weight... | promb            |
| 567 | Reg-Normal_TrainMethod-batch_Momentum-0_Weight... | promb            |
| 571 | Reg-Normal_TrainMethod-batch_Momentum-0_Weight... | promb            |
| 575 | Reg-Normal_TrainMethod-batch_Momentum-0_Weight... | promb            |

[144 rows x 25 columns]

## 2.3 Finding the best parameter set

```
[42]: # Clean up a little bit so it's easier to read
# Also filter out the child runs (the folds) so I can see only the parent runs
# which I logged the averaged metrics there
```

```
params_runs_results = run_df[run_df["tags.mlflow.parentRunId"].isna()].drop(
    columns=[
        "tags.mlflow.parentRunId",
        "metrics.train_loss",
        "metrics.val_loss",
        "metrics.val_r2",
        "tags.mlflow.source.name",
        "tags.mlflow.source.type",
        "tags.mlflow.user",
        "status",
        "experiment_id",
        "run_id",
        "artifact_uri",
    ],
)
params_runs_results.columns
```

```
[42]: Index(['start_time', 'end_time', 'metrics.avg_train_mse', 'metrics.avg_val_r2',
        'metrics.avg_val_mse', 'params.train_method', 'params.momentum',
        'params.model_type', 'params.weight_init_method', 'params.num_epochs',
        'params.learning_rate', 'params.batch_size', 'params.lambda',
        'tags.mlflow.runName'],
        dtype='object')
```

```
[41]: params_runs_results
```

```

[41]:
start_time                                end_time \
3      2025-09-10 13:46:11.582000+00:00 2025-09-10 13:47:10.619000+00:00
7      2025-09-10 13:45:09.971000+00:00 2025-09-10 13:46:11.553000+00:00
11     2025-09-10 13:45:09.095000+00:00 2025-09-10 13:45:09.946000+00:00
15     2025-09-10 13:44:08.302000+00:00 2025-09-10 13:45:09.071000+00:00
19     2025-09-10 13:43:08.525000+00:00 2025-09-10 13:44:08.275000+00:00
..
559    2025-09-10 09:52:14.606000+00:00 2025-09-10 09:53:14.126000+00:00
563    2025-09-10 09:51:44.636000+00:00 2025-09-10 09:52:14.578000+00:00
567    2025-09-10 09:50:20.742000+00:00 2025-09-10 09:51:44.588000+00:00
571    2025-09-10 09:49:03.496000+00:00 2025-09-10 09:50:20.706000+00:00
575    2025-09-10 09:48:40.555000+00:00 2025-09-10 09:49:03.466000+00:00

metrics.avg_train_mse  metrics.avg_val_r2  metrics.avg_val_mse \
3      11.330000        -17.034224        12.738900
7      2.380396         -2.795645         2.683493
11     inf              -inf              inf
15     16.331883        -17.056583        12.754868
19     2.804363         -2.731033         2.638130
..
559    12.015045        -16.081391        12.066079
563    inf              -inf              inf
567    22.888834        -31.405650        22.893884
571    12.026102        -16.094818        12.075706
575    inf              -inf              inf

params.train_method    params.momentum  params.model_type \
3      mini-batch      0.9          Polynomial
7      mini-batch      0.9          Polynomial
11     mini-batch      0.9          Polynomial
15     mini-batch      0.9          Polynomial
19     mini-batch      0.9          Polynomial
..
559    batch           0.0          Normal
563    batch           0.0          Normal
567    batch           0.0          Normal
571    batch           0.0          Normal
575    batch           0.0          Normal

params.weight_init_method  params.num_epochs  params.learning_rate \
3      xavier              500                0.0001
7      xavier              500                0.0010
11     xavier              500                0.0100
15     zero                500                0.0001
19     zero                500                0.0010
..
559    xavier              500                0.0010

```

|     |        |     |        |
|-----|--------|-----|--------|
| 563 | xavier | 500 | 0.0100 |
| 567 | zero   | 500 | 0.0001 |
| 571 | zero   | 500 | 0.0010 |
| 575 | zero   | 500 | 0.0100 |

|     | params.batch_size | params.lambda \ |
|-----|-------------------|-----------------|
| 3   | 50.0              | NaN             |
| 7   | 50.0              | NaN             |
| 11  | 50.0              | NaN             |
| 15  | 50.0              | NaN             |
| 19  | 50.0              | NaN             |
| ..  | ...               | ...             |
| 559 | NaN               | NaN             |
| 563 | NaN               | NaN             |
| 567 | NaN               | NaN             |
| 571 | NaN               | NaN             |
| 575 | NaN               | NaN             |

|     | tags.mlflow.runName                               |
|-----|---------------------------------------------------|
| 3   | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... |
| 7   | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... |
| 11  | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... |
| 15  | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... |
| 19  | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... |
| ..  | ...                                               |
| 559 | Reg-Normal_TrainMethod-batch_Momentum-0_Weight... |
| 563 | Reg-Normal_TrainMethod-batch_Momentum-0_Weight... |
| 567 | Reg-Normal_TrainMethod-batch_Momentum-0_Weight... |
| 571 | Reg-Normal_TrainMethod-batch_Momentum-0_Weight... |
| 575 | Reg-Normal_TrainMethod-batch_Momentum-0_Weight... |

[144 rows x 14 columns]

### 2.3.1 Find the top 5 models based on avg\_val\_r2 and avg\_val\_mse

```
[45]: top5_r2 = params_runs_results.nlargest(5, "metrics.avg_val_r2")
top5_min_valmse = params_runs_results.nsmallest(5, "metrics.avg_val_mse")
top5_r2_df = params_runs_results.loc[top5_r2.index]
top5_min_valmse_df = params_runs_results.loc[top5_min_valmse.index]
```

```
[26]: top5_min_valmse_df[["tags.mlflow.runName", "metrics.avg_val_mse", "metrics.
    ↪avg_val_r2"]]
```

```
[26]: tags.mlflow.runName metrics.avg_val_mse \
531 Reg-Normal_TrainMethod-stochastic_Momentum-0_W... 0.096030
387 Reg-Ridge_TrainMethod-stochastic_Momentum-0_We... 0.101890
111 Reg-Polynomial_TrainMethod-stochastic_Momentum... 0.104705
```

```

99 Reg-Polynomial_TrainMethod-stochastic_Momentum... 0.109274
543 Reg-Normal_TrainMethod-stochastic_Momentum-0_W... 0.113151

```

```

    metrics.avg_val_r2
531         0.863948
387         0.855629
111         0.851565
99         0.845087
543         0.839384

```

```
[27]: top5_r2_df[["tags.mlflow.runName", "metrics.avg_val_mse", "metrics.avg_val_r2"]]
```

```

[27]:          tags.mlflow.runName  metrics.avg_val_mse  \
531  Reg-Normal_TrainMethod-stochastic_Momentum-0_W...  0.096030
387  Reg-Ridge_TrainMethod-stochastic_Momentum-0_We...  0.101890
111  Reg-Polynomial_TrainMethod-stochastic_Momentum...  0.104705
99   Reg-Polynomial_TrainMethod-stochastic_Momentum...  0.109274
543  Reg-Normal_TrainMethod-stochastic_Momentum-0_W...  0.113151

    metrics.avg_val_r2
531         0.863948
387         0.855629
111         0.851565
99         0.845087
543         0.839384

```

### 2.3.2 Log the best parameters

- As top 5 model based on average validation loss (mse) and validation r2 are the same I don't have to analyze further

```
[34]: for p in top5_r2_df.loc[:, "tags.mlflow.runName"]:
      print(p)
```

```

Reg-Normal_TrainMethod-stochastic_Momentum-0_WeightInit-xavier_LR-0.0001
Reg-Ridge_TrainMethod-stochastic_Momentum-0_WeightInit-xavier_LR-0.0001
Reg-Polynomial_TrainMethod-stochastic_Momentum-0_WeightInit-zero_LR-0.0001
Reg-Polynomial_TrainMethod-stochastic_Momentum-0_WeightInit-xavier_LR-0.0001
Reg-Normal_TrainMethod-stochastic_Momentum-0_WeightInit-zero_LR-0.0001

```

## 2.4 Refit the model with entire training set and best parameter set

### 2.4.1 Declare a new class inherited from LinearRegression

- This doesn't have CV because there is no validation set anymore

```
[20]: # This will be copied to the model.py for deployment compatibility
class LinearRegressionNoCV(LinearRegression):
    def fit(self, X_train_ori: Matrix, y_train_ori: Vector) -> None:
```

```

X_train = X_train_ori.copy()
y_train = y_train_ori.copy()
X_train = self.add_intercept(X_train)

# initialize theta
self.weight_init(X_train)
self.prev_step = 0 # for momentum

# one epoch will exhaust the WHOLE training set

for _ in range(self.num_epochs):
    perm = np.random.permutation(X_train.shape[0])

    X_train = X_train[perm]
    y_train = y_train[perm]

    train_loss = self.start_training(X_train, y_train)

print(f"Last Train Loss: {train_loss}")

```

#### 2.4.2 Fit the final model

```

[22]: def train_final_model() -> None:
    from utils import LinearRegressionNoCV

    final_pipe = Pipeline(
        steps=[
            ("preprocess", get_preprocessor()),
            (
                "model",
                LinearRegressionNoCV(
                    regularization=None,
                    momentum=0,
                    train_method="stochastic",
                    weight_init_method="xavier",
                    lr=0.0001,
                    num_epochs=500,
                    batch_size=None,
                    params_for_logging={}, # No need to log, this is final
                    ↪ model with chosen params
                ),
            ),
        ],
    )
    final_pipe.fit(X_train, np.log(y_train.to_numpy()))
    return final_pipe

```

```
final_pipe = train_final_model()
```

Last Train Loss: 0.03125968917514776

```
[23]: signature = mlflow.models.infer_signature(X_train, final_pipe.predict(X_train))
      mlflow.sklearn.log_model(final_pipe, name="final_pipe", signature=signature)
```

```
d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\site-
packages\sklearn\pipeline.py:61: FutureWarning: This Pipeline instance is not
fitted yet. Call 'fit' with appropriate arguments before using other methods
such as transform, predict, etc. This will raise an error in 1.8 instead of the
current warning.
  warnings.warn(
d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\site-
packages\mlflow\types\utils.py:452: UserWarning: Hint: Inferred schema contains
integer column(s). Integer columns in Python cannot represent missing values. If
your input data contains missing values at inference time, it will be encoded as
floats and will cause a schema enforcement error. The best way to avoid this
problem is to infer the model schema based on a realistic data sample (training
dataset) that includes missing values. Alternatively, you can declare integer
columns as doubles (float64) whenever these columns may have missing values. See
`Handling Integers With Missing Values
<https://www.mlflow.org/docs/latest/models.html#handling-integers-with-missing-
values>`_ for more details.
  warnings.warn(
2025/09/12 10:17:37 WARNING mlflow.utils.environment: Failed to resolve
installed pip version. ``pip`` will be added to conda.yaml environment spec
without a version specifier.
```

```
[23]: <mlflow.models.model.ModelInfo at 0x19b371aca50>
```

## 2.5 Comparing the final model with sklearn Linear Regresion

```
[2]: loaded_model = mlflow.sklearn.load_model(
      model_uri="mlartifacts/769694705593400528/models/
      ↪m-21e8645ce11644e696c9fd5b24a762fd/artifacts",
      )
```

```
[55]: y_pred = loaded_model.predict(X_test)
```

```
d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\site-
packages\sklearn\pipeline.py:61: FutureWarning: This Pipeline instance is not
fitted yet. Call 'fit' with appropriate arguments before using other methods
such as transform, predict, etc. This will raise an error in 1.8 instead of the
current warning.
  warnings.warn(
```

```
[56]: from sklearn.linear_model import LinearRegression as SklearnLinearRegression

sklearn_model = SklearnLinearRegression()
sklearn_pipe = Pipeline(steps=[("preprocess", get_preprocessor()), ("model",
    ↪ sklearn_model)])
sklearn_pipe.fit(X_train, np.log(y_train.to_numpy()))
```

```
[56]: Pipeline(steps=[('preprocess',
                        ColumnTransformer(remainder='passthrough',
                                           transformers=[('brand_le',
                                                           Pipeline(steps=[('pre_imputer',
                                                                 SimpleImputer(strategy='most_frequent'))],
                                                                 ('encoder',
                                                                 OrdinalEncoder(handle_unknown='use_encoded_value',
                                                                 unknown_value=nan)),
                                                                 ('post_imputer',
                                                                 SimpleImputer(strategy='most_frequent')))]),
                        ['brand']),
                        ('fuel_le',
                        Pipeline(steps=[('...
                                      SimpleImputer(),
                                      ['mileage']),
                                      ('engine_imputer',
                                       SimpleImputer(strategy='median'),
                                       ['engine']),
                                      ('max_power_imputer',
                                       SimpleImputer(strategy='median'),
                                       ['max_power']),
                                      ('seats_imputer',
                                       SimpleImputer(strategy='most_frequent'),
                                       ['seats'])])),
                        ('scaler',
                        StandardScaler()))],
                        ['year', 'km_driven',
                        'mileage', 'engine',
                        'max_power', 'seats'])]),
                        ('model', LinearRegression()))]
```

```
[57]: y_pred_sklearn = sklearn_pipe.predict(X_test)
```

```
[61]: import pandas as pd

mse_func = loaded_model.steps[-1][1].mse
r_squared_func = loaded_model.steps[-1][1].r_squared

results = {
    "MSE": [
```



```

        mse_func(np.log(y_test.to_numpy()), y_pred),
        mse_func(np.log(y_test.to_numpy()), y_pred_sklearn),
        mse_func(y_test.to_numpy(), np.exp(y_pred)),
        mse_func(y_test.to_numpy(), np.exp(y_pred_sklearn)),
    ],
    "R2": [
        r_squared_func(np.log(y_test.to_numpy()), y_pred),
        r_squared_func(np.log(y_test.to_numpy()), y_pred_sklearn),
        r_squared_func(y_test.to_numpy(), np.exp(y_pred)),
        r_squared_func(y_test.to_numpy(), np.exp(y_pred_sklearn)),
    ],
]
}

results_df = pd.DataFrame(
    results,
    index=[
        "MyModel (log scale)",
        "Sklearn (log scale)",
        "MyModel (original scale)",
        "Sklearn (original scale)",
    ],
)
results_df["MSE"] = results_df["MSE"].apply(lambda x: f"{x:.4f}")
results_df["R2"] = results_df["R2"].apply(lambda x: f"{x:.4f}")

results_df.T

# This shows that my model is very close to sklearn's

```

```

[61]:      MyModel (log scale)  Sklearn (log scale)  MyModel (original scale)  \
MSE              0.0919              0.0918      88113639914.7079
R2              0.8657              0.8657              0.8623

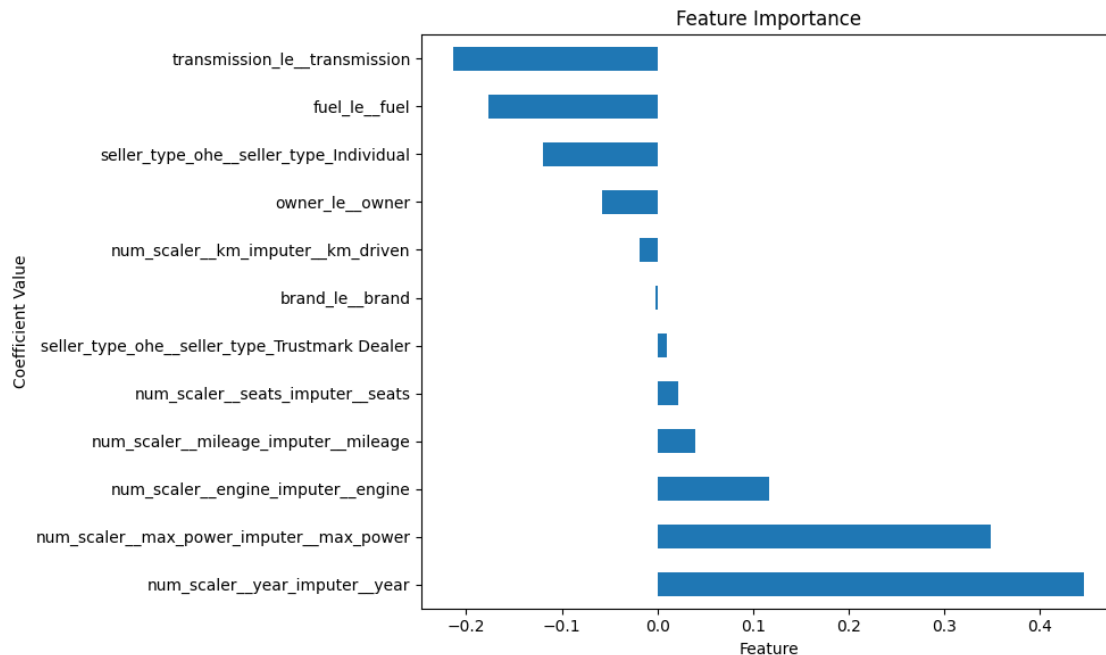
      Sklearn (original scale)
MSE      88821891150.2106
R2              0.8612

```

```

[59]: loaded_model.steps[-1][1].plot_feature_importance(
        feature_names=loaded_model.steps[0][1].get_feature_names_out().tolist(),
    )

```



```
[ ]: # Get categories for transmission and fuel from the fitted OrdinalEncoder
print(loader_model.named_steps["preprocess"].
    ↳named_transformers_["transmission_le"].named_steps["encoder"].categories_)
print(loader_model.named_steps["preprocess"].named_transformers_["fuel_le"].
    ↳named_steps["encoder"].categories_)
print(loader_model.named_steps["preprocess"].named_transformers_["owner_le"].
    ↳named_steps["encoder"].categories_)
# Transmission: Automatic = 0, Manual = 1
# Fuel: Diesel = 0, Petrol = 1
# Owner: First Owner = 0, Second Owner = 1, Third Owner = 2, Fourth & Above,
    ↳Owner = 3
```

```
[array(['Automatic', 'Manual'], dtype=object)]
[array(['Diesel', 'Petrol'], dtype=object)]
[array(['First Owner', 'Second Owner', 'Third Owner',
       'Fourth & Above Owner'], dtype=object)]
```

## 2.6 Looking through the results

### 2.6.1 Clean the mlruns dataframe

```
[66]: run_df.shape
```

```
[66]: (576, 25)
```

```
[67]: # This is dataframe of all runs including child runs (the folds)
run_df.head(5)
```

```
[67]:
```

|   | run_id                           | experiment_id      | status   | \ |
|---|----------------------------------|--------------------|----------|---|
| 0 | 1ad85eaa459048d68b07875036290022 | 769694705593400528 | FINISHED |   |
| 1 | b43dc3b4c63f4cd1ac974a973c39363e | 769694705593400528 | FINISHED |   |
| 2 | 970232f6316e42058b34e6e8cc53b372 | 769694705593400528 | FINISHED |   |
| 3 | d7416a70e47f43cc88f46b47c1d4a0e9 | 769694705593400528 | FINISHED |   |
| 4 | 273614647c184160ae1ff089749e3d6c | 769694705593400528 | FINISHED |   |

|   | artifact_uri                                      | \ |
|---|---------------------------------------------------|---|
| 0 | mlflow-artifacts:/769694705593400528/1ad85eaa4... |   |
| 1 | mlflow-artifacts:/769694705593400528/b43dc3b4c... |   |
| 2 | mlflow-artifacts:/769694705593400528/970232f63... |   |
| 3 | mlflow-artifacts:/769694705593400528/d7416a70e... |   |
| 4 | mlflow-artifacts:/769694705593400528/273614647... |   |

|   | start_time                       | end_time                         | \ |
|---|----------------------------------|----------------------------------|---|
| 0 | 2025-09-10 13:46:50.534000+00:00 | 2025-09-10 13:47:10.501000+00:00 |   |
| 1 | 2025-09-10 13:46:31.540000+00:00 | 2025-09-10 13:46:50.508000+00:00 |   |
| 2 | 2025-09-10 13:46:11.677000+00:00 | 2025-09-10 13:46:31.515000+00:00 |   |
| 3 | 2025-09-10 13:46:11.582000+00:00 | 2025-09-10 13:47:10.619000+00:00 |   |
| 4 | 2025-09-10 13:45:50.331000+00:00 | 2025-09-10 13:46:11.430000+00:00 |   |

|   | metrics.train_loss | metrics.val_loss | metrics.val_r2 | \ |
|---|--------------------|------------------|----------------|---|
| 0 | 11.719401          | 13.077980        | -17.556817     |   |
| 1 | 10.831029          | 12.169875        | -15.929829     |   |
| 2 | 11.439570          | 12.968844        | -17.616027     |   |
| 3 | NaN                | NaN              | NaN            |   |
| 4 | 1.315273           | 2.483097         | -2.523356      |   |

|   | metrics.avg_train_mse | ... | params.weight_init_method | params.num_epochs | \ |
|---|-----------------------|-----|---------------------------|-------------------|---|
| 0 | NaN                   | ... | xavier                    | 500               |   |
| 1 | NaN                   | ... | xavier                    | 500               |   |
| 2 | NaN                   | ... | xavier                    | 500               |   |
| 3 | 11.33                 | ... | xavier                    | 500               |   |
| 4 | NaN                   | ... | xavier                    | 500               |   |

|   | params.learning_rate | params.batch_size | params.lambda | \ |
|---|----------------------|-------------------|---------------|---|
| 0 | 0.0001               | 50.0              | NaN           |   |
| 1 | 0.0001               | 50.0              | NaN           |   |
| 2 | 0.0001               | 50.0              | NaN           |   |
| 3 | 0.0001               | 50.0              | NaN           |   |
| 4 | 0.0010               | 50.0              | NaN           |   |

|   | tags.mlflow.parentRunId          | tags.mlflow.source.type | \ |
|---|----------------------------------|-------------------------|---|
| 0 | d7416a70e47f43cc88f46b47c1d4a0e9 | LOCAL                   |   |

|   |                                  |       |
|---|----------------------------------|-------|
| 1 | d7416a70e47f43cc88f46b47c1d4a0e9 | LOCAL |
| 2 | d7416a70e47f43cc88f46b47c1d4a0e9 | LOCAL |
| 3 | NaN                              | LOCAL |
| 4 | d0e1c94b2067451e9561e5e0ddce109e | LOCAL |

|   | tags.mlflow.source.name                           | \ |
|---|---------------------------------------------------|---|
| 0 | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |   |
| 1 | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |   |
| 2 | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |   |
| 3 | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |   |
| 4 | d:\AT82.03-ML-Car-Price-Prediction\.venv\Lib\s... |   |

|   | tags.mlflow.runName                               | tags.mlflow.user |
|---|---------------------------------------------------|------------------|
| 0 | Fold-2                                            | promb            |
| 1 | Fold-1                                            | promb            |
| 2 | Fold-0                                            | promb            |
| 3 | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... | promb            |
| 4 | Fold-2                                            | promb            |

[5 rows x 25 columns]

```
[73]: run_df_filtered = run_df.drop(
        columns=[
            "tags.mlflow.source.name",
            "tags.mlflow.source.type",
            "tags.mlflow.user",
            "status",
            "experiment_id",
            "artifact_uri",
        ],
    )
run_df_filtered.head(5)
```

| [73]: | run_id                           | start_time                       | \ |
|-------|----------------------------------|----------------------------------|---|
| 0     | 1ad85eaa459048d68b07875036290022 | 2025-09-10 13:46:50.534000+00:00 |   |
| 1     | b43dc3b4c63f4cd1ac974a973c39363e | 2025-09-10 13:46:31.540000+00:00 |   |
| 2     | 970232f6316e42058b34e6e8cc53b372 | 2025-09-10 13:46:11.677000+00:00 |   |
| 3     | d7416a70e47f43cc88f46b47c1d4a0e9 | 2025-09-10 13:46:11.582000+00:00 |   |
| 4     | 273614647c184160ae1ff089749e3d6c | 2025-09-10 13:45:50.331000+00:00 |   |

|   | end_time                         | metrics.train_loss | metrics.val_loss | \ |
|---|----------------------------------|--------------------|------------------|---|
| 0 | 2025-09-10 13:47:10.501000+00:00 | 11.719401          | 13.077980        |   |
| 1 | 2025-09-10 13:46:50.508000+00:00 | 10.831029          | 12.169875        |   |
| 2 | 2025-09-10 13:46:31.515000+00:00 | 11.439570          | 12.968844        |   |
| 3 | 2025-09-10 13:47:10.619000+00:00 | NaN                | NaN              |   |
| 4 | 2025-09-10 13:46:11.430000+00:00 | 1.315273           | 2.483097         |   |

|   | metrics.val_r2 | metrics.avg_train_mse | metrics.avg_val_r2 | \ |
|---|----------------|-----------------------|--------------------|---|
| 0 | -17.556817     | NaN                   | NaN                |   |
| 1 | -15.929829     | NaN                   | NaN                |   |
| 2 | -17.616027     | NaN                   | NaN                |   |
| 3 | NaN            | 11.33                 | -17.034224         |   |
| 4 | -2.523356      | NaN                   | NaN                |   |

|   | metrics.avg_val_mse | params.train_method | params.momentum | params.model_type | \ |
|---|---------------------|---------------------|-----------------|-------------------|---|
| 0 | NaN                 | mini-batch          | 0.9             | Polynomial        |   |
| 1 | NaN                 | mini-batch          | 0.9             | Polynomial        |   |
| 2 | NaN                 | mini-batch          | 0.9             | Polynomial        |   |
| 3 | 12.7389             | mini-batch          | 0.9             | Polynomial        |   |
| 4 | NaN                 | mini-batch          | 0.9             | Polynomial        |   |

|   | params.weight_init_method | params.num_epochs | params.learning_rate | \ |
|---|---------------------------|-------------------|----------------------|---|
| 0 | xavier                    | 500               | 0.0001               |   |
| 1 | xavier                    | 500               | 0.0001               |   |
| 2 | xavier                    | 500               | 0.0001               |   |
| 3 | xavier                    | 500               | 0.0001               |   |
| 4 | xavier                    | 500               | 0.0010               |   |

|   | params.batch_size | params.lambda | tags.mlflow.parentRunId          | \ |
|---|-------------------|---------------|----------------------------------|---|
| 0 | 50.0              | NaN           | d7416a70e47f43cc88f46b47c1d4a0e9 |   |
| 1 | 50.0              | NaN           | d7416a70e47f43cc88f46b47c1d4a0e9 |   |
| 2 | 50.0              | NaN           | d7416a70e47f43cc88f46b47c1d4a0e9 |   |
| 3 | 50.0              | NaN           | NaN                              |   |
| 4 | 50.0              | NaN           | d0e1c94b2067451e9561e5e0ddce109e |   |

|   | tags.mlflow.runName                               |
|---|---------------------------------------------------|
| 0 | Fold-2                                            |
| 1 | Fold-1                                            |
| 2 | Fold-0                                            |
| 3 | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... |
| 4 | Fold-2                                            |

```
[75]: # Separate parent and child runs for easier viewing
parent_runs = run_df_filtered[run_df_filtered["tags.mlflow.parentRunId"].
↳isna()].drop(
    columns=["tags.mlflow.parentRunId", "metrics.train_loss", "metrics.
↳val_loss", "metrics.val_r2"],
)
child_runs = run_df_filtered[~run_df_filtered["tags.mlflow.parentRunId"].
↳isna()].drop(
    columns=["metrics.avg_train_mse", "metrics.avg_val_mse", "metrics.
↳avg_val_r2"],
)
```

```
[80]: parent_runs.head(5)
```

```
[80]:
```

|    | run_id                           | start_time                       | \ |
|----|----------------------------------|----------------------------------|---|
| 3  | d7416a70e47f43cc88f46b47c1d4a0e9 | 2025-09-10 13:46:11.582000+00:00 |   |
| 7  | d0e1c94b2067451e9561e5e0ddce109e | 2025-09-10 13:45:09.971000+00:00 |   |
| 11 | 0718dd6320fc49a4915940752ffec4a8 | 2025-09-10 13:45:09.095000+00:00 |   |
| 15 | c62120a95e5947afb87cd09c71ce1f33 | 2025-09-10 13:44:08.302000+00:00 |   |
| 19 | e8e32659147d4432b4310480005c113f | 2025-09-10 13:43:08.525000+00:00 |   |

|    | end_time                         | metrics.avg_train_mse | \ |
|----|----------------------------------|-----------------------|---|
| 3  | 2025-09-10 13:47:10.619000+00:00 | 11.330000             |   |
| 7  | 2025-09-10 13:46:11.553000+00:00 | 2.380396              |   |
| 11 | 2025-09-10 13:45:09.946000+00:00 | inf                   |   |
| 15 | 2025-09-10 13:45:09.071000+00:00 | 16.331883             |   |
| 19 | 2025-09-10 13:44:08.275000+00:00 | 2.804363              |   |

|    | metrics.avg_val_r2 | metrics.avg_val_mse | params.train_method | \ |
|----|--------------------|---------------------|---------------------|---|
| 3  | -17.034224         | 12.738900           | mini-batch          |   |
| 7  | -2.795645          | 2.683493            | mini-batch          |   |
| 11 | -inf               | inf                 | mini-batch          |   |
| 15 | -17.056583         | 12.754868           | mini-batch          |   |
| 19 | -2.731033          | 2.638130            | mini-batch          |   |

|    | params.momentum | params.model_type | params.weight_init_method | \ |
|----|-----------------|-------------------|---------------------------|---|
| 3  | 0.9             | Polynomial        | xavier                    |   |
| 7  | 0.9             | Polynomial        | xavier                    |   |
| 11 | 0.9             | Polynomial        | xavier                    |   |
| 15 | 0.9             | Polynomial        | zero                      |   |
| 19 | 0.9             | Polynomial        | zero                      |   |

|    | params.num_epochs | params.learning_rate | params.batch_size | params.lambda | \ |
|----|-------------------|----------------------|-------------------|---------------|---|
| 3  | 500               | 0.0001               | 50.0              | NaN           |   |
| 7  | 500               | 0.0010               | 50.0              | NaN           |   |
| 11 | 500               | 0.0100               | 50.0              | NaN           |   |
| 15 | 500               | 0.0001               | 50.0              | NaN           |   |
| 19 | 500               | 0.0010               | 50.0              | NaN           |   |

|    | tags.mlflow.runName                               |
|----|---------------------------------------------------|
| 3  | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... |
| 7  | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... |
| 11 | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... |
| 15 | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... |
| 19 | Reg-Polynomial_TrainMethod-mini-batch_BatchSiz... |

```
[81]: child_runs.head(5)
```

```

[81]:
run_id start_time \
0 1ad85eaa459048d68b07875036290022 2025-09-10 13:46:50.534000+00:00
1 b43dc3b4c63f4cd1ac974a973c39363e 2025-09-10 13:46:31.540000+00:00
2 970232f6316e42058b34e6e8cc53b372 2025-09-10 13:46:11.677000+00:00
4 273614647c184160ae1ff089749e3d6c 2025-09-10 13:45:50.331000+00:00
5 cdaacd5bccfe4126b9021f26ca180e2b 2025-09-10 13:45:30.313000+00:00

end_time metrics.train_loss metrics.val_loss \
0 2025-09-10 13:47:10.501000+00:00 11.719401 13.077980
1 2025-09-10 13:46:50.508000+00:00 10.831029 12.169875
2 2025-09-10 13:46:31.515000+00:00 11.439570 12.968844
4 2025-09-10 13:46:11.430000+00:00 1.315273 2.483097
5 2025-09-10 13:45:50.304000+00:00 2.327555 2.890049

metrics.val_r2 params.train_method params.momentum params.model_type \
0 -17.556817 mini-batch 0.9 Polynomial
1 -15.929829 mini-batch 0.9 Polynomial
2 -17.616027 mini-batch 0.9 Polynomial
4 -2.523356 mini-batch 0.9 Polynomial
5 -3.020422 mini-batch 0.9 Polynomial

params.weight_init_method params.num_epochs params.learning_rate \
0 xavier 500 0.0001
1 xavier 500 0.0001
2 xavier 500 0.0001
4 xavier 500 0.0010
5 xavier 500 0.0010

params.batch_size params.lambda tags.mlflow.parentRunId \
0 50.0 NaN d7416a70e47f43cc88f46b47c1d4a0e9
1 50.0 NaN d7416a70e47f43cc88f46b47c1d4a0e9
2 50.0 NaN d7416a70e47f43cc88f46b47c1d4a0e9
4 50.0 NaN d0e1c94b2067451e9561e5e0ddce109e
5 50.0 NaN d0e1c94b2067451e9561e5e0ddce109e

tags.mlflow.runName
0 Fold-2
1 Fold-1
2 Fold-0
4 Fold-2
5 Fold-1

```

## 2.6.2 Checking Exploding Gradients

- Exploding gradient can be checked from folds with infinity and nan

```
[96]: # Learning rate 0.01 caused overflow in a lot of runs due to exploding
      ↪gradients from large step size
print(np.isinf(child_runs[child_runs["params.learning_rate"] == 0.01]["metrics.
      ↪val_loss"]).sum())
print(np.isinf(child_runs[child_runs["params.learning_rate"] == 0.001]["metrics.
      ↪val_loss"]).sum())
print(np.isinf(child_runs[child_runs["params.learning_rate"] == 0.
      ↪0001]["metrics.val_loss"]).sum())
```

96

4

0

```
[98]: print(np.isnan(child_runs[child_runs["params.learning_rate"] == 0.01]["metrics.
      ↪val_loss"]).sum())
print(np.isnan(child_runs[child_runs["params.learning_rate"] == 0.001]["metrics.
      ↪val_loss"]).sum())
print(np.isnan(child_runs[child_runs["params.learning_rate"] == 0.
      ↪0001]["metrics.val_loss"]).sum())
```

48

0

0

```
[99]: print(np.isinf(child_runs[child_runs["params.train_method"] ==
      ↪"batch"]["metrics.val_loss"]).sum())
print(np.isinf(child_runs[child_runs["params.train_method"] ==
      ↪"mini-batch"]["metrics.val_loss"]).sum())
print(np.isinf(child_runs[child_runs["params.train_method"] ==
      ↪"stochastic"]["metrics.val_loss"]).sum())
```

48

48

4

```
[101]: print(np.isnan(child_runs[child_runs["params.train_method"] ==
      ↪"batch"]["metrics.val_loss"]).sum())
print(np.isnan(child_runs[child_runs["params.train_method"] ==
      ↪"mini-batch"]["metrics.val_loss"]).sum())
print(np.isnan(child_runs[child_runs["params.train_method"] ==
      ↪"stochastic"]["metrics.val_loss"]).sum())
# This is interesting that stochastic has a lot of np.nan instead of inf which
      ↪I will explain in summary section
```

0

0

48



## 2.7 Checking the average performance by features

- I have to filter out the infinity values so the result is calculable so the results will be just some samples of results that don't have extreme values, that means the result in this section might not represent entire experiments but they should still give similar trends and some ideas about the parameters

```
[139]: parent_runs[["params.learning_rate", "metrics.avg_val_mse", "metrics.
        ↪avg_val_r2"]].replace(
        [np.inf, -np.inf],
        np.nan,
    ).dropna().groupby("params.learning_rate").mean()
```

```
[139]:
```

|                      | metrics.avg_val_mse | metrics.avg_val_r2 |
|----------------------|---------------------|--------------------|
| params.learning_rate |                     |                    |
| 0.0001               | 10.673834           | -14.108025         |
| 0.0010               | 6.618113            | -8.367404          |

```
[140]: parent_runs[["params.model_type", "metrics.avg_val_mse", "metrics.avg_val_r2"]].
        ↪replace(
        [np.inf, -np.inf],
        np.nan,
    ).dropna().groupby("params.model_type").mean()
```

```
[140]:
```

|                   | metrics.avg_val_mse | metrics.avg_val_r2 |
|-------------------|---------------------|--------------------|
| params.model_type |                     |                    |
| Lasso             | 8.568102            | -11.128174         |
| Normal            | 8.577094            | -11.139239         |
| Polynomial        | 9.160368            | -11.965349         |
| Ridge             | 8.490184            | -11.017926         |

```
[141]: parent_runs[["params.train_method", "metrics.avg_val_mse", "metrics.
        ↪avg_val_r2"]].replace(
        [np.inf, -np.inf],
        np.nan,
    ).dropna().groupby("params.train_method").mean()
```

```
[141]:
```

|                     | metrics.avg_val_mse | metrics.avg_val_r2 |
|---------------------|---------------------|--------------------|
| params.train_method |                     |                    |
| batch               | 20.526496           | -28.055099         |
| mini-batch          | 4.735236            | -5.701469          |
| stochastic          | 0.280060            | 0.604145           |

```
[142]: parent_runs[["params.weight_init_method", "metrics.avg_val_mse", "metrics.
        ↪avg_val_r2"]].replace(
        [np.inf, -np.inf],
        np.nan,
    ).dropna().groupby("params.weight_init_method").mean()
```

```
[142]:
```

|                           | metrics.avg_val_mse | metrics.avg_val_r2 |
|---------------------------|---------------------|--------------------|
| params.weight_init_method |                     |                    |
| xavier                    | 8.716029            | -11.336868         |
| zero                      | 8.662210            | -11.260702         |

```
[147]: parent_runs[["params.momentum", "metrics.avg_val_mse", "metrics.avg_val_r2"]].
↳replace(
    [np.inf, -np.inf],
    np.nan,
).dropna().groupby("params.momentum").mean()
```

```
[147]:
```

|                 | metrics.avg_val_mse | metrics.avg_val_r2 |
|-----------------|---------------------|--------------------|
| params.momentum |                     |                    |
| 0.0             | 6.576299            | -8.308769          |
| 0.9             | 10.713906           | -14.164217         |

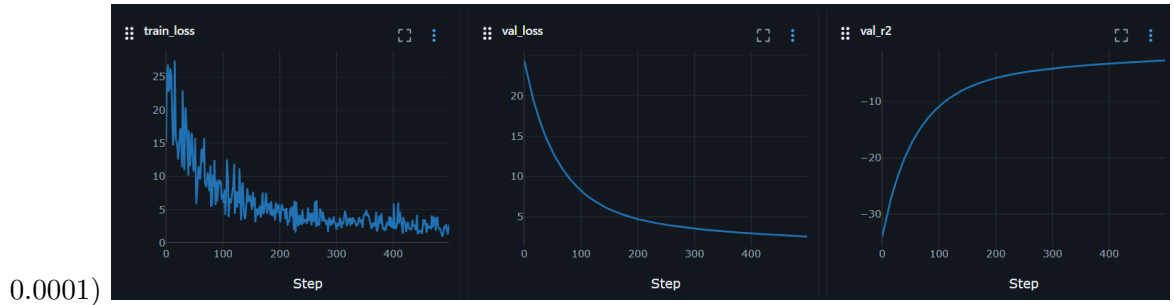
## 2.8 Summary

### 2.8.1 About Parameters

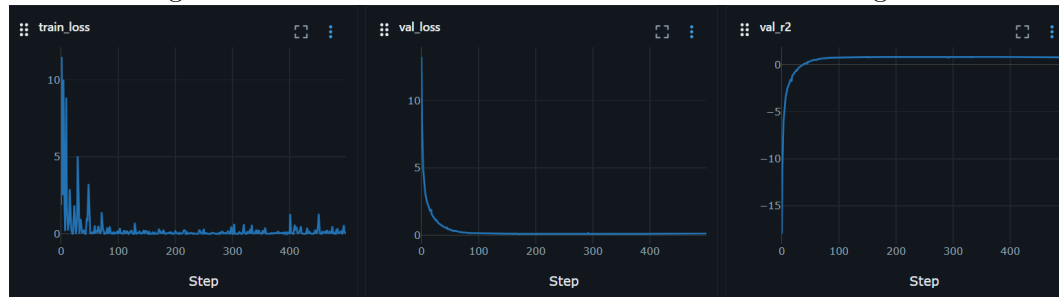
- Learning rate 0.01 is too high and cause most of the models' gradients to explode into infinity.
- Penalty does make the model better overall (a little bit lower MSE)
- Polynomial features reduce the model performance, probably due to too many features was included.
- Xavier weight initialization doesn't help much
- Batch Gradient Descent takes too long to converge, 500 epochs is not enough, resulting in high loss.
- Batch Gradient Descent can stuck in local minima, as this image shows MSE around 20 but the model looks converged (the image below is from Reg-Normal\_TrainMethod-batch\_Momentum-0\_WeightInit-zero\_LR-0.0001)



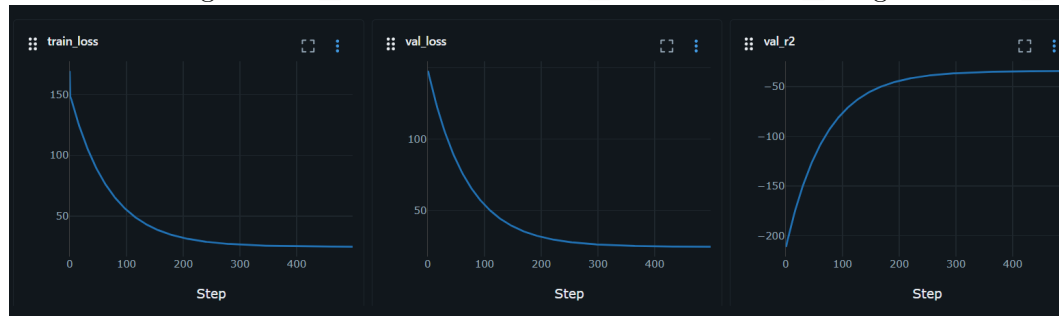
- Mini-batch Gradient Descent makes model training a little bit more shaky for training loss but converge a lot faster than batch gradient descent. (the image below is from Reg-Normal\_TrainMethod-mini-batch\_BatchSize-50\_Momentum-0\_WeightInit-zero\_LR-



- Stochastic Gradient Descent makes model training super shaky, this cause a lot of models' gradients to go into infinity. But with lowest learning rate 0.0001, it can be powerful as it can push model to converge into the lowest MSE (the image below is from Reg-Normal\_TrainMethod-stochastic\_Momentum-0\_WeightInit-zero\_LR-0.0001)



- Momentum somehow make the model converge slower, probably due to the momentum implemented is the most basic form of momentum (the image below is from Reg-Normal\_TrainMethod-batch\_Momentum-0.9\_WeightInit-zero\_LR-0.0001)



## 2.8.2 About Feature Importance

**year**, **max\_power**, **engine** have high positive coefficients, which means if these features increase, the price increases. Next, **transmission** (Manual), **fuel** (Petrol), higher number of **owner** have negative coefficients, meaning if these features increase, the price decreases, which aligns with Pearson correlation in Assignment 1. For **brand**, even though some brands are expensive, it got almost zero coefficient, meaning that the model is ignoring it due to the large range of possible categorical values.

### 2.8.3 MLFlow Screenshot

| Run Name                                                                                    | Created    | Duration |
|---------------------------------------------------------------------------------------------|------------|----------|
| Reg-Polynomial_TrainMethod-mini-batch_BatchSize-50_Momentum-0.9_WeightInit-xavier_LR-0.0001 | 2 days ago | 59.0s    |
| Fold-2                                                                                      | 2 days ago | 20.0s    |
| Fold-1                                                                                      | 2 days ago | 19.0s    |
| Fold-0                                                                                      | 2 days ago | 19.8s    |
| Reg-Polynomial_TrainMethod-mini-batch_BatchSize-50_Momentum-0.9_WeightInit-xavier_LR-0.001  | 2 days ago | 1.0min   |
| Reg-Polynomial_TrainMethod-mini-batch_BatchSize-50_Momentum-0.9_WeightInit-xavier_LR-0.01   | 2 days ago | 0.9s     |
| Reg-Polynomial_TrainMethod-mini-batch_BatchSize-50_Momentum-0.9_WeightInit-zero_LR-0.0001   | 2 days ago | 1.0min   |
| Fold-2                                                                                      | 2 days ago | 19.8s    |
| Fold-1                                                                                      | 2 days ago | 20.4s    |
| Fold-0                                                                                      | 2 days ago | 20.4s    |
| Reg-Polynomial_TrainMethod-mini-batch_BatchSize-50_Momentum-0.9_WeightInit-zero_LR-0.001    | 2 days ago | 59.8s    |
| Reg-Polynomial_TrainMethod-mini-batch_BatchSize-50_Momentum-0.9_WeightInit-zero_LR-0.01     | 2 days ago | 0.9s     |
| Reg-Polynomial_TrainMethod-stochastic_Momentum-0.9_WeightInit-xavier_LR-0.0001              | 2 days ago | 1.3min   |
| Reg-Polynomial_TrainMethod-stochastic_Momentum-0.9_WeightInit-xavier_LR-0.001               | 2 days ago | 1.8min   |

576 matching runs

```
[155]: top20 = params_runs_results.nsmallest(20, "metrics.avg_val_mse")
top20["tags.mlflow.runName"] = top20["tags.mlflow.runName"].str.replace("Reg-", "" , regex=False)
top20["tags.mlflow.runName"] = top20["tags.mlflow.runName"].str.replace("TrainMethod-", "" , regex=False)
top20["tags.mlflow.runName"] = top20["tags.mlflow.runName"].str.replace("WeightInit-", "" , regex=False)
# All model use 50 for mini-batch size so remove that part to make it shorter
top20["tags.mlflow.runName"] = top20["tags.mlflow.runName"].str.replace("BatchSize-50_", "" , regex=False)
top20["tags.mlflow.runName"] = top20["tags.mlflow.runName"].str.replace("LR-", "" , regex=False)
top20[["tags.mlflow.runName", "metrics.avg_val_mse", "metrics.avg_val_r2"]]
```

```
[155]:
```

|     | tags.mlflow.runName                            | metrics.avg_val_mse | \ |
|-----|------------------------------------------------|---------------------|---|
| 531 | Normal_stochastic_Momentum-0_xavier_0.0001     | 0.096030            |   |
| 387 | Ridge_stochastic_Momentum-0_xavier_0.0001      | 0.101890            |   |
| 111 | Polynomial_stochastic_Momentum-0_zero_0.0001   | 0.104705            |   |
| 99  | Polynomial_stochastic_Momentum-0_xavier_0.0001 | 0.109274            |   |
| 543 | Normal_stochastic_Momentum-0_zero_0.0001       | 0.113151            |   |

|     |                                               |          |
|-----|-----------------------------------------------|----------|
| 319 | Ridge_stochastic_Momentum-0.9_xavier_0.001    | 0.113451 |
| 243 | Lasso_stochastic_Momentum-0_xavier_0.0001     | 0.114229 |
| 255 | Lasso_stochastic_Momentum-0_zero_0.0001       | 0.116633 |
| 475 | Normal_stochastic_Momentum-0.9_zero_0.001     | 0.116665 |
| 399 | Ridge_stochastic_Momentum-0_zero_0.0001       | 0.119842 |
| 535 | Normal_stochastic_Momentum-0_xavier_0.001     | 0.121623 |
| 379 | Ridge_mini-batch_Momentum-0_zero_0.001        | 0.123774 |
| 223 | Lasso_mini-batch_Momentum-0_xavier_0.001      | 0.125478 |
| 367 | Ridge_mini-batch_Momentum-0_xavier_0.001      | 0.125891 |
| 511 | Normal_mini-batch_Momentum-0_xavier_0.001     | 0.127729 |
| 235 | Lasso_mini-batch_Momentum-0_zero_0.001        | 0.130447 |
| 43  | Polynomial_stochastic_Momentum-0.9_zero_0.001 | 0.140380 |
| 523 | Normal_mini-batch_Momentum-0_zero_0.001       | 0.140628 |
| 79  | Polynomial_mini-batch_Momentum-0_xavier_0.001 | 0.141509 |
| 547 | Normal_stochastic_Momentum-0_zero_0.001       | 0.142542 |

|     | metrics.avg_val_r2 |
|-----|--------------------|
| 531 | 0.863948           |
| 387 | 0.855629           |
| 111 | 0.851565           |
| 99  | 0.845087           |
| 543 | 0.839384           |
| 319 | 0.839120           |
| 243 | 0.838236           |
| 255 | 0.834802           |
| 475 | 0.835160           |
| 399 | 0.830205           |
| 535 | 0.827998           |
| 379 | 0.824714           |
| 223 | 0.822335           |
| 367 | 0.821734           |
| 511 | 0.819117           |
| 235 | 0.815207           |
| 43  | 0.800400           |
| 523 | 0.801253           |
| 79  | 0.800049           |
| 547 | 0.799052           |

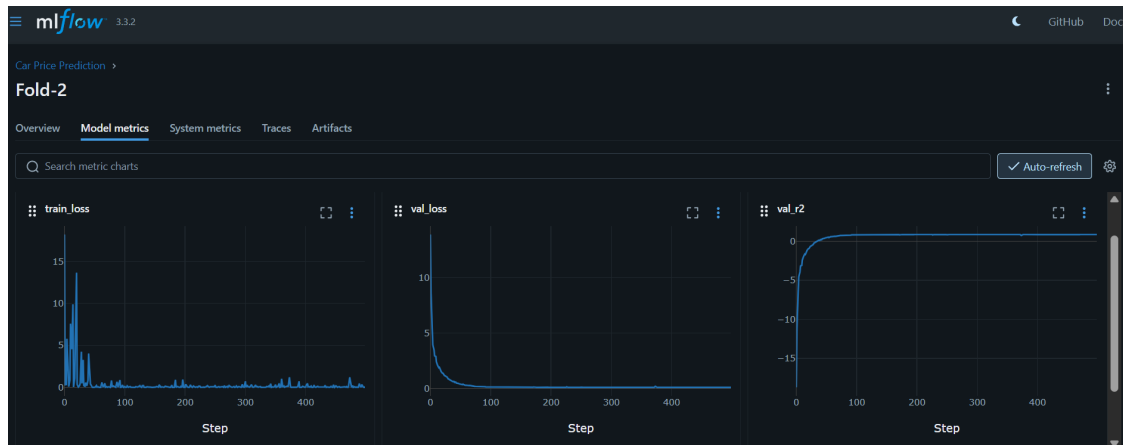
#### 2.8.4 Best model

The best parameter I get is

- Normal Linear Regression (No Penalty)
- Stochastic training method
- No momentum
- Xavier weight initialization
- Learning rate 0.0001

Below is the image showing one of the cv fold trained using this parameter set. It started with very

chaotic training loss after each epoch since this is a stochastic weight update. But it eventually went down pretty fast which result in validation loss soon followed. At around 50 epochs, validation loss and R2 converged into preferable values.



### 2.8.5 Extra

When training with a high learning rate, infinity and nan values occurred in the experiment training loops. Interestingly, the nan values are found in parameter settings that involve Stochastic training method but not Batch and Mini-batch which just have infinity or negative infinity. After some investigations, I found that if there is no early stopping and more epochs, all training methods will eventually become nan instead of infinity. The cause of this is when  $\hat{y}$  becomes infinity, calculating gradient involves multiplying features  $X.T$  with  $(\hat{y} - y_{true})$ , since  $\hat{y}$  becomes  $np.inf$ ,  $(\hat{y} - y_{true})$  also becomes  $np.inf$ , then when multiplying  $X.T$  with error term, if  $X.T$  has no zero, the result will just be  $np.inf$  or  $-np.inf$ , but if there is one zero in  $X.T$ , the result will be  $0 * np.inf = np.nan$ , then updating theta corrupts the theta with nan, then the next training loop will cascade  $np.nan$  everywhere through  $X @ \theta$ .